Space Exploration & Aerospace Engineering

Part XII

Emerging Frontiers

The next era of spaceflight will service, refuel and build things in orbit, use resources from the Moon and asteroids, and move with new kinds of propulsion. Part XII explains the engineering and the economics behind these frontiers.



Chapters 40–42 · pages 735–796 of the complete manual

First Edition · Current to October 2026 · Written in plain language with deep technical detail

ENGINEERING REFERENCE MANUAL

Space Exploration & Aerospace Engineering

A complete, illustrated guide to the space ecosystem — from stratospheric balloons to interstellar probes — explaining how every major class of space system works, how it is engineered, and where artificial intelligence is changing it.

First Edition · Information current to October 2026
A4 format · Fourteen Parts · Fifty-three chapters · Ten appendices
Version 1.0 · October 2026

About this edition

Space Exploration & Aerospace Engineering — An Engineering Reference Manual. First Edition, Version 1.0, October 2026.

This manual was written to be read by students, engineers, analysts, managers and curious readers. Each topic starts with plain-language explanations and builds up to the equations, numbers and design trade-offs that practising engineers use. Sentences are kept short and every technical term is explained in plain words when it first appears; the glossary (Appendix H) and the list of acronyms (Appendix G) collect them. Facts, mission status and programme milestones are current to October 2026; space programmes change quickly, so dates of future events should be read as plans, not certainties.

Figures, photographs and data

Diagrams, charts and graphs were drawn specifically for this manual from published physical relationships and public data. Photographs are from the NASA Image and Video Library and are used under NASA's media usage guidelines (NASA material is generally not subject to copyright in the United States); individual credits appear with each image and in Appendix J. NASA does not endorse this publication. Numerical values (masses, power levels, performance) are representative published figures and may differ slightly between sources and mission phases.

Safety and responsibility

This is an educational reference. It is not a design standard, a flight rule, or a substitute for the requirements, analyses and verification processes of a qualified engineering organisation. Chapters dealing with military and intelligence systems use only openly published information and describe principles, not operational or classified details.

How numbers are written

SI units are used throughout, with common aerospace units (nautical miles, feet, pounds-force) shown where the industry uses them. Large numbers use thin spacing (e.g., 35 786 km). "≈" means approximately, "~" means order-of-magnitude. Dates follow the ISO format (YYYY-MM-DD) in tables.

Typography and production

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This volume is one Part of *Space Exploration & Aerospace Engineering — An Engineering Reference Manual*. Page numbers, chapter numbers and figure numbers are the same as in the complete manual, so cross-references to other chapters (for example “Chapter 24”) and to the appendices refer to the complete book.



PART XII

Emerging Frontiers

The next era of spaceflight will service, refuel and build things in orbit, use resources from the Moon and asteroids, and move with new kinds of propulsion. Part XII explains the engineering and the economics behind these frontiers.

40 In-Space Servicing, Assembly and Manufacturing

41 Space Resources and Asteroid Mining

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Image: NASA/JSC (Terry Virts)

CHAPTER

40 In-Space Servicing, Assembly and Manufacturing



Image: NASA/JSC

Almost every satellite ever launched has been used once and thrown away, retired with working instruments because its propellant ran out or a single part failed. In-space servicing, assembly and manufacturing (ISAM) aims to change that by inspecting, refuelling, repairing, upgrading, moving and disposing of spacecraft in orbit, building structures too large for any rocket fairing, and making products that can only be made in weightlessness. This chapter traces servicing from the Skylab and Hubble spacewalks to the commercial robots docking with communication satellites in 2020–26, explains the engineering of rendezvous, capture and refuelling, and estimates the economics of life extension, debris removal and orbital propellant depots, with the state of every major programme as of October 2026.

IN THIS CHAPTER

40.1 What ISAM Means and Why It Matters
40.2 Sixty Years of Servicing
40.3 Rendezvous, Proximity Operations and Capture
40.4 The Economics of Life Extension and Debris Removal
40.5 Assembling Large Structures in Orbit
40.6 Manufacturing in Space
40.7 Refuelling for Exploration

A geostationary communication satellite costs several hundred million dollars to build, launch and insure, and is designed to work for about 15 years. When it retires, its transponders usually still work: what it has run out of is the few hundred kilograms of propellant needed to stay in its slot. A science observatory can be crippled by one failed gyroscope; a broken solar-array hinge can end a mission worth billions. On Earth, no one would scrap a car because its fuel tank was empty or a tyre was flat. In orbit, until recently, there was no alternative.

In-space servicing, assembly and manufacturing (ISAM) is the set of technologies that treat spacecraft as maintainable, upgradeable and buildable in place. The term was adopted in the United States' *ISAM National Strategy* of April 2022; NASA previously used **OSAM** (on-orbit servicing, assembly and manufacturing), and European documents speak of **in-orbit servicing (IOS)**. The idea is old — astronauts repaired Skylab in 1973 — but for forty years it depended on the Space Shuttle and its crews. What changed after 2019 is that robotic servicers began to dock with commercial satellites as a paid service. In the same period China demonstrated similar capabilities in geostationary orbit, and refuelling became central to plans for returning people to the Moon.

This chapter follows the three strands of ISAM in turn. Servicing extends and repairs existing spacecraft; assembly builds structures larger than any launcher can carry; manufacturing makes things — parts, fibres, crystals, tissues — in orbit. Rendezvous and proximity operations, the common foundation of all three, are treated in detail. The orbital mechanics of relative motion is introduced in Chapter 3, and the security implications of satellites that approach other satellites in Chapter 19.

40.1 What ISAM Means and Why It Matters

Table 40.1 lists the main ISAM activities, from the simplest — looking at another spacecraft from close range — to the most ambitious. They share a ladder of difficulty. **Inspection** needs only safe proximity operations. **Relocation** and **life extension** need a physical connection that can carry thrust loads. **Refuelling** adds leak-tight fluid connections; **repair and upgrade** add dexterous robotics; **assembly** and **manufacturing** require all of these to be repeatable, hundreds of times, without people present.

TABLE 40.1 The main ISAM activities, in rough order of increasing difficulty.

Activity	What it does	Typical client	Status (Oct 2026)
Inspection	Close-range imaging of a satellite or debris object	Any; damaged or unknown objects	Operational (commercial and military)
Relocation and disposal	Tow to a new slot, graveyard orbit or re-entry	Dead or retiring satellites; debris	Demonstrated in GEO (2020, 2022)
Life extension	Attach a vehicle or pod that takes over attitude and orbit control	GEO satellites short of propellant	Commercial since 2020
Refuelling	Transfer propellant through a valve	Prepared clients (port); later unprepared	Demonstrated 2007; commercial demos 2026–27
Repair and upgrade	Replace units, free stuck mechanisms, add payloads	Prepared or high-value clients	Crewed (Hubble); robotic in development
Assembly	Join modules, trusses or mirror segments	Stations, antennas, telescopes	Crewed (ISS); robotic demos planned
Manufacturing	Print, draw, crystallise or grow material in orbit	Parts, fibres, drugs, semiconductors	Demonstrations on ISS and free-flyers

The arguments for ISAM are economic, scientific and environmental:

- **Life extension.** A satellite that keeps earning for five more years defers the cost of a replacement worth hundreds of millions of dollars.

- **Repair and upgrade.** Hubble was launched with a flawed mirror and 1980s electronics; five servicing missions turned it into one of the most productive scientific instruments ever built (Chapter 38).
- **Debris removal.** The objects that threaten long-term sustainability in low Earth orbit are a few thousand large derelicts (Chapter 21); only something that can rendezvous with and grip them can remove them.
- **Size.** A rocket fairing limits a monolithic structure to about 5 m (or 9 m for Starship). Larger telescopes, antennas and solar power stations must be folded or built in orbit.
- **New products.** Weightlessness suppresses convection and sedimentation, enabling purer glasses and crystals; and refuelling in orbit multiplies what one rocket can send beyond it.

40.2 Sixty Years of Servicing

The history of servicing has two halves: four decades in which astronauts did the work, and a commercial robotic era that began in 2020 (Figure 40.1).

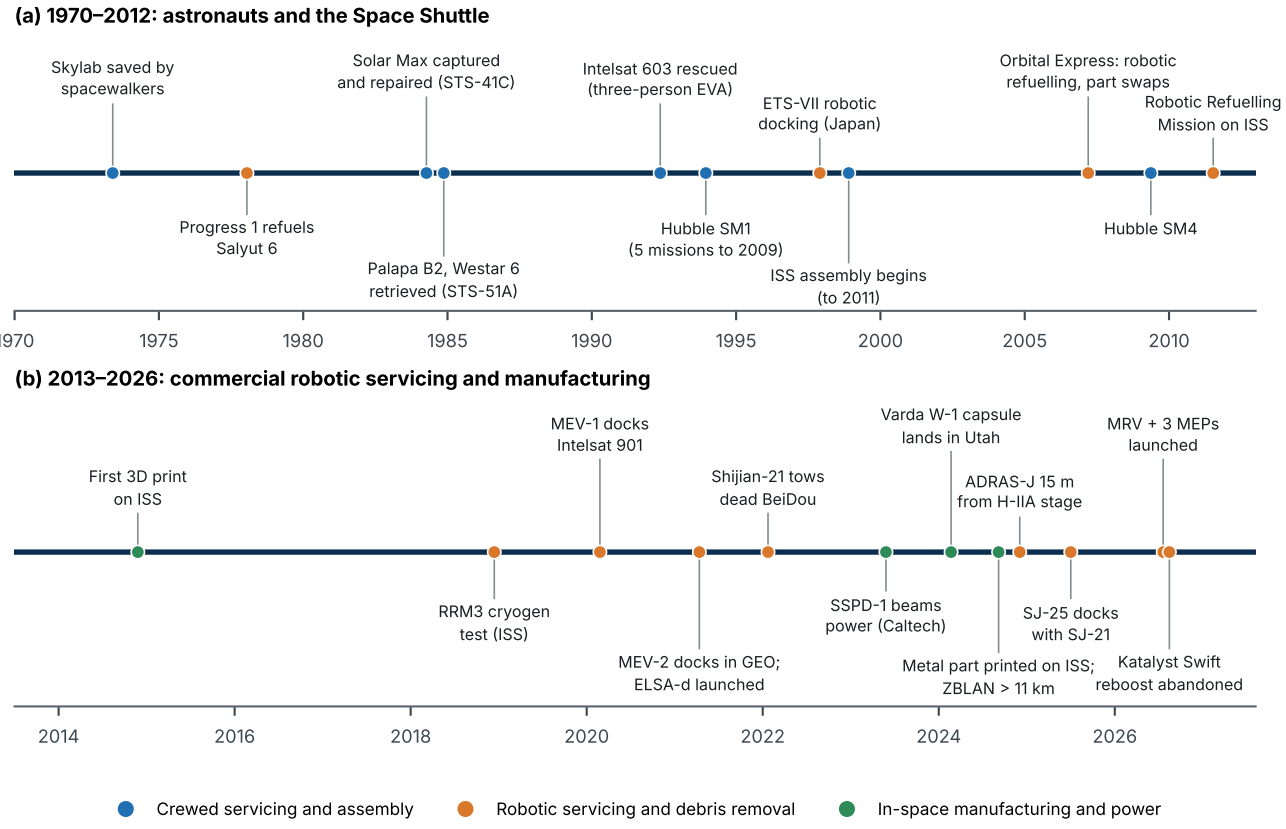


FIGURE 40.1 Milestones of in-space servicing, assembly and manufacturing. (a) Until 2012 nearly all servicing was done by astronauts working from the Space Shuttle. (b) Since 2014 the field has shifted to commercial robotic servicers, debris-removal demonstrators and manufacturing experiments.

40.2.1 Astronauts as Servicers

The first repair was improvised. When Skylab reached orbit in May 1973, its micrometeoroid shield had been torn away and a solar wing was jammed. Its first crew deployed a sunshade through a small airlock and cut the wing free on a spacewalk (Chapter 33). The Space Shuttle then made servicing a planned activity. The **Solar Maximum Mission** satellite had lost its fine pointing in 1980 when fuses in its attitude-control system failed. It was built with a grapple

fixture and replaceable modules precisely so that the Shuttle could fix it. In April 1984 George Nelson flew to it with the Manned Manoeuvring Unit (Figure 40.2) to attach a docking device to a pin on its side. The device failed to latch — an unexpected small fitting was in the way — and Nelson's attempt to steady the satellite by hand set it tumbling. Controllers at NASA Goddard stabilised it overnight, Challenger's robotic arm grappled it, and Nelson and James van Hoften replaced the attitude-control module and repaired an instrument's electronics. Solar Max returned to work until it re-entered in 1989.



FIGURE 40.2 George Nelson, flying the Manned Manoeuvring Unit, approaches the ailing Solar Maximum Mission satellite during STS-41C in April 1984. His attempt to dock by hand failed; the Shuttle's arm captured the satellite instead. Image: NASA/JSC.



FIGURE 40.3 Richard Hieb, Thomas Akers and Pierre Thuot hold the 4.2-tonne Intelsat VI (603) satellite above Endeavour's payload bay on 13 May 1992, minutes after the only three-person capture by hand. Image: NASA/JSC.

Later in 1984, the crew of STS-51A retrieved two communication satellites, Palapa B2 and Westar 6, stranded in useless orbits by failed kick motors. They brought them back to Earth, and both were refurbished and launched again. The most dramatic rescue came in 1992. Intelsat 603 had been stranded in low orbit in 1990, when a wiring error prevented it from separating from its Titan III second stage together with its perigee motor. On Endeavour's first flight, STS-49, two attempts to attach a capture bar failed because the satellite rotated away at the slightest touch. On the third spacewalk three astronauts simply grabbed it by hand (Figure 40.3). This was the only three-person spacewalk ever made. They fitted a new perigee motor, and the satellite reached geostationary orbit and served for many years.

Hubble was designed for servicing from the start, with handrails, standard bolts and drawer-like **orbital replacement units**. Five Shuttle missions between 1993 and 2009 (Chapter 38) replaced every instrument, gyroscope set and battery and corrected the flawed optics. Each began with the Shuttle's arm grasping one of two grapple fixtures on the telescope and berthing it in the payload bay (Figure 40.4). The **International Space Station** was the culmination. Assembled from 1998 to 2011 by Shuttle and Russian launches, it remains a servicing laboratory. There, crews and the Canadarm2 and Dextre robots replace pumps, batteries and solar arrays (Chapter 32).

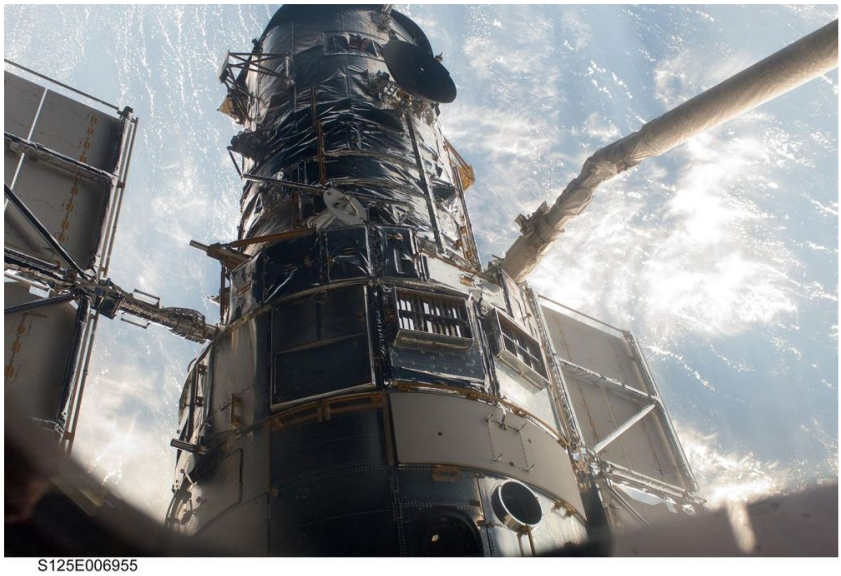


FIGURE 40.4 Hubble held by Atlantis's robotic arm on 13 May 2009 at the start of the fifth and final servicing mission, STS-125. The grapple fixture and handrails were designed in from the start. Image: NASA/JSC.

The lesson of the Shuttle era is mixed. Servicing worked spectacularly on clients designed for it — Solar Max and Hubble — and was heroic but risky on clients that were not. And it was expensive: each Hubble mission required a Shuttle flight costing well over a billion dollars at full cost, so it was never economical for commercial satellites.

40.2.2 Robotic Pioneers

Robotic servicing began with fuel. In January 1978 the uncrewed **Progress 1** docked with the Salyut 6 station and pumped propellant into its tanks — the first transfer of propellant between spacecraft. Progress and China's Tianzhou freighters still refuel the ISS and Tiangong. Japan's **ETS-VII** (1997–99) repeated the automatic docking of two uncrewed satellites, first achieved by the Soviet Kosmos 186 and 188 in 1967. It was also the first uncrewed satellite to test a robotic arm in orbit. NASA's **DART** demonstrator of 2005 (not the later asteroid impactor) showed the dangers: errors in its navigation software made it collide gently with its target. In 2007 DARPA's **Orbital Express** — the Boeing-built ASTRO servicer and Ball's NextSat client — demonstrated nearly everything a servicer needs. This included autonomous rendezvous, capture with a robotic arm, transfer of hydrazine in both directions, and the exchange of a battery and a flight computer.

NASA then tackled a harder problem: refuelling a satellite never designed for it, whose fill-and-drain valves are covered by caps, lock wire and thermal blankets meant to be closed once, on the ground. The **Robotic Refuelling Mission (RRM)** was carried to the ISS on the last Shuttle flight in 2011 (Figure 40.5). It gave Dextre a set of specialised tools that cut lock wire, unscrewed caps and, in 2013, transferred liquid ethanol through a standard valve. RRM3 (2018–19) aimed to store and transfer liquid methane, a cryogenic propellant, but a failed cryocooler forced its methane to be vented.

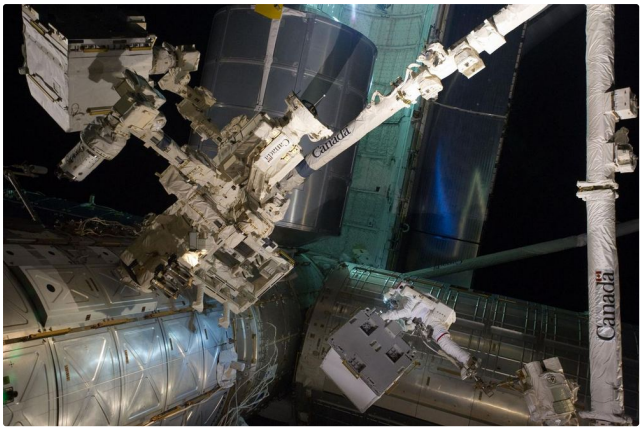


FIGURE 40.5 Astronaut Mike Fossum, riding Canadarm2, carries the Robotic Refuelling Mission module during the final Shuttle mission in July 2011. Dextre later used its tools to open and refuel simulated satellite valves. Image: NASA/JSC.

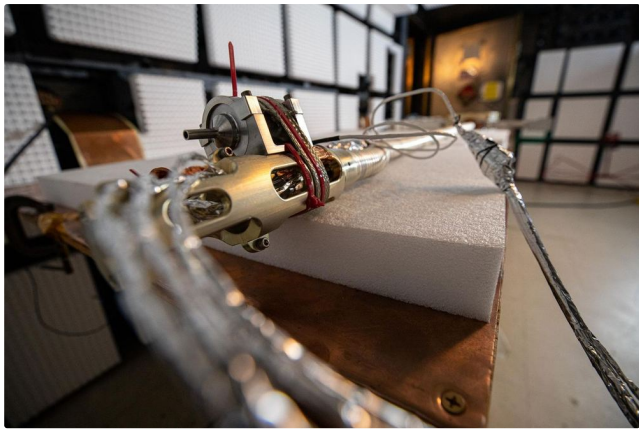


FIGURE 40.6 The hose management assembly built for NASA's Restore-L (later OSAM-1) refuelling servicer, in electromagnetic testing at Kennedy Space Center in 2020. It extended and retracted the propellant hose like a tape measure. Image: NASA/KSC (NASA/Cory Huston).

RRM led to **Restore-L**, renamed **OSAM-1**: a servicer that would grasp the Landsat 7 satellite and refuel it. It would also use a separate arm called SPIDER to assemble a 3 m antenna and manufacture a 10 m beam (Figure 40.6). Its cost estimate grew from $626–753 million to more than $2 billion, and in March 2024 NASA cancelled it, citing "continued technical, cost, and schedule challenges" and "a broader community evolution away from refueling unprepared spacecraft". About $1.5 billion had been appropriated. A companion mission to 3D-print beams in orbit, OSAM-2 (Archinaut One), had been closed out in 2023 before flight.

40.2.3 Commercial Life Extension

The first commercial servicer avoided refuelling altogether. Northrop Grumman's **Mission Extension Vehicle (MEV)** is about 2.3 t at launch and built for a 15-year life. It docks with a client, then provides all of the combined vehicle's attitude and orbit control with its own electric thrusters. It needs nothing special on the client. It inserts a probe into the satellite's liquid apogee engine (the large rocket motor used once to reach geostationary orbit) and expands a tip inside the nozzle. It then pulls itself in until stanchions press against the launch-adapter ring (see Figure 40.8).

MEV-1 docked with Intelsat 901, which had been parked in the graveyard orbit about 300 km above GEO, on 25 February 2020. This was the first docking of two commercial satellites, and MEV-1 returned Intelsat 901 to service in April 2020. After five years it moved Intelsat 901 back to the graveyard orbit, undocked on 9 April 2025 and went on to extend the life of Optus D3, from which it separated again in mid-2026. **MEV-2** docked with Intelsat 10-02 directly in its operational slot in April 2021; its planned five-year stay was extended in 2024 by several more years. Intelsat reportedly paid about $13 million per year for the service.

Northrop's second generation launched on a Falcon 9 on 21 July 2026: the **Mission Robotic Vehicle (MRV)**, together with three **Mission Extension Pods (MEPs)**. The MRV carries the robotic arms and tools of DARPA's Robotic Servicing of Geosynchronous Satellites (RSGS) programme. Each MEP is a small propulsion module that the MRV will attach to a client's launch-adapter ring. The pod then keeps the satellite on station for up to about eight years while the MRV moves on.

The MRV needs about a year to raise itself to geostationary orbit with electric propulsion; early customers include Optus and SES (which absorbed Intelsat in 2025). After checkout, RSGS operations are to pass to the US Space Force.

40.2.4 China's Geostationary Servicers

China has developed comparable capabilities, with less publicity. **Shijian-21**, launched in October 2021, docked with the dead BeiDou-2 G2 navigation satellite in December 2021. In January 2022 it towed the satellite about 3 000 km above the geostationary belt — far higher than the usual 300 km graveyard — before returning to GEO. **Shijian-25**, launched in January 2025, approached Shijian-21 and the two appeared merged in commercial optical tracking from about 2 July 2025, separating again by late November. The pairing was widely reported as the first refuelling in geostationary orbit, but China has not described what was transferred. Because the same rendezvous, grappling and towing capabilities could disable another nation's satellite, these missions are watched closely (Chapter 19).

40.2.5 Debris Removal and Inspection

Removing debris is harder than servicing a satellite: the target was never meant to be approached, may be tumbling, has no lights, reflectors or radio, and nobody pays it to cooperate. Japan's **Astroscale** has worked up to it step by step (Table 40.2). **ELSA-d** (launched March 2021) released and recaptured its own small client with a magnet in August 2021, but thruster failures in 2022 ended its more ambitious tests. **ADRAS-J** was launched on an Electron rocket in February 2024 under a JAXA contract. It made the first close inspection of a large piece of real debris: an 11 m, 3 t H-IIA upper stage left in orbit in 2009. It flew around the stage at 50 m in mid-2024, once aborting automatically when the target's attitude did not match predictions. In late November 2024 it closed to about 15 m from the planned capture point. **ADRAS-J2**, under a JAXA contract worth about ¥12 billion (about $80 million), is to capture the same stage with a robotic arm and lower it for re-entry. A launch on Isar Aerospace's Spectrum rocket in Japan's fiscal year 2027 (up to April 2028) was contracted in September 2026.

TABLE 40.2 Robotic servicing and debris-removal missions (selected).

Mission	Operator	Launch	Client	Result (as of Oct 2026)
Orbital Express	DARPA (Boeing, Ball)	2007	Own client (NextSat)	Capture, hydrazine transfer, battery and computer swap
MEV-1	Northrop Grumman	2019	Intelsat 901, then Optus D3	Intelsat 901 2020–25; Optus D3 2025–26
MEV-2	Northrop Grumman	2020	Intelsat 10-02 (GEO)	Docked in slot 2021; stay extended
ELSA-d	Astroscale	2021	Own client (17 kg)	Magnetic capture 2021; ended 2024
Shijian-21	China	2021	Dead BeiDou-2 G2	Towed 3 000 km above GEO, 2022
ADRAS-J	Astroscale / JAXA	2024	H-IIA upper stage	Inspected down to 15 m
Shijian-25	China	2025	Shijian-21	Docked 2025; refuelling reported, unconfirmed

Mission	Operator	Launch	Client	Result (as of Oct 2026)
Katalyst LINK	Katalyst / NASA	2026	Swift observatory	Lost attitude control; reboost abandoned
MRV + 3 MEPs	Northrop Grumman	2026	GEO comsats	Raising orbit to GEO
Otter	Starfish Space	2026	Defunct US satellites	Commissioning; inspection for NASA
ADRAS-J2	Astroscale / JAXA	FY2027	H-IIA upper stage	Capture and de-orbit planned
ClearSpace-1	ESA (OHB, ClearSpace)	≈ 2029	PROBA-1 (95 kg)	Capture with four arms planned

Europe's **ClearSpace-1**, led by OHB with the Swiss company ClearSpace, will capture ESA's 95 kg PROBA-1 satellite, launched in 2001, with four arms and de-orbit it; ESA lists the launch for 2029. In the USA, **Starfish Space** launched its first full-size Otter servicer on a SpaceX rideshare on 1 October 2026. For NASA it is to approach up to four defunct US satellites in early 2027, without touching them, ahead of planned life-extension work for Intelsat in GEO and de-orbit services for the Space Development Agency. Its smaller Otter Pup 2 demonstrator, in orbit since June 2025, had not yet docked with a client.

A reminder of how unforgiving the field is came in 2026. NASA's Swift gamma-ray observatory, launched in 2004, was decaying towards re-entry. In September 2025 NASA gave the start-up **Katalyst Space** a $30 million contract to boost it. This was a first attempt to reboost a satellite not designed for servicing, on a schedule of months rather than years. Its 425 kg LINK spacecraft reached orbit on the final Pegasus XL launch on 3 July 2026, but three weeks later two of its three reaction wheels failed and it lost attitude control. It came within 12–15 km of Swift before the reboost was abandoned in August; LINK re-entered on 25 September, and Swift is expected to re-enter around the end of 2026.

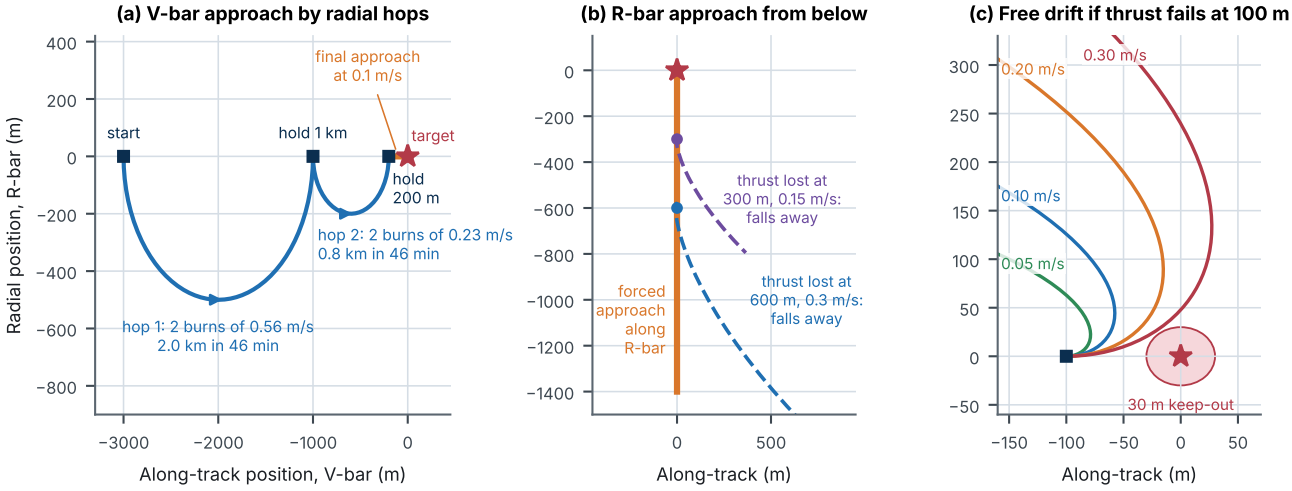
40.3 Rendezvous, Proximity Operations and Capture

Every ISAM activity starts with **rendezvous and proximity operations (RPO)**: bringing two spacecraft from hundreds of kilometres apart to contact, safely. The far-range phases — launch into a slightly lower or higher orbit, phasing, and orbit matching — use the methods of Chapter 3. Within a few kilometres the relative motion is described by the Clohessy–Wiltshire equations in the target's local frame. In this frame x is radial (along the **R-bar**, the line through the target and Earth's centre; positive away from Earth) and y is along the target's velocity (the **V-bar**). Their solutions are counter-intuitive: a forward push raises a spacecraft and makes it drift backwards, and a spacecraft below the target drifts forwards.

40.3.1 Approach Strategies

Operators approach along one of two straight lines through the target (Figure 40.7). On the **V-bar** a chaser can rest at any distance ahead of or behind the target, because it then shares the target's orbit. This makes the V-bar ideal for **hold points**, where the approach pauses while controllers check sensors and give the go-ahead. The chaser moves between hold points by **radial hops**. A

small burn towards Earth makes it dip below the target's orbit, travel forwards and return to the V-bar half an orbit later. There an equal and opposite burn stops it. The ISS's European ATV and Japanese HTV, and today's Cygnus and Dragon, all use variants of these patterns.



Clohessy–Wiltshire solutions for a target in a 420 km circular orbit (period 93 min). Radial axis up = away from Earth; target flies to the right. Axes not to equal scale.

FIGURE 40.7 Approach strategies computed from the Clohessy–Wiltshire equations for a target in a 420 km orbit. (a) A V-bar approach: two radial hops carry the chaser from 3 km to a hold point 200 m behind the target, followed by a slow straight-line approach. (b) An approach up the R-bar from below: if thrust is lost, orbital mechanics makes the chaser fall away. (c) If thrusters fail during a V-bar approach from 100 m, a slow chaser stops short of the target while a fast one sweeps past it — the basis of "passively safe" approach speeds.

The **R-bar** approach climbs towards the target from directly below (or descends from above). Here orbital mechanics provides **natural braking**. A chaser below the target is in a lower orbit, so the tidal term in the equations pulls it down and away. To hold position on the R-bar it must thrust continuously towards the target. If the thrust stops, it falls away and drifts ahead, as in panel (b). The Shuttle approached the ISS this way, and so do servicers wanting to reach a client's underside. The final metres are flown along a narrow **approach corridor** at a few centimetres per second, with the docking interface pointing at the chaser.

The key design rule is **passive safety**: at every point, if the chaser's thrusters stopped working, its free drift must not hit the target. Panel (c) shows why approach speeds must be low. A chaser closing at 0.1 m/s from 100 m behind on the V-bar, with its thrusters suddenly dead, coasts forward only 42 m. Orbital mechanics then lifts it and carries it back. At 0.3 m/s it sweeps over the target within 34 m. Approach paths are surrounded by **keep-out spheres** that the chaser may enter only after explicit approval.

EXAMPLE: HOPPING ALONG THE V-BAR

A servicer must move from a hold point 1 km behind a client in a 420 km orbit to a point just behind it. The mean motion is $n = \sqrt{\mu/r^3} = 1.126 \times 10^{-3}$ rad/s, so the orbital period is 93.0 min.

1. **Radial hop.** For a radial burn Δv, the Clohessy–Wiltshire solution gives $x = -(\Delta v/n)\sin nt$ and $y = (2\Delta v/n)(1 - \cos nt)$. After half an orbit the chaser is back on the V-bar, $4\Delta v/n$ further forward. For a 1 000 m hop, $\Delta v = nL/4 = 1.126 \times 10^{-3} \times 1\,000 / 4 =$ **0.28 m/s**; an equal burn stops it, so the hop costs 0.56 m/s and takes 46.5 min. The path dips 250 m below the V-bar.
2. **Straight-line approach.** Flying the same 1 000 m straight along the V-bar requires continuous radial thrust to cancel the Coriolis term, $a_x = 2n\dot{y}$. Integrated over the approach this costs $2nL =$ **2.25 m/s** whatever the speed — four times the hop's cost, which is why straight lines are kept for the last few hundred metres.
3. **Passive safety.** A chaser drifting forward at speed v along the V-bar travels at most $0.478\ v/n$ before reversing. In LEO at 0.1 m/s that is 42 m. In GEO, where $n = 7.29 \times 10^{-5}$ rad/s, the same speed would carry it 655 m. To stop short of a client 100 m away, a GEO servicer must approach at less than 1.5 cm/s. Hops are cheap there (0.036 m/s per kilometre) but each one takes 12 hours.

40.3.2 Sensors and Navigation

A servicer measures the client's relative position and attitude — its **pose** — with a sequence of sensors that hand over as the range shrinks. Relative GNSS or ground tracking works at tens of kilometres. Visible and infrared cameras see the client first as a point and then as a shape. Below a few hundred metres, lidar or time-of-flight cameras measure range and three-dimensional shape directly. Cooperative clients such as the ISS carry retroreflectors and radio beacons. Unprepared clients do not, so the servicer must estimate pose from images alone, using the client's known geometry. It must do this under harsh lighting that changes every few minutes. Thermal infrared cameras help in eclipse and against glinting surfaces. The guidance software runs on board because ground loops are too slow and too intermittent. It must recognise when measurements stop making sense and abort — as ADRAS-J did when the rocket stage's attitude departed from predictions.

40.3.3 Capturing a Client

How a servicer grips its client depends on whether the client was **prepared** — fitted before launch with a grapple fixture, docking plate or refuelling port — or **unprepared**. Figure 40.8 compares the main options. Prepared interfaces are simple and reliable: the Shuttle's and ISS's arms snare a grapple pin with three wires, and Astroscale's clients carry a ferromagnetic plate. Most satellites in orbit, though, were never prepared. So the commercial servicers have learned to grip features that almost every satellite has anyway. These are the **launch-adapter ring** and its bolt holes, by which the satellite was clamped to its rocket, and the nozzle of the **liquid apogee engine** that raised it to GEO. Both are strong enough to carry towing loads and are located on the satellite's axis.

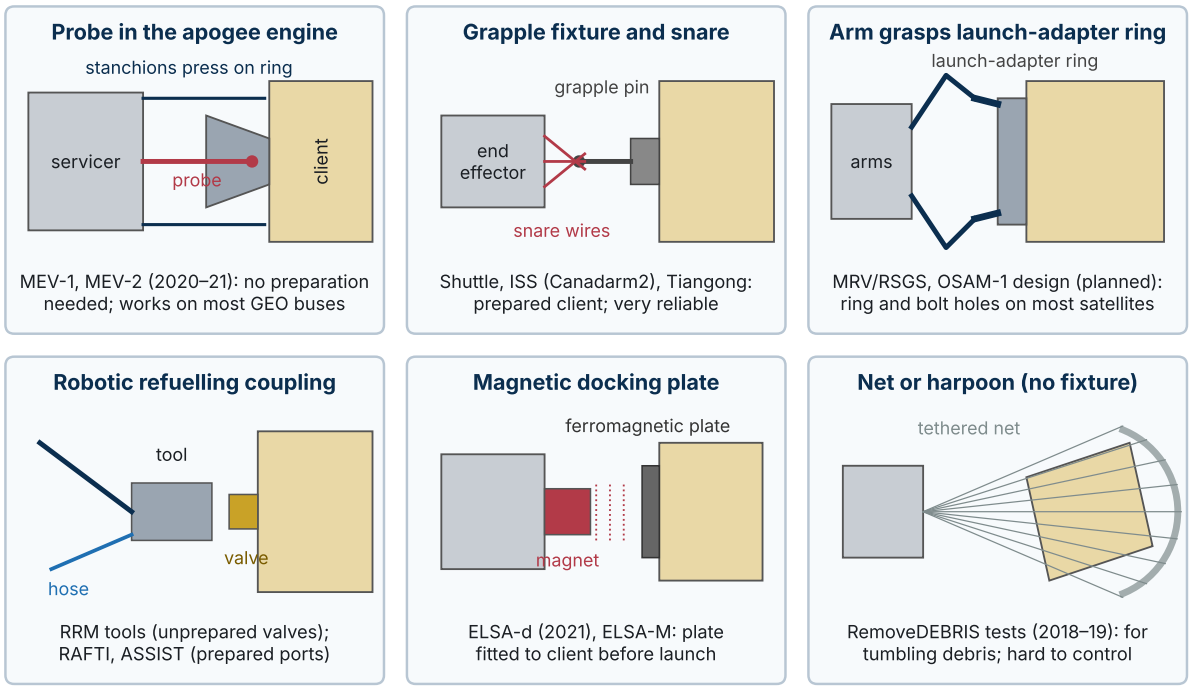
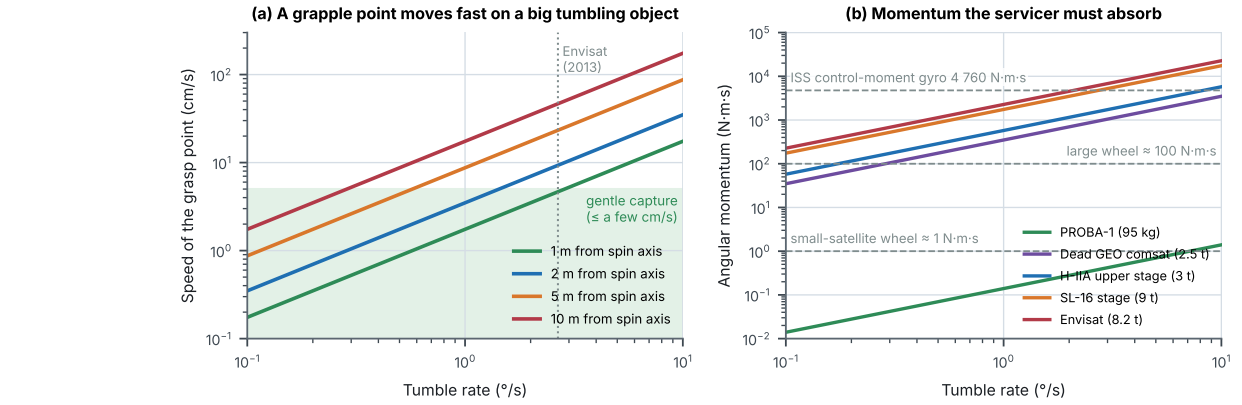


FIGURE 40.8 Six ways for a servicer to grip a client. The apogee-engine probe, grapple fixture and magnetic plate have all worked in orbit, and refuelling couplings were tested by Orbital Express and on the ISS; prepared interfaces (grapple fixture, magnetic plate, refuelling port) need equipment fitted before launch, while the apogee-engine probe and robotic ring grasp (planned for the MRV) work on most satellites already in orbit. Nets and harpoons, tested by RemoveDEBRIS in 2018–19, can catch tumbling debris but give poor control afterwards.

Tumbling is the main obstacle to capturing debris. Without attitude control, objects are spun up over years by solar radiation pressure, gravity-gradient torques and eddy currents induced by Earth's magnetic field. Envisat, the 8.2 t ESA satellite that failed in 2012, was measured by laser ranging spinning once every 135 s in 2013 — about 2.7° per second. A grasp point 5 m from the spin axis of such an object moves at about 23 cm/s (Figure 40.9), far too fast to seize gently. So a servicer must either fly in synchrony with the rotation, matching the spin while hovering, or slow the target down first. It can slow the target, for example, by eddy-current braking, plume impingement or repeated gentle contacts. After capture, the stack's angular momentum must be absorbed. An H-IIA upper stage (moment of inertia about 3.3×10^4 kg·m^2) tumbling at 1° per second carries about 580 N·m·s. This is about six times what a large reaction wheel (about 100 N·m·s) can store, so servicers use thrusters to detumble the combined vehicle.



v = ωr; H = Iω about the largest principal axis. Moments of inertia approximated from mass and dimensions (stages as uniform cylinders; Envisat ≈ 1.3 × 10⁵ kg·m²). Envisat rate from laser ranging.

FIGURE 40.9 Capturing a tumbling object. (a) The speed of a grasp point grows with tumble rate and with its distance from the spin axis; only slowly rotating targets can be seized directly. (b) The angular momentum of large debris objects at typical tumble rates exceeds the storage of reaction wheels, so thrusters must stop the rotation after capture.

40.3.4 Robotic Arms and Tools

Servicing robots descend from the Shuttle's arm and from the ISS's **Canadarm2** and **Dextre** (Chapter 32). But they must work without astronauts nearby, often with a time delay of a quarter of a second or more through geostationary relays. The arms that FREND and RSGS developed for GEO servicing have seven joints, like a human arm, so that they can reach around obstacles, and carry force–torque sensors at the wrist. Their controllers use **compliance** (also called impedance control). Instead of driving the hand rigidly to a position, they make it behave like a spring and damper. Contact forces then stay small even if the client is not exactly where the cameras think it is. Free-flying arms also disturb their own base: every arm motion rotates the servicer slightly in the opposite direction, and the control system must account for this coupling.

Tools turn arms into technicians. NASA's RRM tools were a wire cutter, a multifunction tool that unscrews caps, a nozzle tool that mates with standard fill-and-drain valves, and a camera tool for looking inside blanketed areas. They showed that even lock-wired valves could be opened robotically. Teleoperation from the ground remains normal for complex tasks, with operators watching stereo video and a simulation that predicts the arm's motion through the time delay; routine motions are increasingly automated.

40.3.5 Refuelling Interfaces and Standards

Refuelling an unprepared satellite is possible but slow, because the robot must reproduce work done by hand on the ground. The industry is therefore converging on **refuelling ports** fitted before launch (Table 40.3), which open, seal and verify the connection in one robotic operation, together with standard **grapple fixtures** for capture. Agreeing on such interfaces — and on how close servicers may approach other operators' satellites — is the work of industry groups. One is **CONFERS** (Consortium for Execution of Rendezvous and Servicing Operations), set up with DARPA support in 2017. Its principles fed into the international standard ISO 24330:2022 on rendezvous, proximity operations and on-orbit servicing.

TABLE 40.3 Capture and refuelling interfaces and standards.

Interface	Developer	Type	Notes
Standard fill-and-drain valve	Satellite industry	Unprepared	Opened robotically by RRM tools; slow, multi-step
Probe and ring docking	Northrop Grumman (MEV, MEP)	Unprepared	No fluid transfer; servicer takes over propulsion
RAFTI port	Orbit Fab	Prepared	Accepted by the US Space Force (2024) as a refuelling interface
GRIP	Orbit Fab	Servicer side	Docks with RAFTI-equipped clients; flight-qualified 2025
ASSIST	GMV / ESA	Prepared	Docking and refuelling; servicer tool plus passive client fixture
Magnetic docking plate	Astroscale	Prepared	For capture and de-orbit at end of life
ISO 24330:2022	ISO (CONFERS input)	Standard	Programmatic principles and practices for RPO and servicing

Refuelling demonstrations on prepared clients were scheduled for 2026–27. Astroscale US's 300 kg APS-R servicer is to transfer hydrazine to one of the Space Force's Tetra-5 satellites in GEO. Orbit Fab planned to fly its GRIP system to service the same satellites. As of October 2026, neither had been reported as launched.

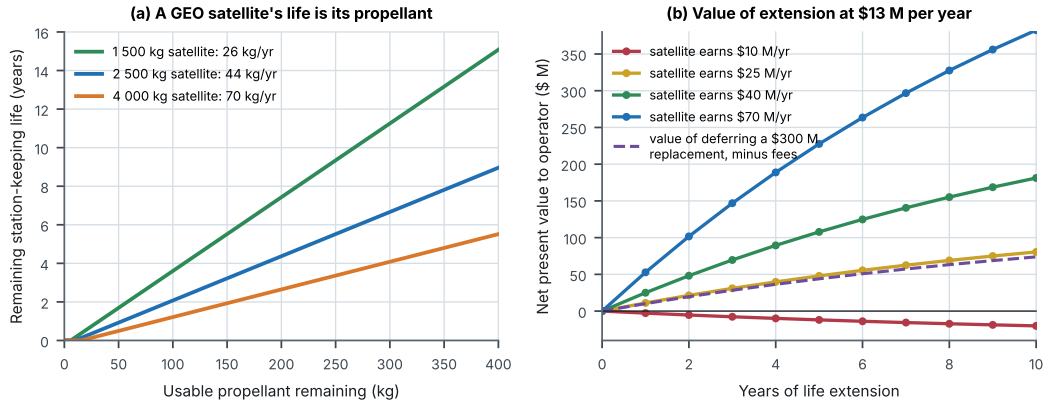
40.3.6 Safety, Law and Trust

Servicing raises questions that ordinary satellites do not. Under the Outer Space Treaty, a state keeps **jurisdiction and control** over an object it registered — including debris — indefinitely, so removing another nation's dead satellite or rocket stage needs that state's consent. Liability for damage in orbit depends on proving fault, which is hard when two spacecraft touch on purpose. Servicers must convince their clients' insurers and regulators that they will not turn a working satellite into a tumbling wreck. MEV-1 made the first docking with Intelsat 901 in the graveyard orbit, where an accident would not have endangered the GEO belt. And because the same technology can inspect, disable or move another country's satellite, transparency about intentions and approaches — what CONFERS calls being a "responsible actor" — matters as much as engineering (Chapter 19).

40.4 The Economics of Life Extension and Debris Removal

40.4.1 What a Satellite-Year Is Worth

A geostationary satellite drifts north–south under lunar and solar gravity and east–west under Earth's slightly elliptical equator, and needs about 50 m/s of Δv per year to stay in its slot. With chemical thrusters of 290 s specific impulse, a 2 500 kg satellite therefore burns about 44 kg of propellant a year. It must also keep about 10 kg in reserve to raise itself to the graveyard orbit at the end (Figure 40.10). With 100 kg of usable propellant left, it has about two years to live — and because propellant gauging is uncertain, operators often retire satellites with a margin that a servicer could have used.



(a) Δv 50 m/s per year, chemical Isp 290 s, 11 m/s reserved for disposal 300 km above GEO. (b) Fee ≈ $13 M/yr (reported MEV price); discount rate 8 %; net revenue and replacement cost illustrative.

FIGURE 40.10 Economics of life extension. (a) Remaining life of a GEO satellite as a function of its usable propellant. (b) Net present value to an operator of extending a satellite's life at the reported MEV price of about $13 million a year: worthwhile if the satellite still earns well, or if extension defers an expensive replacement.

Whether extension pays depends on what the satellite still earns. A satellite with 40 transponders leased at about $1 million each per year earns $40 million a year; paying $13 million a year for five more years gives the operator a present value of about $108 million at an 8 % discount rate. Even if the satellite would otherwise be replaced immediately, deferring a $300 million replacement by five years is worth about $96 million in present value, $44 million more than the fees. Conversely, a satellite earning less than its servicing fee is not worth extending. The market is limited by the number of high-value satellites approaching the end of their propellant at any time — some tens per year. It is also limited by competition from cheaper new satellites and from LEO constellations.

EXAMPLE: A TUG IN GEOSTATIONARY ORBIT

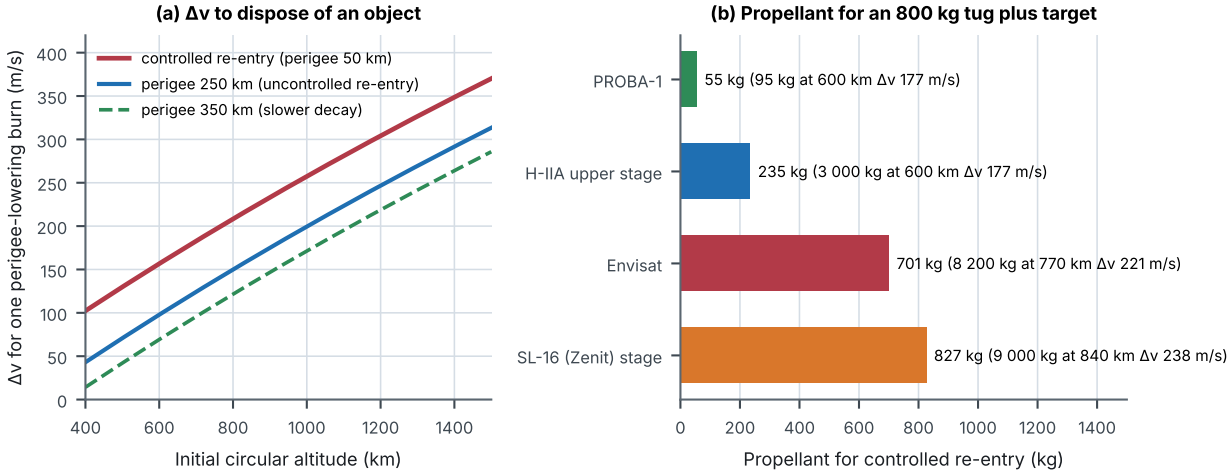
A servicer of 2 330 kg (MEV class) docks with a 2 500 kg satellite whose tanks are empty. The combined mass is 4 830 kg.

1. **Disposal.** Raising the stack 300 km above GEO (r = 42 164 km, v = 3.075 km/s) by a Hohmann transfer takes two burns of 5.44 m/s, 10.9 m/s in total. With chemical thrusters (Isp 300 s, v_e = 2 942 m/s) this uses $m_p = m_0(1 - e^{-\Delta v/v_e}) = 4\,830 \times (1 - e^{-10.9/2\,942}) = \mathbf{17.8\ kg}$; with Hall-effect thrusters of Isp 1 700 s, only **3.2 kg**.
2. **Station keeping.** Keeping the stack in its slot for five years needs 5 × 50 = 250 m/s. That is **72 kg** of xenon at 1 700 s, but **393 kg** of chemical propellant at 300 s — almost the entire life-extension budget. This is why servicers use electric propulsion.
3. **Relocation.** To drift along the GEO arc at 1° per day, the orbit's semi-major axis must change by $\Delta a = \frac{2}{3}\,(\dot{\lambda}/n)\,a$ = 78 km, costing 2.8 m/s to start and 2.8 m/s to stop. Moving 90° to a new client takes three months and about 6 m/s. A single 15-year servicer can therefore serve several clients in turn, as MEV-1 has done.

40.4.2 Paying to Remove Debris

Debris removal has a harder business case: nobody earns revenue from a dead rocket stage. The Δv to dispose of an object grows with altitude (Figure 40.11). A controlled re-entry lowers the perigee to about 50 km so that the object falls into an empty ocean area. It needs about 157 m/s from 600 km and 218 m/s from 840 km, plus about 20 m/s for rendezvous and attitude control. For a tug of 800 kg, de-orbiting the 3 t H-IIA stage needs about 235 kg of propellant, and a 9 t Soviet-

era SL-16 stage about 830 kg, before any margin for a second attempt. One tug can therefore remove one large object, or perhaps a few small satellites — which is why proposals for multi-target removal favour attaching cheap de-orbit kits to several objects in the same orbital plane.



Single impulsive burn from a circular orbit. (b) Isp 300 s; +20 m/s for rendezvous and attitude margin; target masses and altitudes approximate.

FIGURE 40.11 Disposing of debris. (a) Δv to lower the perigee of an object from a circular orbit, for a controlled re-entry or to accelerate natural decay. (b) Propellant needed by an 800 kg tug to de-orbit four candidate objects with a controlled re-entry.

Current removal missions are demonstrations funded by space agencies (Table 40.4): ADRAS-J2 costs about ¥12 billion for one object, and ESA's 2020 contract for ClearSpace-1 was worth €86 million. Studies suggest that removing five to ten of the most dangerous large objects per year would stabilise the LEO population (Chapter 21), which at today's prices means several hundred million dollars a year. Possible funding models include government procurement as a public good and removal fees built into launch licences. Another is a requirement that new satellites carry a docking plate, so that a cheap commercial tug can remove them if they fail.

TABLE 40.4 Prices and budgets of selected servicing and debris-removal missions.

Mission or service	Customer	Value	Notes
MEV life extension	Intelsat	≈ $13 M per year (reported)	Five-year term per client
OSAM-1 (cancelled 2024)	NASA	≈ $1.5 B appropriated; > $2 B estimate	Refuelling and assembly demonstration
ADRAS-J2	JAXA	≈ ¥12 B (≈ $80 M)	Capture and de-orbit one rocket stage
ClearSpace-1	ESA	€86 M (2020 contract)	Capture and de-orbit PROBA-1
Swift reboost (LINK)	NASA	$30 M	Abandoned 2026
Otter inspection (SSPICY)	NASA	$15 M	Up to four defunct US satellites (2027)
Otter de-orbit services	Space Development Agency	$52.5 M	Proliferated LEO constellation

40.5 Assembling Large Structures in Orbit

40.5.1 Lessons from the ISS

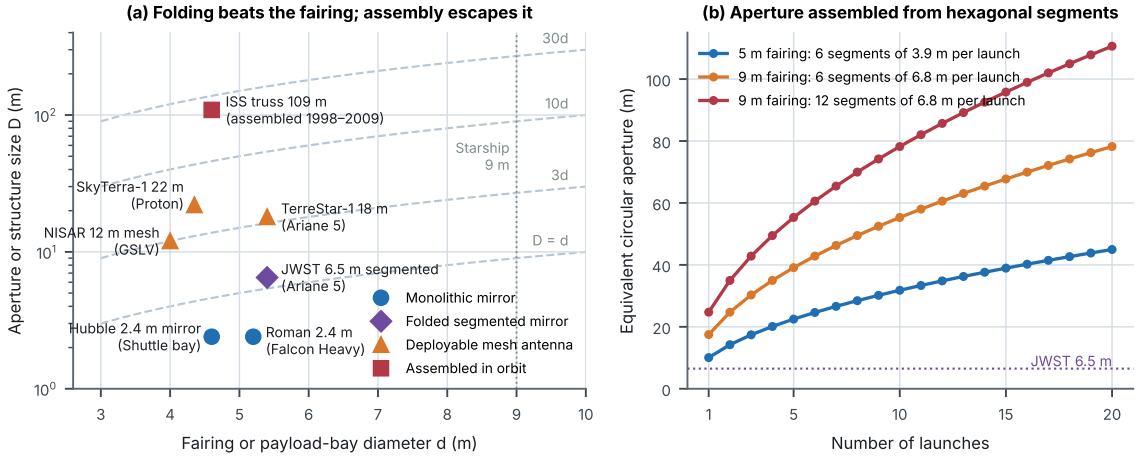
The ISS is the largest structure ever assembled in space: about 420 t, with a 109 m truss carrying eight solar wings, each 35 m long. It was built from components launched on more than 40 flights between 1998 and 2011. The Shuttle's arm lifted truss segments out of the payload bay and handed them to Canadarm2, which positioned them for bolting; spacewalkers then connected the power, data and coolant lines that robots could not (Figure 40.12). Hundreds of hours of spacewalks went into assembly (Chapter 33). The station's design shows what makes assembly possible. It uses standard mechanical interfaces (the segment-to-segment attach system and the common berthing mechanism). Its structures were tested together on the ground before launch wherever possible, and its connectors were designed for gloved hands or robot end-effectors.



FIGURE 40.12 The Starboard One (S1) truss segment newly installed on the ISS in October 2002, during the STS-112 assembly mission; Canadarm2 is visible at left. Each truss segment was berthed by robotic arms and connected by spacewalkers. Image: NASA/JSC.

40.5.2 Beyond the Fairing

Every launch vehicle limits the size of what it carries to the diameter of its fairing — about 5 m for most large rockets, 7 m for New Glenn and 9 m for Starship. Designers escape this limit in three ways (Figure 40.13). A **monolithic** mirror must fit inside the fairing: Hubble's and Roman's are 2.4 m. **Folding** gains a factor of up to about five. JWST's 6.5 m segmented mirror was folded into a 5.4 m fairing. The gold-plated mesh reflectors of mobile communication satellites such as TerreStar-1 (18 m) and SkyTerra-1 (22 m), and NISAR's 12 m radar antenna, unfurl like umbrellas. Beyond that, structures must be **assembled**, as the ISS was.



(a) Nominal fairing/bay outer diameters. (b) Hexagons with flat-to-flat width 0.85 × usable fairing diameter (4.6 m or 8 m); D = √(4A/π); gaps, edge losses and support structure ignored.

FIGURE 40.13 Aperture size versus launcher. (a) Monolithic, folded and mesh structures stay within a few fairing diameters; only assembly escapes the limit. (b) Equivalent aperture of a telescope assembled from hexagonal segments, versus number of launches, for illustrative packing assumptions.

Assembly trades launch volume for complexity. In the illustrative case of panel (b), a 20 m telescope needs about four launches of segments in 5 m fairings. It needs two in a 9 m fairing, or one if twelve segments can be stacked inside it. The challenges lie in precision. Mirror segments must be aligned to a small fraction of a wavelength — tens of nanometres for JWST, far finer for a telescope searching for Earth-like planets. So they need actuators and wavefront sensing like JWST's (Chapter 38), and the supporting truss must be stiff and thermally stable. Robotic assembly has been demonstrated on Earth and in small experiments; NASA's in-orbit demonstrations — OSAM-1's SPIDER antenna assembly and OSAM-2's printed beams — were cancelled before flight. NASA's early studies of the Habitable Worlds Observatory assume that it will be serviceable, with standard interfaces and an orbit reachable by servicers, so that instruments could be replaced as Hubble's were. Modular spacecraft with standard mechanical, electrical and data interfaces between their blocks — sometimes called "Lego-like" — are a related idea. A damaged module could be swapped and a new payload added without replacing the whole satellite.

40.5.3 Space-Based Solar Power

The most ambitious assembly concept is a **space solar power station**, collecting sunlight continuously in GEO and beaming it to Earth as microwaves. The physics sets its scale. To deliver 1 GW with about 10 % end-to-end efficiency from sunlight at 1.36 kW/m^2 requires a collecting area of about 7 km^2. That means a structure of thousands of tonnes even at 1 kg/m^2 — far beyond any foldable spacecraft. Caltech's **SSPD-1** demonstrator, launched in January 2023, showed the first steps. Its MAPLE experiment, a lightweight array of microwave transmitters steered electronically, beamed power to receivers in orbit. It also sent a detectable beam to a receiver on the roof of a laboratory at Caltech. Commercial viability would require launch costs, mass per unit area and robotic assembly far beyond today's.

40.6 Manufacturing in Space

40.6.1 Why Make Things in Orbit?

Manufacturing in orbit offers three things that are hard to get on Earth. The first is **weightlessness**, which suppresses buoyant convection and the settling of heavier components, so that crystals grow more evenly and mixtures do not separate. The second is **containerless processing**, in which a liquid can solidify without touching walls that would contaminate or seed it. The third is an unlimited **high vacuum**. Against these stand the costs of launch and return, a few cubic metres of space on a station, and the need to automate almost everything. In-space manufacturing therefore targets two niches: products so valuable per kilogram that they justify the round trip (Table 40.5), and parts made in space for use in space, which save the cost of launching spares.

TABLE 40.5 Candidate products for in-space manufacturing.

Product	Why microgravity helps	Status (2026)	Example
Spare parts (polymer)	Print on demand instead of stocking spares	ISS printers since 2014	Made In Space (Redwire) printers
Metal parts	Strong tools and parts for long missions	First metal print 2024	ESA/Airbus printer in Columbus
ZBLAN optical fibre	Less crystallisation, so lower optical loss	> 11 km drawn on ISS (2024)	Flawless Photonics
Protein and drug crystals	Larger, more uniform crystals	Routine ISS research; capsule returns	Merck, Varda (ritonavir)
Bioprinted tissue	Soft tissue holds shape without scaffolds	Meniscus printed 2023	Redwire BioFabrication Facility
Semiconductor crystals	Purer crystals grown from gas or melt	Furnace operated in orbit 2025	Space Forge ForgeStar-1
Large structures	Unlimited length; no launch loads	Studies; demos cancelled	OSAM-2 beam printing

40.6.2 Printing in Orbit

The first 3D printer in space, built by Made In Space for NASA, reached the ISS in 2014 and printed its first part — a faceplate for its own extruder — in November (Figure 40.14). Its test coupons, compared with identical ones printed on the ground, showed that extrusion works normally in weightlessness. Later in 2014 it printed a ratchet wrench from a file sent up from the ground. A commercial printer followed in 2016, and the company, now part of Redwire, printed hundreds of parts. Metal is harder: molten metal must be controlled without gravity, and powder beds cannot be used because loose powder floats. ESA and Airbus's printer, installed in the Columbus module in 2024 (Figure 40.15), melts stainless-steel wire with a laser. It produced its first part that summer. Samples were returned to Earth for testing against parts printed on the ground.

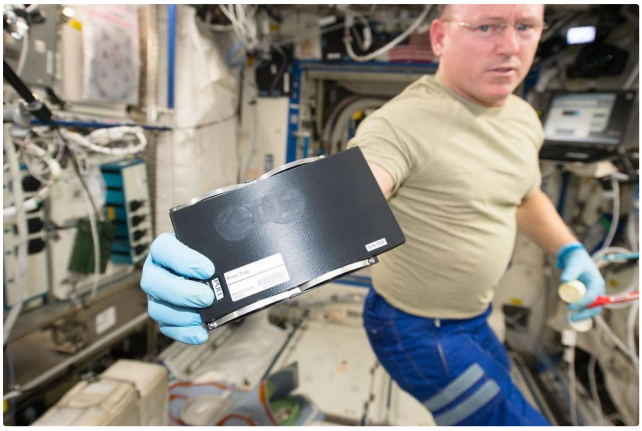


FIGURE 40.14 Barry Wilmore holds a test coupon made by the first 3D printer in space, aboard the ISS in December 2014. Comparing such coupons with identical ones made on the ground showed that printing works in weightlessness. Image: NASA/JSC (Barry Wilmore).



FIGURE 40.15 Jeanette Epps configures the ESA metal 3D printer in the Columbus module in August 2024. It melts stainless-steel wire with a laser inside a sealed chamber. Image: NASA/JSC.

40.6.3 Fibres, Crystals and Tissue

ZBLAN is a fluoride glass (named after its zirconium, barium, lanthanum, aluminium and sodium components) that in theory transmits infrared light far better than silica. But on Earth tiny crystals form as it is drawn into fibre, and they scatter light. In weightlessness, without convection and sedimentation, fewer crystals form. In 2024 the start-up Flawless Photonics drew more than 11 km of ZBLAN fibre on the ISS, the longest run yet; whether its quality justifies the cost of production in orbit is still being assessed.

Protein crystals grown in weightlessness are often larger and more regular. This helps determine molecular structures by X-ray diffraction and can improve drug formulation. Experiments on the ISS helped Merck develop a more uniform crystalline suspension of its cancer drug pembrolizumab. **Bioprinting** benefits because soft tissues printed from cells and gels do not collapse under their own weight. Redwire's BioFabrication Facility on the ISS printed a human knee meniscus in 2023, a first step towards printing tissues for transplantation.

40.6.4 Free-Flying Factories

Stations offer crew help but little throughput. **Varda Space Industries** builds small free-flying factories: a satellite bus carries a process module and a re-entry capsule, which returns its product at almost orbital speed and lands under a parachute. W-1 grew crystals of the HIV drug ritonavir. After months waiting for permission to land, it came down in Utah in February 2024. This was the first re-entry licensed under the US FAA's newer Part 450 rules, and the first time the FAA had licensed a commercial spacecraft to land on US soil. Its successors land at the Koonibba Test Range in South Australia: W-2 in February 2025 and W-3 in May 2025. W-5 followed in January 2026 with the first heat shield made entirely of Varda's own carbon ablator, and W-6 in May 2026. Each capsule is also a **hypersonic test bed**, carrying re-entry experiments for the US Air Force Research Laboratory and, on W-6, NASA — an important source of revenue alongside pharmaceutical work. In the UK, **Space Forge**'s ForgeStar-1, launched in June 2025, ignited a plasma at about 1 000 °C in its miniature furnace in December 2025. This was a step towards growing semiconductor crystals in orbit and returning them on later missions.

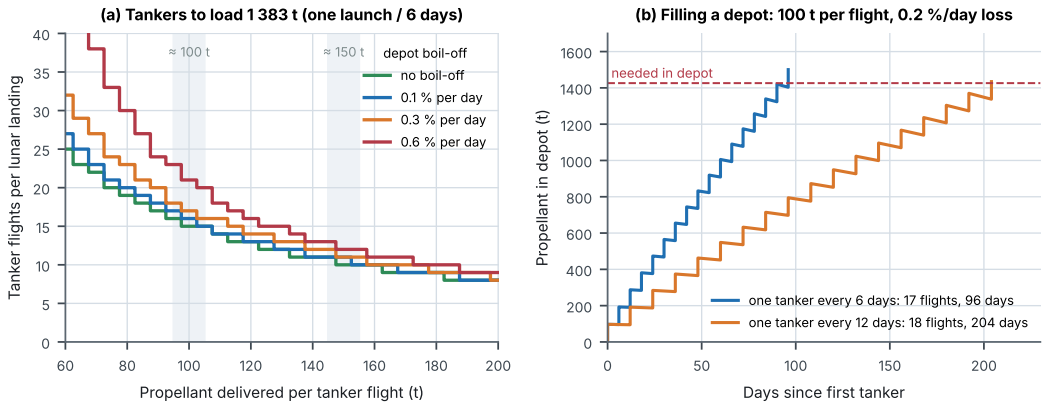
The economics of all these products depends on the price of the round trip. At a rideshare price of a few thousand dollars per kilogram to orbit, plus a capsule and recovery, only very valuable products are plausible candidates. They must be worth hundreds of thousands of dollars per kilogram — drug crystals, seed crystals and specialised fibre. Lower launch costs and reusable capsules would widen the range.

40.7 Refuelling for Exploration

Refuelling in orbit matters most for exploration. A rocket that arrives in low Earth orbit has used most of its propellant. If it can be refilled there, it can carry far more payload to the Moon or Mars than its launch vehicle alone could send. Both Artemis landers depend on this. SpaceX's Starship HLS is filled in LEO from a depot ship loaded by a series of tanker launches. Blue Origin's Blue Moon Mk2 is to be refuelled by a separate transporter vehicle.

Cryogenic propellants make this hard. Liquid oxygen boils at −183 °C and methane at −162 °C, so heat leaking into a tank boils them away continuously — **boil-off** — unless the tank is shaded, insulated or actively cooled. In weightlessness the liquid floats as blobs and does not sit over the outlet, so transfer needs a small settling acceleration or capillary devices. Transfer lines must first be chilled, wasting some propellant as gas. And the amount in a tank is hard to measure without gravity. RRM3's methane experiment failed before it could test these effects. In March 2024, on Starship's third flight, SpaceX transferred liquid oxygen between tanks inside one ship. The first ship-to-ship transfer, between two Starships docked in orbit, was planned for no earlier than late 2026; Starship first reached orbit on 28 September 2026. NASA restructured Artemis in 2026 so that Artemis III, in 2027, will rehearse rendezvous and docking with the landers in Earth orbit, with the first landing on Artemis IV.

How many tanker flights a landing needs depends on assumptions that SpaceX has not published in detail. Public estimates range from about 10 (SpaceX) to the "high teens" (NASA officials). Figure 40.16 shows how the number depends on how much each tanker delivers and how fast the depot loses propellant.



Assumed lander: 120 t dry + 15 t payload Δv 8.9 km/s after leaving LEO (to NRHO landing ascent) Isp 375 s → 1 383 t of propellant. 97 % efficiency for each transfer. Illustrative; not SpaceX figures.

FIGURE 40.16 Propellant transfer for a lunar lander. (a) Tanker flights needed to load the lander, versus propellant delivered per tanker, for different depot boil-off rates. (b) Propellant in the depot during a campaign with one tanker every 6 or 12 days: slower launch rates lose more to boil-off.

EXAMPLE: HOW MANY TANKERS FOR A LUNAR LANDING?

Assume (these are not SpaceX figures) a lander of 120 t dry mass carrying 15 t, with Raptor engines of Isp 375 s (v_e = 3 677 m/s). After leaving LEO it needs about 3.6 km/s to reach near-rectilinear halo orbit (NRHO), 2.65 km/s to land and 2.65 km/s to return to NRHO: 8.9 km/s in total.

1. Mass ratio: $e^{8\,900/3\,677}$ = 11.25, so the lander must leave LEO with 135 × 11.25 = 1 518 t, of which **1 383 t** is propellant.
2. If each tanker delivers 100 t, and each transfer (tanker to depot, depot to lander) loses 3 %, the lander needs 1 383 / (0.97 × 0.97 × 100) = 14.7. This rounds up to **15 tanker flights** with no boil-off.
3. With boil-off of 0.2 % per day in the depot and one tanker every 6 days, the campaign takes **17 flights over 96 days**; at one every 12 days, 18 flights over 204 days. Add the depot and the lander themselves, and a single landing needs about 19–20 launches.
4. If each tanker could deliver 150 t, the count falls to about **11**. The answer is most sensitive to tanker payload and launch rate — which is why rapid reuse is central to the plan.

AI AND AUTONOMY FOR SERVICING

Servicing is one of the most demanding applications of space autonomy. Approaching an unprepared, tumbling object requires the servicer to estimate its pose from camera images in real time. Convolutional neural networks trained on synthetic images now estimate pose from a single image. They have been tested in international satellite pose-estimation competitions run by ESA and Stanford. But the gap between rendered training images and real lighting remains the main difficulty. Onboard guidance must decide when to abort — ADRAS-J's autonomous abort in 2024 is an example — and learning-based planners are being studied for grasping tumbling targets and for scheduling multi-client missions. On the ground, machine-learning models monitor the health of the servicer and client and flag anomalies such as unexpected attitude behaviour (Chapters 45, 48 and 49).

KEY TAKEAWAYS

- ISAM spans inspection, relocation, life extension, refuelling, repair, assembly and manufacturing; each step up the ladder adds harder requirements for capture, fluid transfer and dexterity.
- Astronauts serviced Solar Max, Intelsat 603 and Hubble and assembled the ISS. Since 2020 commercial robots have docked with satellites as a paid service (MEV-1, MEV-2; MRV and MEPs launched July 2026). China has towed and probably refuelled satellites in GEO.
- Rendezvous uses V-bar hops and R-bar approaches designed to be passively safe. Final approaches are slow — centimetres per second, slower still in GEO. Capture of unprepared or tumbling clients relies on launch-adapter rings, engine nozzles and careful detumbling.
- Life extension pays when a satellite earns more than the roughly $13 million per year fee or when it defers a replacement; debris removal remains a government-funded demonstration market costing about $80–100 million per object.
- Assembly escapes the fairing limit that caps monolithic and folded structures at a few fairing diameters; NASA's robotic assembly demonstrations were cancelled, but the ISS proves the principle.
- Manufacturing in orbit has moved from ISS printers to free-flying factories with return capsules; refuelling in orbit, still undemonstrated between two vehicles, is the key to Artemis landers that need 10–20 tanker flights each.

REVIEW QUESTIONS

1. List the main ISAM activities in order of difficulty and explain what each additional step requires of the servicer and the client.
2. Why did servicing Solar Max and Hubble succeed while the first attempts to capture Intelsat 603 failed? What does this imply for designing future satellites?
3. How does the Mission Extension Vehicle attach to a client that was never designed for servicing, and why does it use electric propulsion?
4. Using the Clohessy–Wiltshire equations, calculate the Δv for a 2 km radial hop along the V-bar in a 500 km orbit, and the time it takes.
5. Explain "passive safety" in rendezvous. Why must approaches in GEO be slower than in LEO?
6. A dead satellite with a moment of inertia of 5×10^4 kg·m^2 tumbles at 2° per second. What is its angular momentum, and how could a servicer remove it after capture?
7. A GEO satellite of 3 000 kg has 80 kg of usable propellant. Estimate its remaining life with chemical thrusters of 290 s, keeping 11 m/s for disposal.
8. Why do debris-removal missions struggle to find a business case, and what funding models have been proposed?
9. Explain why folding structures cannot grow beyond a few fairing diameters, and outline the engineering challenges of assembling a 20 m telescope in orbit.
10. Explain the difficulties of transferring cryogenic propellants in orbit and estimate how the number of tanker flights for a lunar landing changes if each tanker delivers 150 t instead of 100 t.

CHAPTER

41 Space Resources and Asteroid Mining

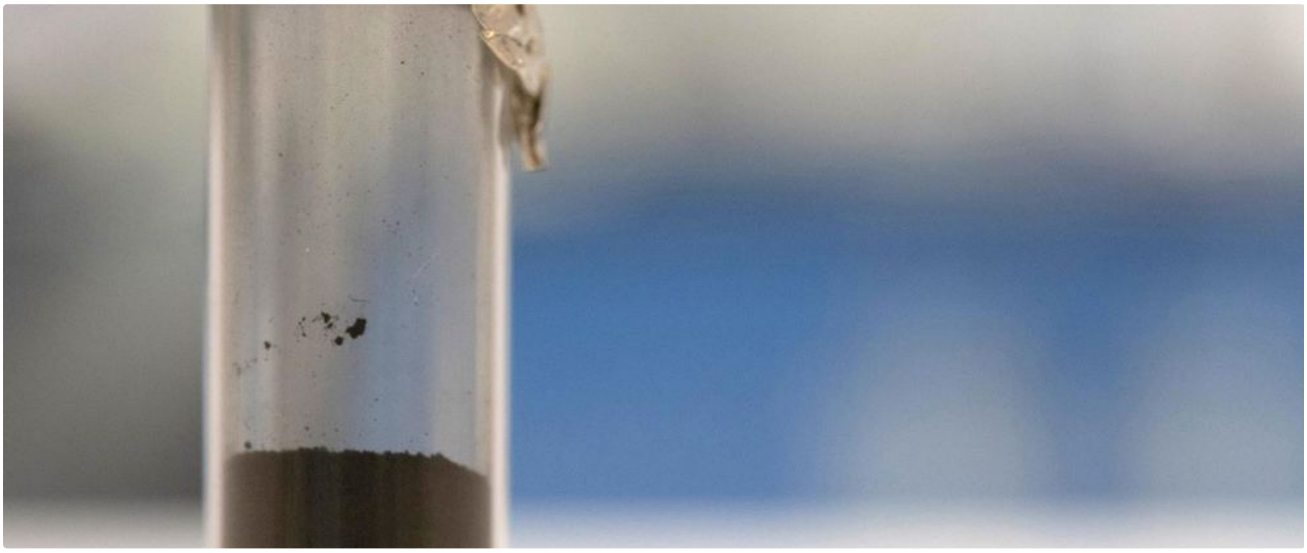


Image: NASA/GSFC

Every kilogram of propellant, water or shielding used beyond Earth orbit must be lifted out of a deep gravity well at great cost. Space resources — ice and oxygen-rich rock on the Moon, carbon dioxide and ice on Mars, water and metals in asteroids — promise to cut that cost by making what is needed where it is needed. This chapter quantifies the "gear ratio" that makes local production attractive, surveys what is known about each resource, and explains the processes that turn regolith and ice into water, oxygen, propellant, metals and building material, with their energy costs. It then reviews the prospecting missions from LCROSS to VIPER, the rise and fall of asteroid-mining companies, the economics of selling space metals on Earth, and the laws that decide who may own what is extracted.

IN THIS CHAPTER

41.1 Why Use Space Resources?
41.2 What Is Out There
41.3 Turning Rock and Ice into Products
41.4 Prospecting: Finding the Ore
41.5 The Business of Space Mining
41.6 Law and Policy

Polar explorers hauled everything on sledges until they learned to hunt and to build shelters from snow. Spaceflight is still in the sledge-hauling era: every Apollo crew carried all its oxygen, water and propellant from Earth. Because of the exponential rocket equation the penalty is severe — with the assumptions used in this chapter, each kilogram placed on the lunar surface requires about 130 kg of rocket on the launch pad.

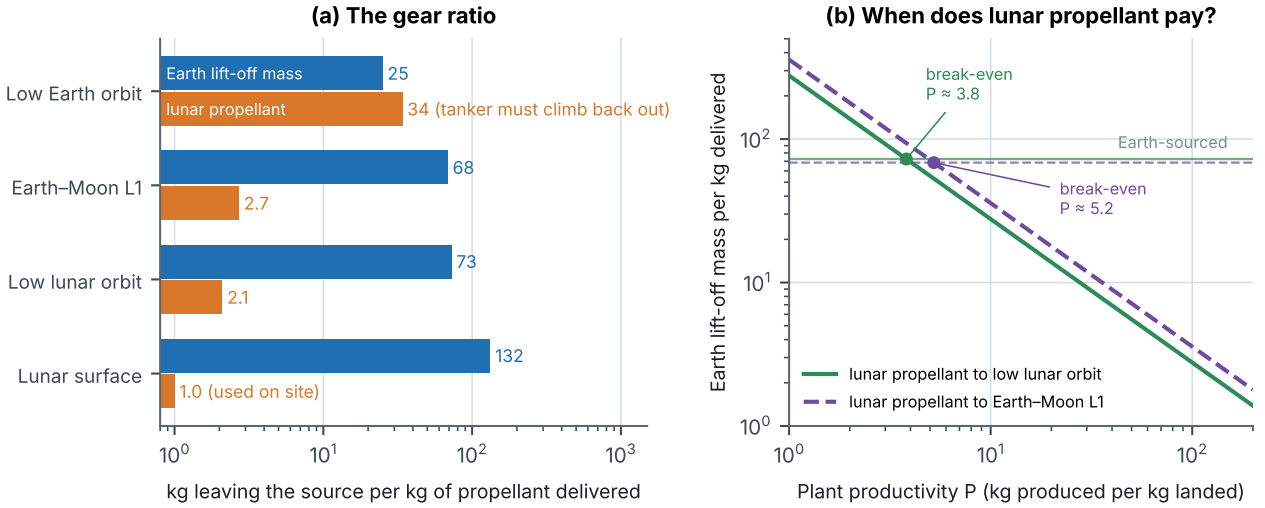
In situ resource utilisation (ISRU) — from the Latin *in situ*, "in place" — means producing useful materials from what is found at the destination. The most valuable products are mundane. One is **oxygen**, about four-fifths of the propellant mass of hydrogen–oxygen and methane–oxygen rockets. Another is **water**, for drinking, growing food and radiation shielding. A third is **regolith**, the loose rock and dust covering the Moon, Mars and asteroids, for building landing pads, roads and shelters. Further off lie metals for building in space, and precious metals and helium-3 for Earth.

This chapter covers the arithmetic that makes space resources attractive, the resources themselves, the chemistry and energy cost of extracting them, prospecting, the mining companies and the law. Landers and rovers are covered in Chapters 27 and 28, propellant transfer in orbit in Chapter 40.

41.1 Why Use Space Resources?

41.1.1 The Gear Ratio

The rocket equation (Chapter 34), $\Delta v = v_e \ln(m_0/m_f)$, makes the mass needed grow exponentially with the velocity change. Reaching low Earth orbit (LEO) takes about 9.4 km/s including losses, so even a large launcher delivers only about 4 % of its lift-off mass to orbit. From LEO, a further 4.0 km/s reaches low lunar orbit (LLO) and 5.9 km/s the lunar surface. The ratio of mass leaving Earth to mass delivered is the **gear ratio** of a destination (Figure 41.1 a).



Launcher: 4 % of lift-off mass reaches LEO. Earth-sourced: expendable LOX/LH_2 stages, Isp 450 s, dry mass 10 % of propellant; Δv LEO→L1 3.8, →LLO 4.0, →lunar surface 5.9 km/s. Moon-sourced: reusable LOX/LH_2 tanker (10 t dry, 30 t loaded) keeps propellant to return; Δv surface→LLO 1.9, →L1 2.5, →LEO 2.8 km/s (aerobraking; 5.9 km/s back). Plant and tanker mass in (b) landed from Earth.

FIGURE 41.1 The gear ratio. (a) Mass leaving its source per kilogram of propellant delivered: Earth lift-off mass for Earth-launched propellant, and lunar propellant for a reusable lunar tanker. (b) Earth lift-off mass per kilogram delivered to low lunar orbit or L1, against the productivity of a lunar plant.

With hydrogen–oxygen stages of specific impulse 450 s whose empty mass is 10 % of their propellant, each kilogram delivered to LLO needs 2.9 kg in LEO and about 73 kg at lift-off. Each kilogram on the lunar surface needs 5.3 kg in LEO and 132 kg at lift-off. Propellant made on the Moon starts in a far shallower gravity well — escape velocity 2.4 km/s against Earth's 11.2 km/s,

with no atmosphere. A reusable tanker that carries lunar propellant to LLO, keeping enough to fly back down, burns only about 1.1 kg for every kilogram it delivers: a gear ratio of 2.1 instead of 73. Delivery to the Earth–Moon L1 point, a proposed depot site, costs 2.7.

The comparison reverses in LEO. A tanker can brake into LEO in the atmosphere (aerobraking), needing only about 2.8 km/s, but then needs the full 5.9 km/s to return. Each kilogram delivered costs 34 kg of lunar propellant — worse than launching from Earth — or about 5 if the tanker refuels with lunar propellant at L1 on the way back. Lunar propellant is most valuable close to the Moon.

41.1.2 When Does Local Production Pay?

A lunar plant is not free: its excavators, reactors, power system and tanker must be landed from Earth at 132 kg of lift-off mass per kilogram. The figure of merit is the plant's **productivity** P, the mass of product made over its life divided by its landed mass. Lunar propellant delivered to LLO beats Earth-launched propellant once P exceeds about 3.8, and at L1 once it exceeds about 5.2 (Figure 41.1 b). The question is whether a lunar plant can survive abrasive dust, temperatures from 40 K to 390 K and years without maintenance for long enough.

EXAMPLE: THE LEVERAGE OF A LUNAR-ORBIT TANKER

A reusable tanker with an empty mass of 10 t carries 30 t of liquid oxygen and hydrogen from a polar plant to a depot in low lunar orbit and returns. Its engines have I_{sp} = 450 s (v_e = 4 413 m/s); each leg needs Δv = 1.9 km/s.

1. Mass ratio per leg: $R = e^{1\,900/4\,413}$ = 1.538.
2. Ascent propellant (10 t of tanker plus 30 t of cargo): 40 × (1.538 − 1) = 21.5 t. To return, the empty tanker keeps 10 × 0.538 = 5.4 t, so it delivers 30 − 5.4 = **24.6 t** per trip.
3. Lunar production per trip: 21.5 + 30 = 51.5 t — a gear ratio of 51.5 / 24.6 = **2.09**.
4. Delivering 24.6 t from Earth instead would need 24.6 × 2.90 = 71.5 t in LEO, or about **1 790 t** of launch vehicle at lift-off.
5. Suppose the plant, its power supply and the tanker land 40 t from Earth (5 290 t at lift-off) and produce 100 t of propellant a year for 10 years (P = 25). They deliver 1 000 / 2.09 = 478 t to LLO, at 5 290 / 478 = **11 kg** of Earth lift-off mass per kilogram, against 73 kg for Earth-launched propellant — a sixfold gain. At P = 3.8 the advantage vanishes.

The same logic applies to Mars, where an ascent needs about 4–5 km/s (Chapter 29), and to other products: a tonne of regolith piled over a habitat as shielding need not be launched.

41.2 What Is Out There

The resources differ in concentration by a factor of a billion — from oxygen, nearly half of every rock, to helium-3, a few parts per billion of lunar soil. Figure 41.4 compares the elemental make-up of typical feedstocks, and Table 41.1, at the end of this section, summarises the resources.

41.2.1 Lunar Regolith

The Moon is covered by **regolith**: rock fragments, glass and dust produced by four billion years of meteorite impacts. It is about 4–5 m deep on the dark volcanic plains (**maria**) and 10–15 m in the older **highlands**. It is made of oxides: oxygen is 41–45 % of the mass, bound to silicon, aluminium, iron, calcium, magnesium and titanium. Highland soils are rich in aluminium and calcium. Mare soils carry up to about 20 % FeO. In the "high-titanium" basalts sampled by Apollo 11 and 17, they carry 10 % or more of the iron–titanium oxide **ilmenite** ($FeTiO_3$), the favourite feedstock for oxygen extraction (Figure 41.2). Volcanic glass beads such as Apollo 17's orange soil (Figure 41.3) are also rich in these oxides.

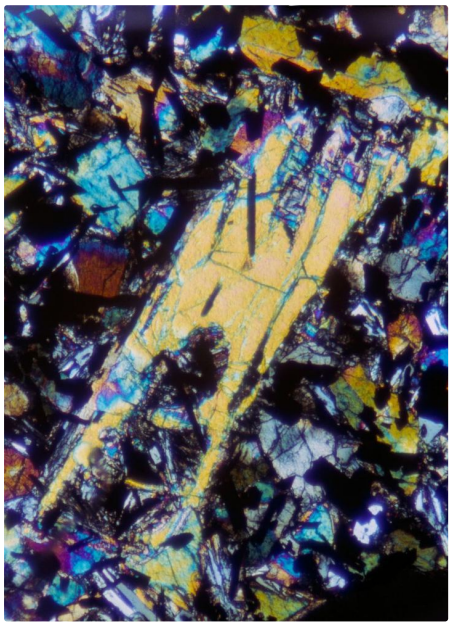
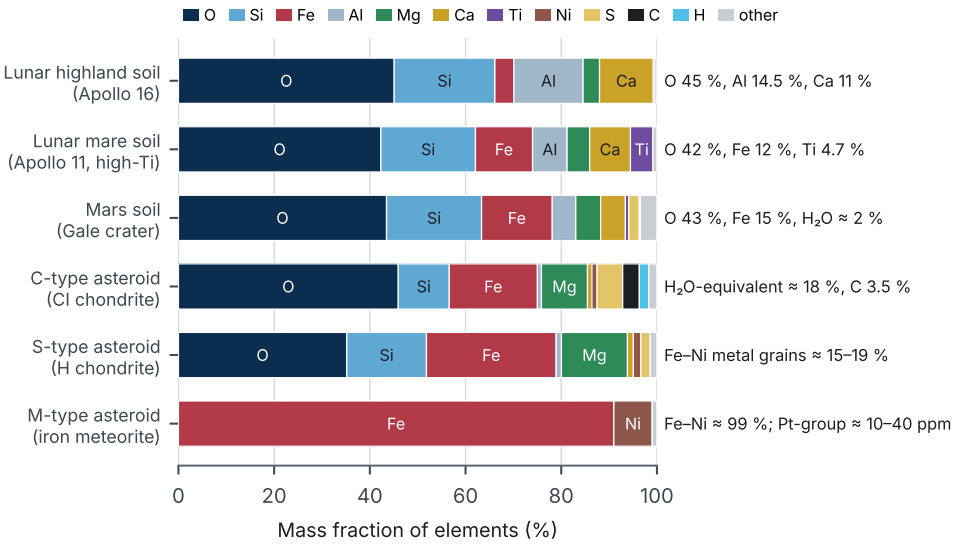


FIGURE 41.2 Photomicrograph of a thin slice of Apollo 11 basalt 10022. Black grains are ilmenite, light blue and white grains are plagioclase and the brightly coloured grains are pyroxene. Image: NASA/JSC.



FIGURE 41.3 The orange soil discovered by Apollo 17 at Shorty crater in December 1972 — tiny beads of volcanic glass rich in titanium and iron. Image: NASA/JSC.



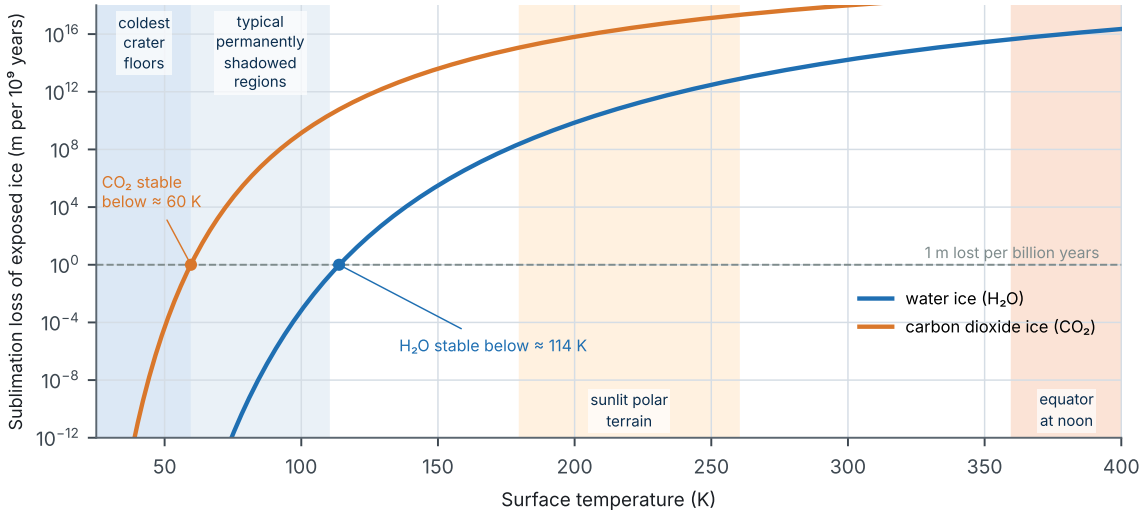
Element fractions from average oxide analyses (lunar soils 64501-type and 10084; MSL Rocknest with ≈ 2 % H_2O), CI (Lodders 2003), H-chondrite and iron-meteorite averages. Representative values only.

FIGURE 41.4 Elemental composition by mass of lunar highland and mare soils, Martian soil and the meteorite types matching C-, S- and M-type asteroids.

Regolith also holds **solar-wind** atoms implanted into grain surfaces, hydrogen at tens of parts per million (ppm). On the western near side, the **Procellarum KREEP Terrane** is enriched in potassium (K), rare-earth elements (REE) and phosphorus (P). But its rare-earth concentrations, though hundreds of times those of primitive meteorites, are far below those of rich ores on Earth.

41.2.2 Ice at the Poles

The Moon's spin axis is tilted only 1.54° to the ecliptic, so crater floors near the poles never see the Sun (Chapter 28). In these **permanently shadowed regions (PSRs)** the Diviner radiometer on NASA's Lunar Reconnaissance Orbiter (LRO) measures temperatures below 40 K, and about 25 K in the coldest places. Water molecules from comets, asteroids, volcanic gases or the solar wind hop across the surface until they land in such a **cold trap** and freeze. Figure 41.5 shows why. Exposed ice at 114 K loses about a metre per billion years by sublimation, but at 150 K it loses hundreds of kilometres. Carbon dioxide ice needs temperatures below about 60 K.



Free sublimation into vacuum (Hertz–Knudsen) with ice vapour pressure from Murphy & Koop (2005) and CO_2 from Clausius–Clapeyron; a regolith cover slows the loss further.

FIGURE 41.5 Sublimation loss of exposed water and carbon dioxide ice in vacuum. Water ice survives for billions of years only below about 110–115 K, on permanently shadowed crater floors; CO_2 ice only in the coldest.

The evidence accumulated over three decades:

- **Lunar Prospector** (1998) found excess hydrogen at both poles with a neutron spectrometer: hydrogen nuclei slow the neutrons leaking from the surface.
- NASA's Moon Mineralogy Mapper on India's **Chandrayaan-1** (2008–09) detected hydroxyl and water bound in minerals across much of the surface; a 2018 reanalysis found the infrared signature of water ice exposed in some PSRs.
- **LCROSS** (9 October 2009) crashed a spent 2.3 t Centaur upper stage into Cabeus crater and flew through the plume. It measured 5.6 ± 2.9 % water by mass in the ejecta, together with hydrogen sulphide, ammonia, sulphur dioxide, carbon dioxide and methanol — useful, but also contaminants that a water plant must remove.
- **LRO** (since 2009) mapped cold traps and found ultraviolet hints of patchy frost. **ShadowCam** on Korea's Danuri orbiter images PSRs by light scattered from sunlit terrain. In 2026 it found no widespread surface ice above about 20–30 % by mass — only a few patches 20–50 m across with probably more than 10 %.

- **Chandrayaan-3**'s ChaSTE probe measured 82 °C on a Sun-facing slope at 69° S in 2023, implying that steep pole-facing slopes at high latitudes may keep ice near the surface even outside PSRs.

Cold traps, including countless "micro cold traps" a few metres across, cover about 40 000 km^2, 60 % of it in the south. Their ice content is uncertain by orders of magnitude. If the top metre of all of them held LCROSS-like 5.6 % ice at a regolith density of 1 500 kg/m^3, the total would be about 3 billion tonnes. That is enough for a thousand tonnes of propellant a year for millions of years. The real deposits are probably patchy and buried, which is why prospecting on the ground matters.

41.2.3 Helium-3: A Realistic Assessment

Helium-3, a light isotope carried by the solar wind, is often cited as a reason to mine the Moon because its fusion with deuterium releases few neutrons. The arithmetic is daunting. Lunar regolith contains on average about 4 parts per billion (ppb) of helium-3, and 10–20 ppb in the richest titanium-bearing mare soils, where ilmenite retains it best. A litre of the gas weighs 0.135 g, so at 10 ppb about 13 t of regolith must be dug and heated to several hundred degrees Celsius per litre — 34 t at 4 ppb. Deuterium–helium-3 fusion also needs plasma temperatures several times higher than the deuterium–tritium reaction, which no power plant has yet used to produce net electricity.

The near-term market is **cryogenics**: dilution refrigerators that cool quantum computers to millikelvin temperatures need helium-3. Terrestrial supply, mostly from the decay of tritium, is tiny, and it sells for about US$ 2 500–3 000 per litre — some US$ 20 million per kilogram. The start-up **Interlune** has agreements to supply 3 litres to the US Department of Energy by April 2029 and up to 10 000 litres a year in 2028–37 to the cryostat maker Bluefors. At 10 ppb, 10 000 litres a year means processing 135 000 t of regolith a year, 15 t every hour: a credible niche product, not an energy source.

41.2.4 Mars

Mars offers two resources almost everywhere: an atmosphere that is 95 % carbon dioxide at a mean surface pressure of about 0.6 kPa, and water. The polar caps hold enough ice to cover the planet about 20 m deep, but lie far from the low latitudes where landing is easiest. Closer to hand, Phoenix scraped into ice centimetres below the soil at 68° N in 2008. Orbital radar has mapped buried mid-latitude glaciers. Fresh craters expose ice within a metre of the surface as far from the poles as about 40° (Figure 41.6). Even soil in Gale crater releases about 2 % water when heated.

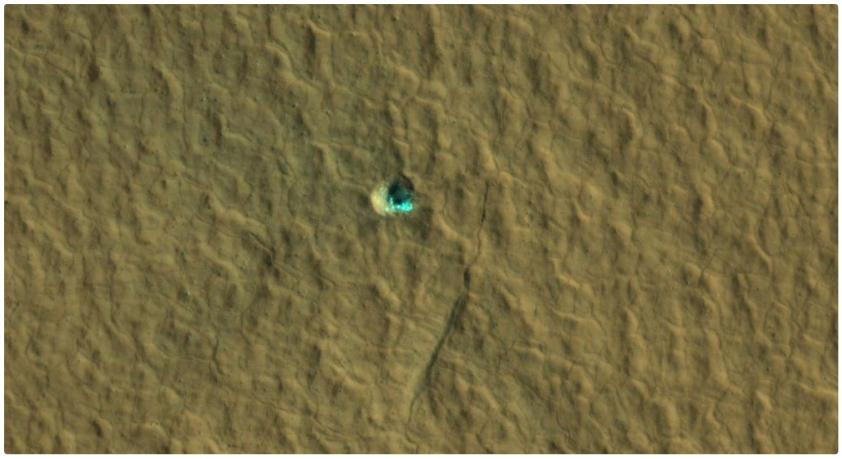


FIGURE 41.6 A small, fresh impact crater on Mars that has exposed bright subsurface ice, surrounded by polygon-patterned ground, imaged by the HiRISE camera on Mars Reconnaissance Orbiter. Craters like this are used to map shallow ice near candidate landing sites. Image: NASA/JPL (NASA/JPL-Caltech/University of Arizona).

41.2.5 Asteroids

Asteroids are classified by their reflectance spectra; three classes matter most:

- **C-type** (carbonaceous) asteroids are dark, dominate the outer main belt and match carbonaceous chondrites. The most primitive, CI chondrites, hold water equivalent to 13–20 % of their mass, mostly in clays, plus 3–4 % carbon. Samples from Ryugu and Bennu are similar clay- and carbon-rich material, though Ryugu's grains hold only about 7 % water (Chapter 29).
- **S-type** (stony) asteroids dominate the inner belt and the known near-Earth population. They match ordinary chondrites, which in the H group contain 15–19 % iron–nickel metal; NEAR Shoemaker confirmed such a composition at the 34 km Eros in 2000–01 (Figure 41.7).
- **M-type** asteroids have spectra and radar echoes suggestive of metal. Iron meteorites, their likely counterparts, are about 90 % iron and 5–10 % nickel. They hold 10–40 ppm of **platinum-group metals** (platinum, palladium, rhodium, iridium, osmium, ruthenium). This is several times the grade of the richest ores on Earth, which hold a few grams per tonne.

The largest M-type, **16 Psyche**, about 220 km across, was thought to be the bare iron core of a protoplanet. Its density of 3 400–4 100 kg/m^3 suggests instead perhaps 30–60 % metal by volume, mixed with rock. NASA's Psyche spacecraft (Figure 41.8), launched in October 2023 and driven by Hall-effect thrusters, flew past Mars in May 2026 and will arrive in 2029. Headline claims that Psyche is "worth" quintillions of dollars are meaningless: such supply would collapse metal prices.

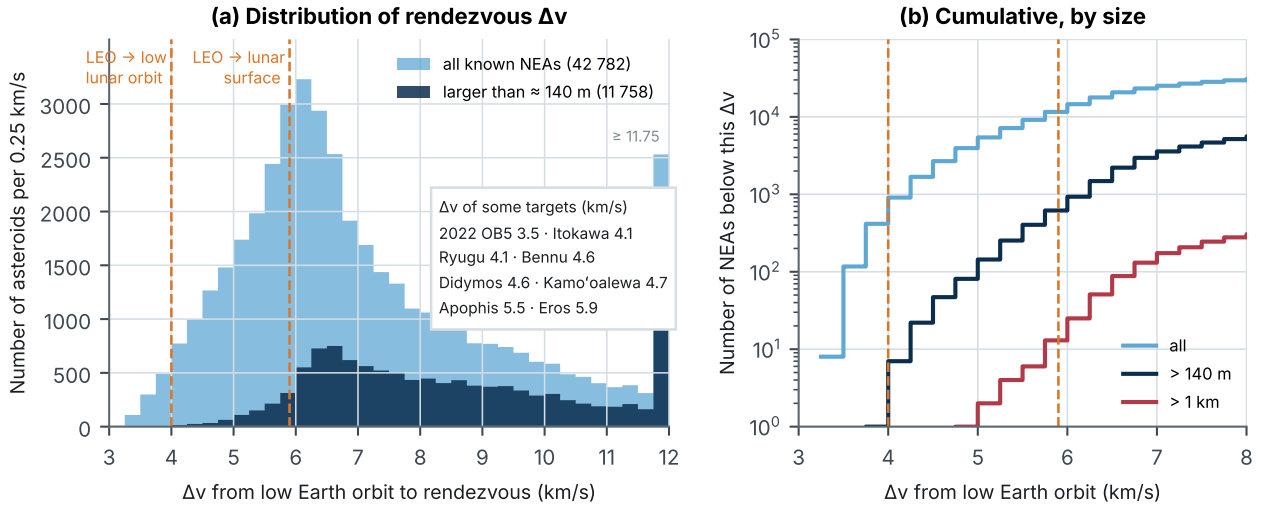


FIGURE 41.8 NASA's Psyche spacecraft in a clean room near Kennedy Space Center in June 2023, during final assembly before its launch to the metal-rich asteroid 16 Psyche. Image: NASA/KSC (NASA/Frank Michaux).

FIGURE 41.7 Mosaic of the northern hemisphere of the S-type asteroid Eros, 34 km long, imaged by NEAR Shoemaker in 2000. Image: NASA/JPL (NASA/JPL/JHUAPL).

For mining, the important asteroids are the **near-Earth asteroids (NEAs)**, whose orbits come within 1.3 astronomical units of the Sun. On 19 October 2026 the JPL Small-Body Database listed 42 782, including 875 larger than about 1 km and 11 758 larger than about 140 m; new surveys will multiply the number (Chapter 26).

Figure 41.9 estimates the Δv to reach each one, in the spirit of Eugene Shoemaker and Eleanor Helin (1978). It uses an idealised two-burn transfer from a 400 km orbit to the asteroid's farthest point from the Sun. About 900 known NEAs need less Δv than a trip to low lunar orbit (4.0 km/s), and about 14 600 less than 6 km/s, close to the cost of landing on the Moon. The catch is size: only 7 of the easiest are larger than 140 m, and only about 25 objects larger than 1 km lie below 6 km/s. Their Earth-like orbits also mean that launch opportunities recur only every few years or decades.



42 782 NEAs from the JPL Small-Body Database (19 Oct 2026). Idealised two-impulse transfer from a 400 km orbit to the asteroid's aphelion, plane change split optimally; phasing and launch windows ignored. Size from absolute magnitude H (≤ 22 ≈ 140 m, ≤ 17.75 ≈ 1 km, albedo 0.14). Real missions usually need more Δv; Earth and lunar flybys can reduce it.

FIGURE 41.9 Accessibility of the 42 782 known near-Earth asteroids. (a) Estimated Δv from low Earth orbit to rendezvous. (b) Number of NEAs below a given Δv, by size. Dashed lines: Δv from LEO to low lunar orbit and to the lunar surface.

TABLE 41.1 The main resources beyond Earth.

Resource	Where	Typical concentration	Main products	Key uncertainty
Polar water ice	Lunar cold traps (≈ 40 000 km²)	0 to > 10 %; LCROSS 5.6 %	Water, H_2, O_2	Amount, depth, patchiness
Oxides in regolith	Whole Moon	O 41–45 %, Fe ≤ 15 %, Al ≤ 15 %	O_2, Fe, Si, Al, Ti	Energy per kg
Solar-wind volatiles	Lunar regolith, esp. Ti-rich mare	H tens of ppm; ^{3}He ≈ 4–20 ppb	H_2O, ^{3}He	Very low grade
Atmospheric CO_2	All of Mars	95 % of a 0.6 kPa atmosphere	O_2, CO, CH_4 (with H_2)	Power, dust
Water on Mars	Poles, buried ice, soil	≈ 2 % in soil to near-pure ice	Water, H_2	Depth and access
Carbonaceous asteroids	Outer main belt; some NEAs	Up to ≈ 18 % water-equivalent, C 3–4 %	Water, carbon, organics	Access, anchoring
Metallic asteroids	Rare; Psyche; a few NEAs	Fe–Ni ≈ 90 %+; Pt-group 10–40 ppm	Metals	Composition, market

41.3 Turning Rock and Ice into Products

Three process chains dominate ISRU planning (Figure 41.10): water from lunar polar ice; oxygen and metals from ordinary regolith; and oxygen and methane from Martian carbon dioxide and ice. Table 41.2 compares the main processes.

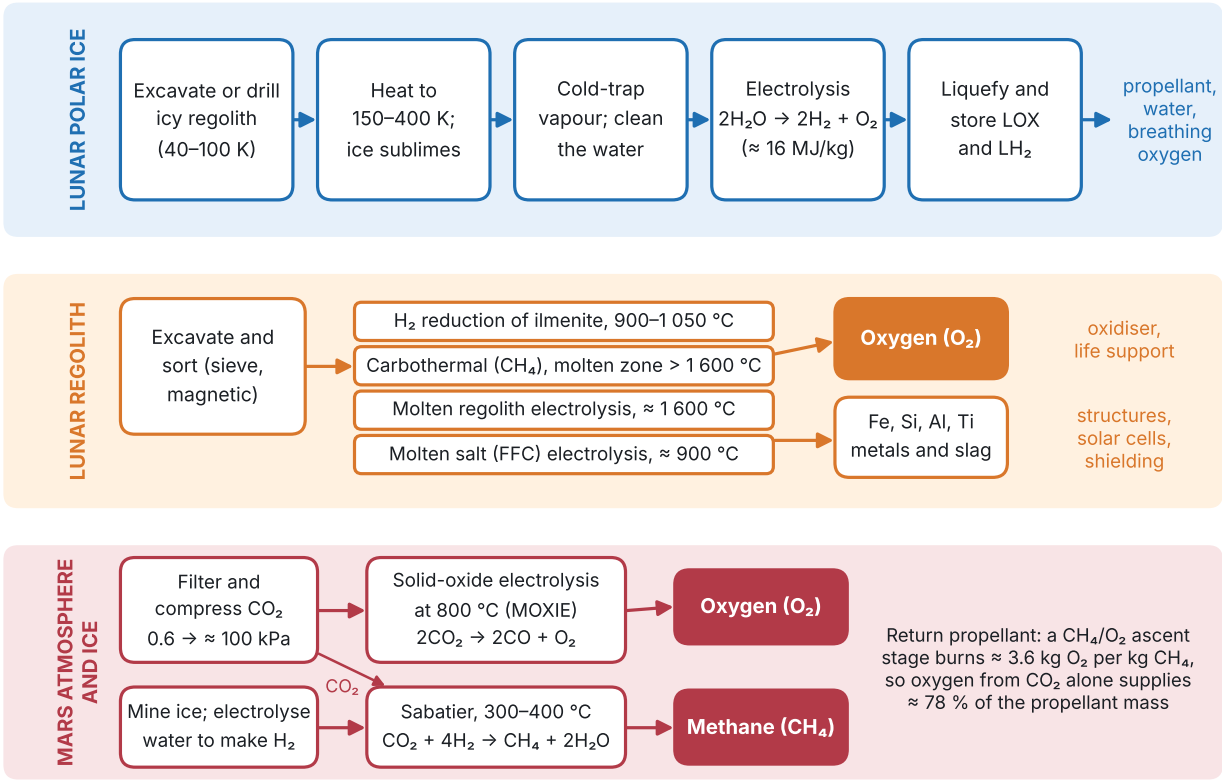


FIGURE 41.10 The three main ISRU process chains: lunar polar ice, ordinary lunar regolith, and the Martian atmosphere and ice.

TABLE 41.2 Main ISRU processes. Maximum oxygen yields are theoretical, for typical lunar soils.

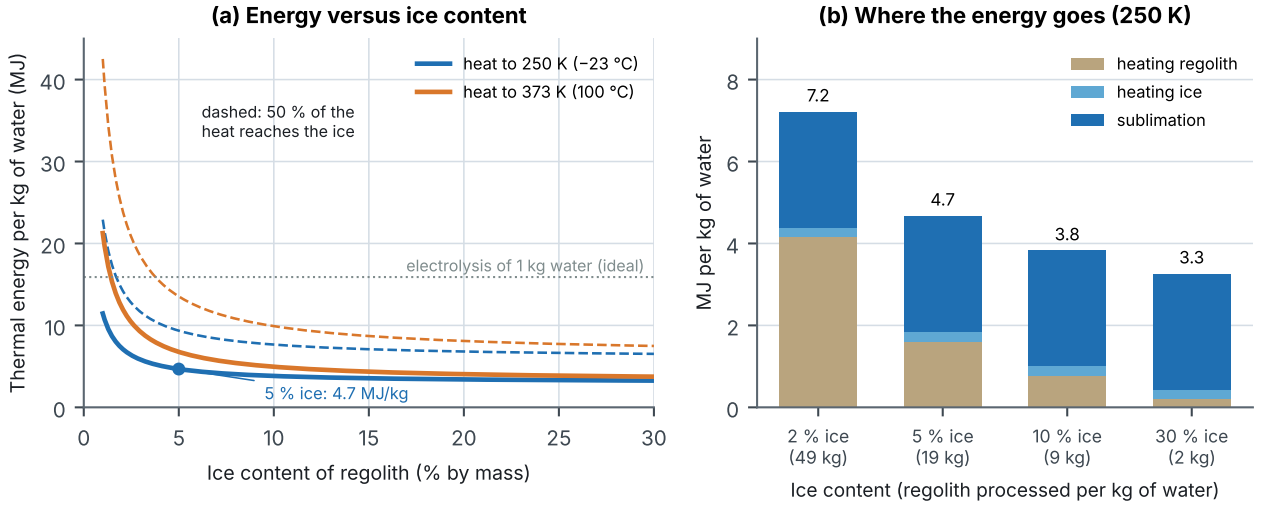
Process	Feedstock	Temperature	Products	Max. O_2 (kg/t)	Status (Oct 2026)
Ice extraction + electrolysis	Icy polar regolith	−120 to +100 °C	H_2O, H_2, O_2	Depends on ice	Lab tests; prospecting from 2027
H_2 reduction of ilmenite	Mare soil, volcanic glass	900–1 050 °C	O_2 (via H_2O), Fe, TiO_2	9–34	Field tests in Hawaii (2008–10)
Carbothermal reduction	Any regolith	> 1 600 °C (melt zone)	O_2 (via H_2O), slag	250–290	Vacuum demonstration 2023
Molten regolith electrolysis	Any regolith	≈ 1 600 °C	O_2, Fe, Si, Al	420–450	Lab; Blue Alchemist
Molten salt (FFC) electrolysis	Any regolith	≈ 900–950 °C	O_2, metal alloy powder	400–430	Lab (ESA, Metalysis)
Solid-oxide CO_2 electrolysis	Mars atmosphere	≈ 800 °C	O_2, CO	—	Flown: MOXIE, 122 g O_2
Sabatier reaction	CO_2 + H_2	300–400 °C	CH_4, H_2O	—	Used on ISS from 2010

41.3.1 Water from Icy Regolith

The simplest chain digs polar regolith, heats it to drive off the ice, freezes the vapour on a cold surface, purifies it and splits it. Machines must dig soil at 40–100 K, where ice-cemented regolith may be as hard as weak rock, in darkness. Prototypes such as NASA's **RASSOR** and **IPEx** use counter-rotating bucket drums whose digging forces cancel, so a light robot can excavate in one-

sixth gravity without being pushed backwards (Figure 41.14). Alternatives heat the ground in place, with heated drills or — in one "thermal mining" proposal — mirrors on a crater rim that reflect sunlight onto icy ground beneath a dome.

The thermal energy has two main terms: warming the regolith that comes with the ice, and the latent heat of sublimation (2.83 MJ/kg). Figure 41.11 shows that at 30 % ice sublimation dominates and about 3.3 MJ is needed per kilogram of water; at 2 % ice, 49 kg of rock must be warmed per kilogram of water and the total more than doubles. Real plants lose heat and vapour, so the practical figure may be twice the ideal.



Regolith starts at 40 K; specific heat from Hayne et al. (2017), ice c ≈ 7.7·T J/(kg·K), sublimation 2.83 MJ/kg. Solid lines assume all heat reaches the feedstock and all vapour is captured.

FIGURE 41.11 Thermal energy to extract water from icy regolith. (a) Energy per kilogram of water versus ice content; dashed lines assume half the heat reaches the feedstock. (b) Breakdown at 250 K: at low ice content, most energy heats rock.

Electrolysis, $2H_2O \rightarrow 2H_2 + O_2$, needs at least 15.9 MJ per kilogram of liquid water — more than the extraction. It yields hydrogen and oxygen in the mass ratio 1 : 8, while engines burn them at about 1 : 6, leaving surplus oxygen for life support. Oxygen liquefies at 90 K, which radiators can reach near the poles; hydrogen needs 20 K and active refrigeration (Chapter 40).

EXAMPLE: POWER FOR A POLAR WATER PLANT

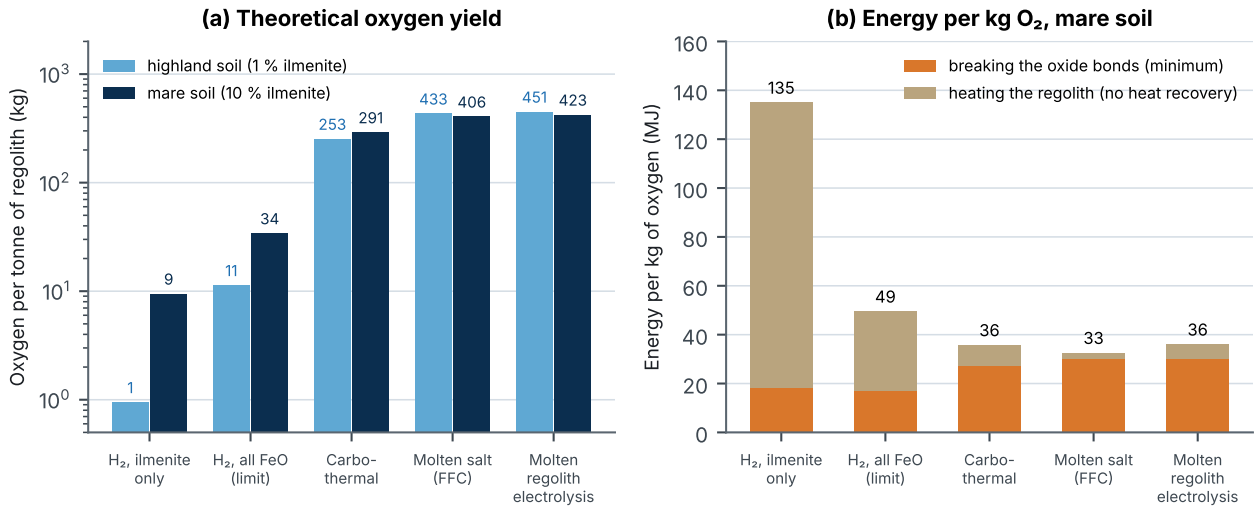
A plant mines regolith containing 5 % ice by mass and must produce 100 t of water a year. The regolith starts at 40 K and is warmed to 250 K.

1. Regolith processed: 19 kg per kilogram of water, so **1 900 t a year**, or 5.2 t a day.
2. Ideal thermal energy (Figure 41.11): warming the regolith 1.61 MJ, warming the ice 0.24 MJ, sublimation 2.83 MJ — **4.7 MJ** per kilogram of water; 9.4 MJ/kg if only half the heat reaches the feedstock.
3. Electrolysis at 70 % efficiency: 15.9 / 0.7 = 22.7 MJ/kg. Total about **32 MJ** (8.9 kWh) per kilogram of water.
4. Annual energy: 100 000 kg × 32 MJ = 3.2 TJ. Divided by the 3.16×10^7 s in a year, this is an average of **about 100 kW**, day and night, before liquefaction and losses. That is the output that NASA's 2025 directive set for a lunar fission reactor around 2030. However, the 2026 draft procurement for the first unit, Lunar Reactor-1, asks for at least 20 kWe.
5. Products: 11.2 t of hydrogen and 88.8 t of oxygen. At a mixture ratio of 6 : 1 the hydrogen supports 78 t of propellant, leaving 22 t of oxygen for breathing or an oxygen-rich engine.

41.3.2 Oxygen from Ordinary Regolith

Ice exists only at the poles, but oxygen is everywhere. Freeing it from the strong bonds of silicates and oxides takes at least the enthalpy of formation of the oxides. This is 18 MJ per kilogram of oxygen for ilmenite and about 30 MJ/kg for complete reduction of typical soil. The soil must also be heated to 900–1 600 °C. Figure 41.12 compares four processes.

- **Hydrogen reduction.** Hot hydrogen removes the oxygen bound to iron, $\mathrm{FeTiO_3 + H_2 \rightarrow Fe + TiO_2 + H_2O}$, at 900–1 050 °C; the water is electrolysed and the hydrogen recycled. Only iron oxide reacts, so the yield is about 1 % of the soil mass unless the ilmenite is first concentrated magnetically or electrostatically. NASA field tests on Mauna Kea, Hawaii, in 2008 and 2010 ran small hydrogen-reduction and carbothermal plants.
- **Carbothermal reduction.** Methane passed over molten regolith above about 1 600 °C reduces silicates too, giving carbon monoxide and hydrogen that are converted back into methane and water; yields reach 25–30 % of the regolith mass. In April 2023 a NASA demonstration with a laser-heated Sierra Space reactor extracted oxygen from simulant this way in vacuum for the first time.
- **Molten regolith electrolysis (MRE).** As in aluminium smelting, a current through regolith melted at about 1 600 °C releases oxygen at an inert anode, while iron and silicon collect at the cathode. MRE frees almost all the oxygen but needs anodes that survive hot, corrosive melts. Blue Origin's **Blue Alchemist**, partly funded by a US$ 34.7 million NASA award, has made solar cells from simulant this way and passed its critical design review in 2025.
- **Molten salt (FFC Cambridge) electrolysis.** A regolith pellet forms the cathode in molten calcium chloride at about 950 °C; oxygen ions migrate to the anode, leaving metal alloy powder. Researchers working with ESA and Metalysis extracted 96 % of the oxygen from a simulant in 2020; inert anodes are again the key development.



Typical Apollo 16 and Apollo 11 soil oxides; 90 % of ilmenite reduced; reaction energy = enthalpy of formation of the reduced oxides (298 K). Process temperatures 1 300 K (H_2), 1 200 K (FFC), 1 900 K (molten); fusion 0.48 MJ/kg. Real plants add electrical and thermal losses, and can recover much of the sensible heat.

FIGURE 41.12 Oxygen from lunar regolith. (a) Theoretical yield per tonne of highland and mare soil (log scale). (b) Minimum energy per kilogram of oxygen from mare soil, without heat recovery.

EXAMPLE: OXYGEN FROM ILMENITE

A hydrogen-reduction plant processes high-titanium mare soil containing 10 % ilmenite by mass and reduces 90 % of it at 1 300 K.

1. Ilmenite per tonne: 100 kg, or 100 000 / 151.7 = 659 mol of $FeTiO_3$, of which 593 mol is reduced.
2. Each mole gives one mole of water and, after electrolysis, half a mole of O_2: 593 × 16.0 g = **9.5 kg of oxygen per tonne** (0.95 %). The 1.2 kg of hydrogen is recycled.
3. Chemical energy: 288 kJ per mole of oxygen atoms, or 171 MJ per tonne — **18 MJ per kilogram of O_2.**
4. Heating a tonne of regolith from 250 K to 1 300 K takes about 1 110 MJ, or 117 MJ per kilogram of O_2 — six times the chemistry. Recovering 75 % of the heat from the spent soil cuts the total from 135 to 47 MJ/kg; concentrating the ilmenite to 50 % first cuts the heating term five times.
5. To make 10 t of oxygen a year the plant must dig and process about **1 050 t of soil**, or 2.9 t a day. This is why low-yield processes need beneficiation, and why ice is attractive where it exists.

41.3.3 Oxygen and Methane on Mars

On Mars the feedstock comes to the plant. NASA's **Mars Oxygen In-Situ Resource Utilisation Experiment (MOXIE)** was a 17 kg unit inside Perseverance (Figure 41.13). It compressed Martian air from about 0.6 kPa to near 100 kPa and passed it over solid-oxide electrolysis cells at 800 °C. There a current split carbon dioxide: $2CO_2 \rightarrow 2CO + O_2$. In 16 runs between April 2021 and August 2023, by day and night and in every season, it made 122 g of oxygen. It produced up to 12 g an hour, at 98 % purity or better.

NASA estimates that a crewed Mars ascent vehicle would need about 25 t of oxygen. Made over the 14 months before a crew arrives, that needs 2.4 kg an hour — 200 times MOXIE's best rate — and tens of kilowatts. Methane comes from the **Sabatier reaction**, $CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O$, over a nickel or ruthenium catalyst at 300–400 °C (used on the ISS from 2010), with hydrogen from

Martian ice or brought from Earth. A methane–oxygen engine burns about 3.6 kg of oxygen per kilogram of methane, so oxygen from the air alone supplies 78 % of the propellant — one reason SpaceX's Starship burns methane.



FIGURE 41.13 Engineers lower MOXIE into the body of the Perseverance rover at JPL before launch. The experiment made the first oxygen produced on another planet. Image: NASA/JPL (NASA/JPL-Caltech).

41.3.4 Building with Regolith

By mass, the largest use of lunar material may be the least glamorous. This means landing pads that stop rocket exhaust blasting regolith across a base, **berms** (walls of piled soil) against ejecta, and roads. It also means a few metres of soil over habitats, against radiation and a day–night temperature swing of nearly 300 K. In June 2025 NASA's RASSOR robot built a berm about 0.9 m high from simulant at Kennedy Space Center (Figure 41.15).

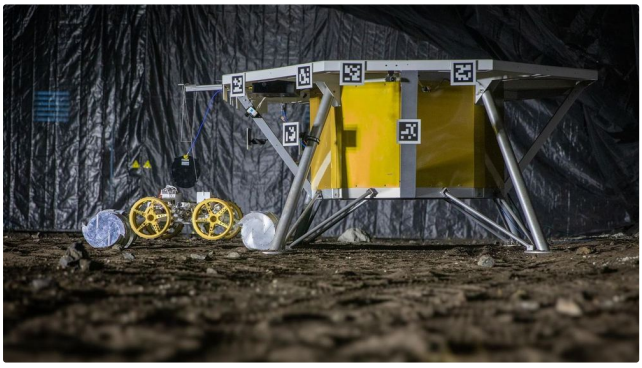


FIGURE 41.14 NASA's ISRU Pilot Excavator (IPEx), an excavator and dump truck in one, during a simulated lunar mission in a regolith testbed at Kennedy Space Center in August 2024. Image: NASA/KSC (NASA/Frank Micheaux).



FIGURE 41.15 RASSOR, with counter-rotating bucket drums at each end, building a berm of simulated lunar regolith at Kennedy Space Center in June 2025. Image: NASA/KSC (NASA/Frank Michaux).

Sintering fuses grains with focused sunlight, lasers or microwaves; **melting** makes glass and cast basalt; **binders** such as sulphur (melting at 115 °C) make concrete without water; and **3D printing** builds walls layer by layer. ICON, which printed NASA's simulated Mars habitat for the CHAPEA missions (Chapter 30), is developing laser vitrification of lunar soil under a US$ 57.2 million NASA contract running to 2028. China plans to test solar-concentrator brick-making on its Chang'e 8 lander around 2028.

41.3.5 Power: The Hidden Requirement

Every process is limited by power: 100 t of water a year needs about 100 kW continuously. Ridges near the south pole, such as the rim of Shackleton crater, are sunlit for 80–90 % of the year. But inside the cold traps, and through the 14-day lunar night elsewhere, nuclear power is the practical choice. NASA and the US Department of Energy renewed their Fission Surface Power partnership in January 2026. They aim for a 100 kW reactor ready to launch by about 2030 and to run for at least 10 years. Its waste heat could warm regolith.

41.4 Prospecting: Finding the Ore

Miners distinguish a **resource**, known to exist, from a **reserve**, which can be extracted at a profit with proven technology; no space resource is yet a reserve. Orbital data show where hydrogen and cold traps are, but not the form, depth and purity of the ice; these must be measured on the ground (Table 41.3).

TABLE 41.3 Lunar prospecting missions.

Mission	Agency	Year	Method	Result or status (Oct 2026)
Lunar Prospector	NASA	1998–99	Neutron spectrometer	Hydrogen excess at both poles
Chandrayaan-1	ISRO, NASA	2008–09	Infrared spectra, radar	Surface OH/H_2O; ice in some PSRs
LCROSS	NASA	2009	Impactor and shepherding probe	5.6 ± 2.9 % water in Cabeus
LRO; ShadowCam on Danuri	NASA; KARI	2009–; 2022–	Thermal, UV, neutron mapping; PSR imaging	Cold-trap maps; no extensive surface ice
Lunar Trailblazer	NASA	2025	Infrared and thermal mapping	Lost the day after launch
PRIME-1 (on IM-2)	NASA	2025	1 m drill, mass spectrometer	Lander tipped over; no drilling
VIPER	NASA	late 2027	Rover, 1 m drill, spectrometers	Revived 2025; on Blue Moon MK1
Chang'e 7	CNSA	2027	Orbiter, lander, rover, hopper	Missed 2026 launch window
PROSPECT (on IM-4)	ESA	2027	1 m drill and gas laboratory	In preparation
LUPEX (Chandrayaan-5)	ISRO, JAXA	≈ 2028	Lander and rover with drill	In development
Chang'e 8	CNSA	≈ 2028	ISRU and construction tests	In development

41.4.1 Bad Luck in 2025

Two missions launched on one Falcon 9 in February 2025 showed how hard the work is. NASA's US$ 94 million **Lunar Trailblazer** carried infrared and thermal mappers to map the form and abundance of polar water (Figure 41.16). Contact was lost the day after launch, probably because misaligned solar arrays let its batteries drain. The IM-2 lander *Athena* carried NASA's **PRIME-1**: the TRIDENT drill, to bring up regolith from 1 m depth in 10 cm steps, and the MSolo mass spectrometer to analyse the released gases (Figure 41.17). *Athena* landed on 6 March 2025 at 84.8° S near Mons Mouton but came to rest on its side in a crater. It operated for only about 12 hours instead of 10 days. TRIDENT never drilled, and MSolo mostly recorded gases vented by the lander itself.



FIGURE 41.16 NASA's Lunar Trailblazer in a clean room at Lockheed Martin in 2023, after the Lunar Thermal Mapper (black box, upper right) was installed. The orbiter was lost the day after its launch in February 2025. Image: NASA/JPL (Lockheed Martin Space).

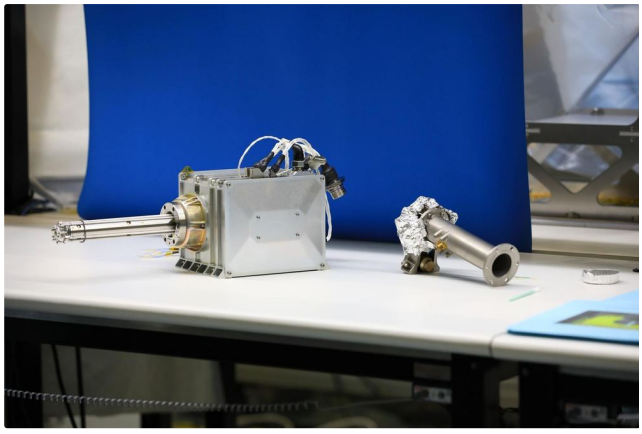


FIGURE 41.17 Parts of the MSolo mass spectrometer, a modified commercial instrument, at Kennedy Space Center in 2020. MSolo flew on the IM-2 lander as part of PRIME-1. Image: NASA/KSC (NASA/Glenn Benson).

41.4.2 Next Steps on the Ground

NASA's half-tonne **VIPER** rover carries a neutron spectrometer, a near-infrared spectrometer, MSolo and a TRIDENT drill to map polar ice (Chapter 28). Cancelled in July 2024 over cost growth, it was revived in September 2025 with a task order worth up to US$ 190 million. Under this order Blue Origin is to land it near Mons Mouton in late 2027 on the second Blue Moon MK1. That flight depends on the first MK1 flight, delayed to early 2027 after a New Glenn booster exploded during a static-fire test in May 2026.

China's **Chang'e 7** — orbiter, lander, rover and a small "hopper" designed to jump into a shadowed crater and analyse its water — missed its August 2026 launch window and is expected in 2027. ESA's **PROSPECT** drill and gas laboratory, first developed for Russia's Luna 27, will fly on Intuitive Machines' IM-4 in 2027. The ISRO–JAXA **LUPEX** (Chandrayaan-5) lander and rover is planned for around 2028, as is China's **Chang'e 8**, which will test resource use and construction.

The stakes rose in 2026, when NASA paused the Gateway station and announced a permanent base near the south pole, starting with frequent robotic landings from 2027. Whether the base can live partly off the land depends on what these drills find.

41.5 The Business of Space Mining

41.5.1 The First Wave

Asteroid mining became a business in April 2012, when **Planetary Resources** was unveiled with backing from Google's Larry Page and Eric Schmidt and the film director James Cameron. It planned to find water-rich NEAs with small telescopes and sell the water as propellant in space. **Deep Space Industries (DSI)** followed in 2013, and Luxembourg's space-resources initiative of 2016 invested in Planetary Resources and agreed to co-fund DSI. With no customers for propellant in space, investors lost patience: Planetary Resources, after two test satellites, was bought by the blockchain company ConsenSys in October 2018, and DSI by Bradford Space in January 2019.

41.5.2 The Second Wave

A second generation started with smaller, cheaper missions (Table 41.4).

- **AstroForge** (founded 2022) aims to mine platinum-group metals from metallic NEAs. Brokkr-1, launched in April 2023 to test refining asteroid-like material in orbit, could not be commanded reliably. Odin, a 100 kg, US$ 3.5 million spacecraft that rode with IM-2 in February 2025 to fly past the possibly metallic asteroid 2022 OB5, was declared lost in March. DeepSpace-2, designed to land on 2022 OB5, was completed in June 2026 and is to ride on IM-3 in 2027.
- **Karman+** (Denver) raised US$ 20 million in 2025 to rendezvous with an NEA and extract about 1 kg of material around 2027, aiming to sell water and regolith to servicing tugs.
- **TransAstra** proposes "optical mining", using concentrated sunlight to drive volatiles out of an asteroid enclosed in a bag, and reported testing a 1 m capture bag inside the ISS's Bishop airlock in October 2025.
- **Interlune** plans to harvest lunar helium-3 (see above).

TABLE 41.4 Space-mining companies.

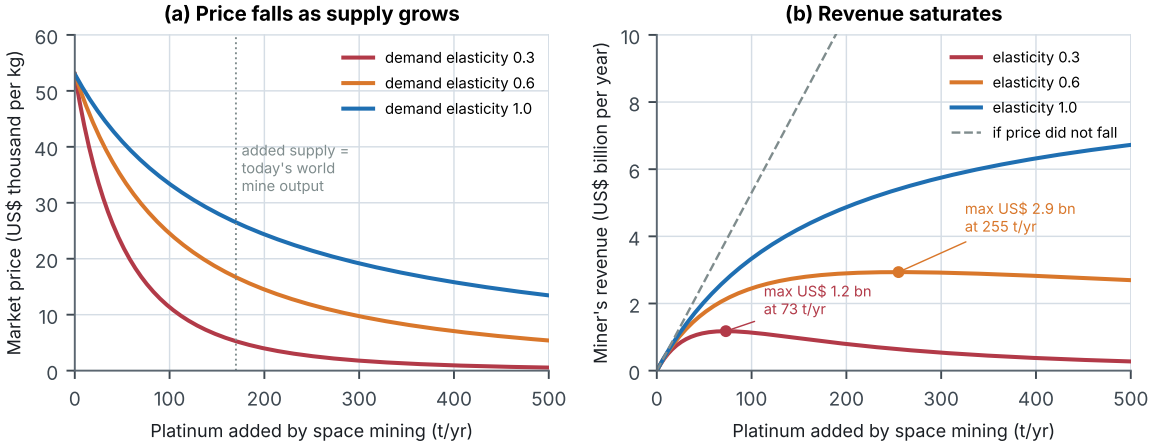
Company	Founded	Target	Missions flown	Outcome or status (Oct 2026)
Planetary Resources	2012 (public)	Water from NEAs	Arkyd-3R (2015), Arkyd-6 (2018)	Acquired by ConsenSys, 2018
Deep Space Industries	2013	Water from NEAs	None of its own	Acquired by Bradford Space, 2019
AstroForge	2022	Platinum-group metals	Brokkr-1 (2023), Odin (2025)	Both failed; DeepSpace-2 due 2027
Karman+	early 2020s	Water, regolith for use in space	None yet	Demonstration targeted 2027
TransAstra	2015	Volatiles by optical mining	Capture-bag test on ISS (2025)	Development
Interlune	2020	Lunar helium-3	None yet	Contracts for delivery from 2028–29

41.5.3 Who Would Buy?

A space resource can be sold in two markets. **In space**, the competitor is material launched from Earth: propellant in lunar orbit is worth what it costs to deliver from Earth, thousands of dollars per kilogram, but that market barely exists yet. **On Earth**, the competitor is terrestrial mining, and the product must be worth enough per kilogram to pay for its return.

Platinum is the usual example. World mine supply is about 170 t a year, and the price is volatile: in 2026 it reached a record of about US$ 2 900 per troy ounce in January, fell to about US$ 1 540 in July and stood near US$ 1 650 (about US$ 53 000 per kilogram) in mid-October. Figure 41.18 uses a simple constant-elasticity demand model, in which a 1 % increase in supply lowers the price by $1/\varepsilon$ %, where ε is the **price elasticity of demand**. If demand is inelastic ($\varepsilon = 0.3$), the miner's revenue peaks at only about US$ 1.2 billion a year, at 73 t a year; at $\varepsilon = 0.6$ the peak is US$ 2.9 billion at 255 t a year. A single 100 m metallic asteroid of bulk density 5 000 kg/m^3 with 20 ppm of

platinum-group metals would hold about 50 t of them. If it were all platinum and sold in one year, that would be enough to cut the price by between about a quarter and more than a half. Space mining for Earth markets caps its own revenue.



Illustrative constant-elasticity demand $P = P_0(1 + X/S_0)^{(-1/\varepsilon)}$ with P_0 = US$ 53 000/kg (≈ US$ 1 650 per troy ounce, mid-October 2026; the price ranged from about US$ 1 540 to 2 920 per ounce during 2026) and S_0 = 170 t/yr (world mine supply, WPIC 2025).

FIGURE 41.18 Why space metals cannot be valued at today's prices. (a) Platinum price as space-mined supply is added, for three demand elasticities. (b) The miner's annual revenue, which peaks when demand is inelastic.

The most plausible early business is therefore service to government exploration — data, delivery, demonstrations and, eventually, propellant sold in lunar orbit.

41.6 Law and Policy

41.6.1 Treaties

The **Outer Space Treaty** of 1967, ratified by more than 110 states, is the foundation of space law (Chapters 1 and 19). Article I guarantees the freedom to explore and use outer space. Article II declares that outer space, "including the Moon and other celestial bodies, is not subject to national appropriation by claim of sovereignty, by means of use or occupation, or by any other means". Article VI makes states responsible for authorising and supervising their companies. The treaty is silent on whether materials taken from a celestial body can be owned, though nobody objected when Apollo and Luna brought samples home.

The 1979 **Moon Agreement** declared the Moon and its natural resources "the common heritage of mankind" and called for an international regime to govern their exploitation once it became feasible. It entered into force in 1984 but was never joined by the United States, Russia or China. Since Saudi Arabia left in January 2024 it has 17 parties. Several of them, such as Australia and the Netherlands, have also signed the Artemis Accords.

41.6.2 National Laws and the Artemis Accords

Since 2015 several states have let their companies own extracted resources while disclaiming sovereignty over the body itself — like owning the fish but not the sea (Table 41.5). The United States led with the **Commercial Space Launch Competitiveness Act** of November 2015, followed by **Luxembourg** (2017), the **United Arab Emirates** (a 2019 space law and 2023 space-resources

regulations) and **Japan** (2021). Under Japan's law the government licensed ispace in 2022 and again in 2024 to transfer ownership of lunar regolith to NASA — the first licences issued under the law.

In a symbolic programme of December 2020, NASA agreed to buy small regolith samples, collected and photographed on the Moon and left in place. The sellers were Lunar Outpost for US$ 1, ispace Japan and ispace Europe for US$ 5 000 each and Masten Space Systems for US$ 15 000. By October 2026 no sale had been completed: ispace's landers crashed in April 2023 and June 2025, Masten went bankrupt in 2022, and Lunar Outpost's rover was aboard the toppled IM-2 lander.

TABLE 41.5 Law and policy on space resources.

Instrument	Year	Parties or scope	Position on resources
Outer Space Treaty	1967	> 110 states	No national appropriation (Art. II); silent on extracted resources
Moon Agreement	1979	17 states	"Common heritage of mankind"; future international regime
US Commercial Space Launch Competitiveness Act	2015	United States	Citizens may own and sell recovered resources
Luxembourg space-resources law	2017	Luxembourg	"Space resources are capable of being appropriated"
UAE space law and regulations	2019, 2023	United Arab Emirates	Licensing of resource activities
Japan Space Resources Act	2021	Japan	Licensed operators own what they extract
Artemis Accords	2020–	76 signatories (Sept 2026)	Extraction is not in itself appropriation; safety zones
COPUOS working group principles	2021–	UN Legal Subcommittee	Non-binding principles in draft (2026)

The **Artemis Accords**, launched by the United States in October 2020 with eight founding signatories, had 76 signatories by late September 2026. Section 10 affirms that "the extraction of space resources does not inherently constitute national appropriation under Article II", and Section 11 introduces **safety zones** around operations to avoid harmful interference. China and Russia, leaders of the rival International Lunar Research Station, have not signed.

41.6.3 The United Nations Working Group

In 2021 the Legal Subcommittee of the UN **Committee on the Peaceful Uses of Outer Space (COPUOS)** set up a working group. This Working Group on Legal Aspects of Space Resource Activities is chaired by the Australian law professor Steven Freeland. In September 2026 it circulated draft non-binding "Initial Recommended Principles": extraction in accordance with the Outer Space Treaty would not by itself be national appropriation, and benefit sharing would be voluntary. Open questions remain about how large and lasting a safety zone may be, and who may work in the few square kilometres of the best cold traps. Another is how to protect pristine ice — a record of water in the inner Solar System — from contamination.

AI FOR PROSPECTING AND MINING

Machine-learning classifiers map minerals and hydration from orbital spectra and fuse them with temperature and topography to rank candidate ice sites. A physics-based deep-learning denoiser has turned noisy LRO images of shadowed craters into boulder maps for planning rover routes. Networks also classify the spectra of tens of thousands of asteroids observed by Gaia. On the surface, delays and darkness demand autonomy. It is needed for perception under headlights, digging that adapts to unknown soil, and fleets of excavators and haulers. It is also needed for fault detection in plants that must run unattended for years (Chapters 46, 48 and 51).

KEY TAKEAWAYS

- The gear ratio drives ISRU. A kilogram on the lunar surface costs about 130 kg at Earth lift-off, and one in low lunar orbit about 73 kg. But lunar propellant delivered to low lunar orbit costs only about 2.1 kg of lunar production.
- A lunar plant pays only if it makes several times its own landed mass (3.8 times for delivery to low lunar orbit) before it fails.
- Oxygen is everywhere (41–45 % of lunar regolith; Mars's CO_2 atmosphere); water ice is trapped in polar cold traps below about 110 K, in uncertain and probably patchy amounts.
- Water extraction needs 3–7 MJ of heat per kilogram plus at least 15.9 MJ/kg for electrolysis; oxygen from regolith needs 18–30 MJ/kg plus heat. Power is the limiting resource.
- MOXIE made 122 g of oxygen on Mars, but a crewed ascent needs 200 times its rate. Prospecting suffered losses in 2025; VIPER, Chang'e 7 and PROSPECT are due in 2027.
- Selling space metals on Earth collapses their price; the first asteroid miners failed, and the second wave has yet to succeed in space.
- The Outer Space Treaty bans national appropriation but is silent on extracted resources; national laws and the 76-signatory Artemis Accords allow ownership.

REVIEW QUESTIONS

1. Explain the "gear ratio" of a destination. Using the rocket equation with I_{sp} = 450 s, show why a kilogram in low lunar orbit costs about 2.9 kg in low Earth orbit.
2. Why does lunar propellant cost more to deliver to low Earth orbit than to low lunar orbit?
3. A 15 t lunar plant makes 20 t of propellant a year. How long must it run to pass the break-even productivity of 3.8 for delivery to low lunar orbit?
4. Using Figure 41.5, explain why water ice survives only in permanently shadowed craters.
5. Estimate the energy to extract 1 kg of water from regolith containing 2 % ice, and compare it with the energy to electrolyse the water.
6. Compare hydrogen reduction of ilmenite with molten regolith electrolysis.
7. MOXIE produced up to 12 g of oxygen an hour. How long would one MOXIE take to make the 25 t needed for a crewed ascent, and how many would be needed to make it in 14 months?
8. How much regolith must be processed for 1 litre of helium-3 at 4 ppb? Why is helium-3 more likely to be sold for cryogenics than for fusion?
9. Using the model of Figure 41.18, explain why the revenue from selling asteroid platinum on Earth reaches a maximum. Why might water sold in lunar orbit be a better business?
10. How do the Outer Space Treaty, the Moon Agreement and the Artemis Accords treat extracted resources?

42 CHAPTER Advanced Propulsion



Image: NASA/GRC

Chemical rockets have reached the limit set by the energy stored in their propellants, so travelling faster or farther needs another source of energy or a way to avoid carrying propellant at all. This chapter shows, with worked numbers, why exhaust velocity and power decide every advanced concept. It covers high-power Hall and ion thrusters and the power-limited trajectories they fly; nuclear thermal rockets from NERVA to the cancelled DRACO; space fission reactors and NASA's SR-1 Freedom nuclear-electric mission planned for 2028; solar, electric and laser sails, including Breakthrough Starshot; fusion and antimatter; scramjets and rotating detonation engines; and tethers. It ends by explaining why "reactionless" drives and warp drives fail the test of physics.

IN THIS CHAPTER

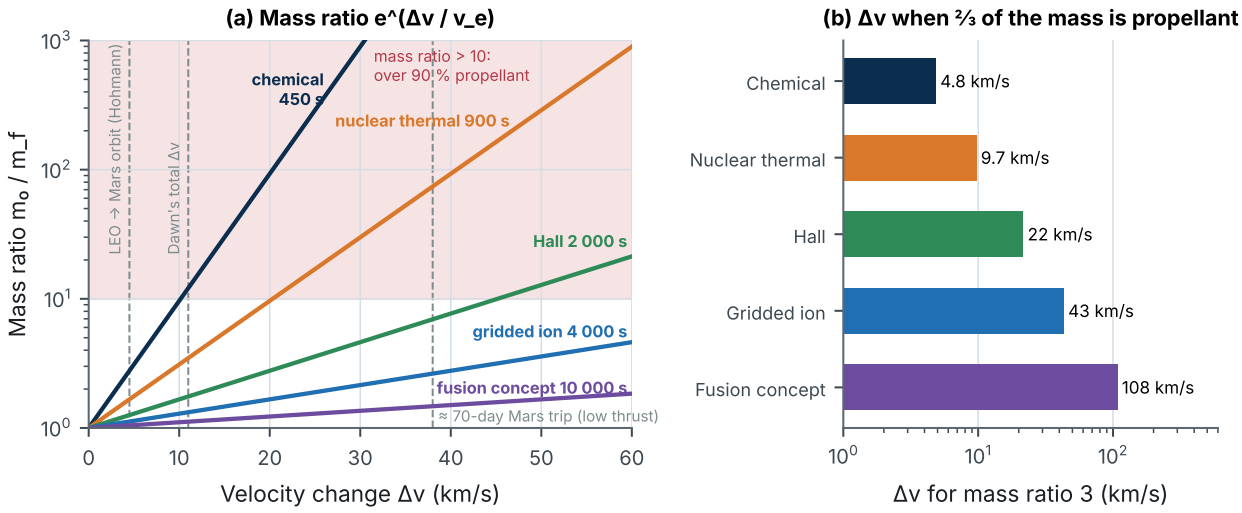
42.1 The Tyranny of the Rocket Equation
42.2 Advanced Electric Propulsion
42.3 Nuclear Thermal Propulsion
42.4 Nuclear Electric Propulsion and Space Reactors
42.5 Sailing on Light
42.6 Beamed Energy and the Interstellar Challenge
42.7 Breathing Air and Detonating Fuel
42.8 Propellantless and Exotic Ideas

Every spacecraft that has left Earth has done so on chemical rockets, whose energy is fixed by chemistry. The best practical combination, hydrogen and oxygen, gives an exhaust velocity of about 4.4 km/s (Chapter 35). The rocket equation turns that limit into an exponential penalty for every extra kilometre per second of velocity change. A crewed round trip to Mars, a fast mission to the outer planets or a probe to another star needs one of two things. One is a much faster exhaust; the other is a way of pushing a spacecraft without carrying its reaction mass along.

Advanced propulsion splits into two families. The first raises the exhaust velocity with energy from somewhere other than the propellant — a nuclear reactor, solar arrays, fusion or antimatter. The second dispenses with propellant and pushes on sunlight, laser beams, the solar wind, a planet's magnetic field or the atmosphere. Some fly by the thousand, others have never left the laboratory, and a few contradict the laws of physics. This chapter sorts them by exhaust velocity, thrust, power per kilogram and maturity.

42.1 The Tyranny of the Rocket Equation

The **rocket equation**, $\Delta v = v_e \ln(m_0/m_f)$, says that the velocity change Δv depends on the exhaust velocity v_e and on the **mass ratio**. This is the starting mass m_0 divided by the mass m_f left when the propellant is gone. Because the mass ratio is $e^{\Delta v/v_e}$, it grows exponentially once Δv exceeds the exhaust velocity (Figure 42.1). A hydrogen–oxygen stage needs a mass ratio of 2.75 to go from low Earth orbit (LEO) into Mars orbit on the cheapest trajectory (4.5 km/s). But it would need 12 for the 11 km/s that the ion-propelled Dawn accumulated (Chapter 24). Doubling the exhaust velocity takes the square root of the mass ratio; multiplying it by ten turns an impossible mission into a routine one.



Ideal rocket equation, v_e = g₀ I_sp. Reference Δv values: Hohmann transfer from 400 km LEO to a 1-sol Mars orbit (this chapter's model); Dawn ≈ 11 km/s; 70-day trip from the power-limited model in this chapter.

FIGURE 42.1 The rocket equation for five exhaust velocities. (a) Mass ratio against velocity change. (b) The velocity change each technology gives when two-thirds of the vehicle is propellant.

42.1.1 Energy, Power and Thrust

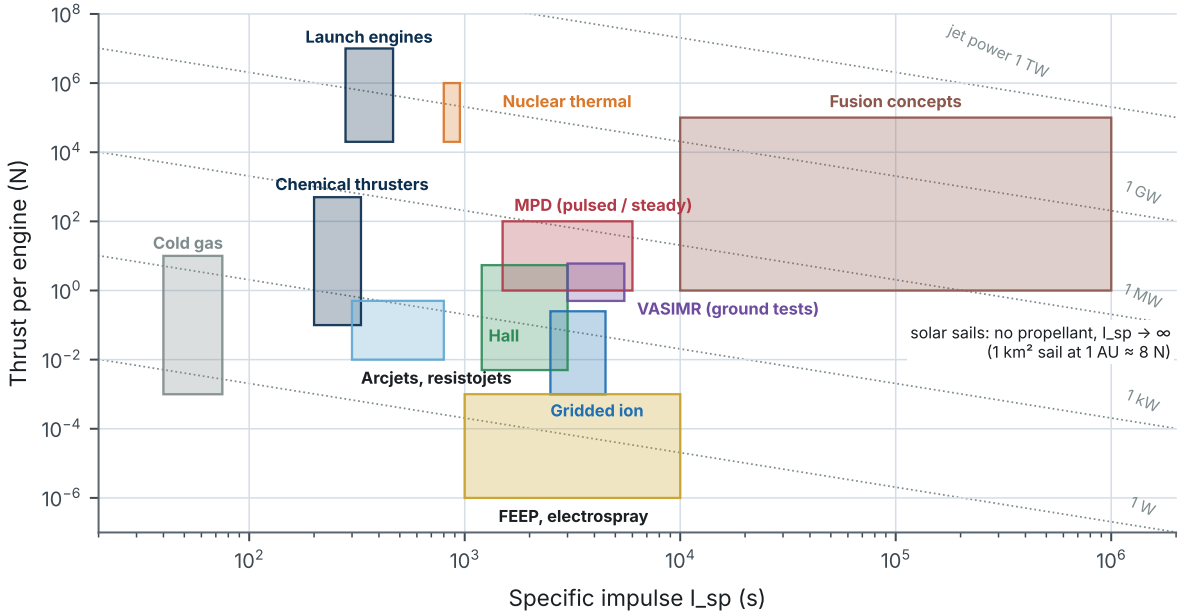
A faster exhaust has a price: the power it carries away, the **jet power**, is

$$P_{jet} = \tfrac{1}{2}\dot{m}v_e^2 = \tfrac{1}{2}Fv_e,$$

so, for a given thrust F, doubling the exhaust velocity doubles the power needed. A chemical rocket carries its power in its propellant — each F-1 engine of the Saturn V had a jet power near 9 GW. But an electric thruster must be fed by solar arrays or a reactor, and that power plant has mass. A thruster that turns a fraction η of its electrical input P into jet power gives

$$F = \frac{2\eta P}{v_e} = \frac{2\eta P}{g_0 I_{sp}},$$

so a 12 kW thruster at 60 % efficiency and 2 600 s produces only about 0.56 N (Figure 42.2).



Indicative ranges of flown or ground-tested devices (fusion: design studies only).
Dotted lines: jet power P = F g_0 I_sp / 2 — high thrust at high I_sp needs enormous power.

FIGURE 42.2 Thrust and specific impulse of propulsion technologies. Dotted lines join points of equal jet power: more thrust at higher specific impulse demands power that only large reactors, fusion or beamed energy could supply.

Advanced propulsion is therefore a contest of energy sources and their **specific power** α, the power per kilogram of power and propulsion system. For a mission that thrusts for a time t, the best exhaust velocity of a power-limited rocket is of the order of the **characteristic velocity** $v_{ch} = \sqrt{2\eta\alpha t}$. A faster exhaust would need a heavier power plant than the propellant it saves. With α = 50 W/kg, η = 0.6 and one year of thrusting, $v_{ch} \approx 43$ km/s (Isp ≈ 4 400 s). With α = 10 W/kg it falls to 19 km/s (Isp ≈ 2 000 s), the territory of Hall thrusters. Table 42.1 summarises the options. Technology readiness levels (TRLs, Chapter 4) run from 1 (principle observed) to 9 (proven in operation).

TABLE 42.1 Advanced propulsion at a glance (October 2026). TRLs are approximate, judged from flight and test history.

Technology	Isp (s)	Thrust	Energy source	TRL (approx.)	Examples
Chemical (reference)	300–465	N to MN	Propellant	9	RL10, Raptor
Nuclear thermal	850–950	10–1 000 kN	Fission reactor	5–6	NERVA XE-Prime, RD-0410
Hall thruster	1 200–3 000	mN to 5 N	Electric	9 (≤ 5 kW), 6–7 (12 kW)	Starlink, Psyche, AEPS
Gridded ion	2 500–4 500	1–250 mN	Electric	9	Dawn, BepiColombo, NEXT-C
MPD, VASIMR	1 500–6 000	1–100 N	Electric, 0.1–10 MW	4–5	VX-200, laboratory MPDs
FEEP, electrospray	1 000–10 000	µN to mN	Electric, watts	9	LISA Pathfinder, Enpulsion
Solar sail	no propellant	≈ 8 µN/m² at 1 AU	Sunlight	6–7	IKAROS, LightSail 2, ACS3

Technology	Isp (s)	Thrust	Energy source	TRL (approx.)	Examples
Electric or magnetic sail	no propellant	μN–mN (studies)	Solar wind	2–3	ESTCube-1 (not deployed)
Laser sail	no propellant	≈ 670 N per 100 GW	Ground laser	2	Breakthrough Starshot
Fusion	10^4–10^6	N–kN (studies)	Fusion reactions	1–3	Daedalus, Direct Fusion Drive
Electrodynamic tether	no propellant	0.1–1 N	Orbital energy, solar power	5–6	PMG, TSS-1R
Scramjet	1 000–4 000	kN	Fuel and air	6	X-43A, X-51A

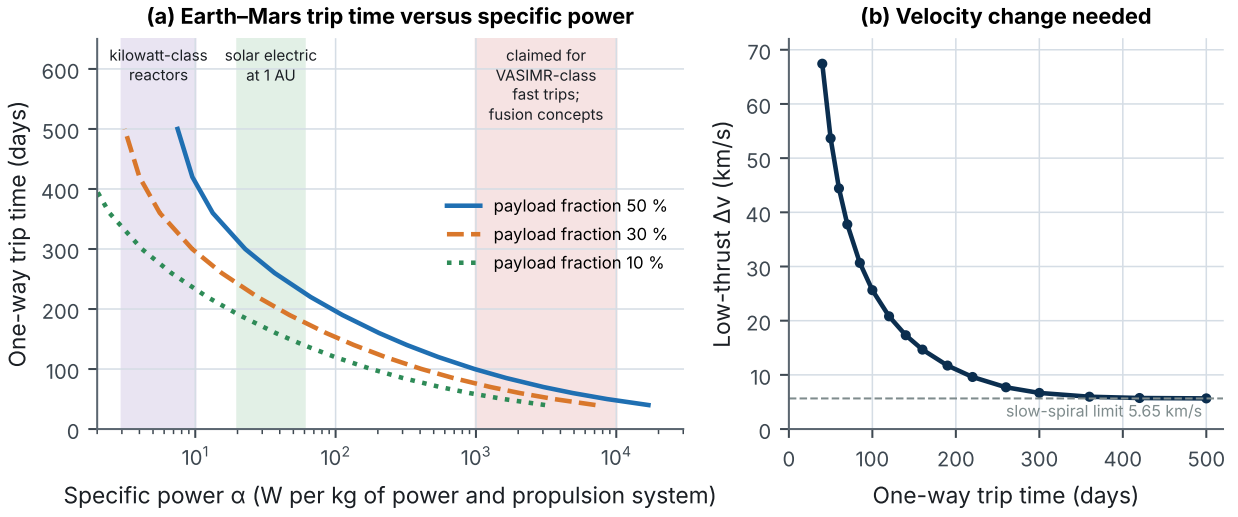
42.2 Advanced Electric Propulsion

Electric propulsion is already mass-produced — thousands of Hall thrusters fly on Starlink satellites, and ion and Hall engines have carried Dawn, BepiColombo and Psyche (Chapters 6 and 24). The frontier is power: tens to hundreds of kilowatts per spacecraft.

42.2.1 Power-Limited Trajectories

A power-limited spacecraft thrusts for months (Figure 42.3), and the ideal strategy is to vary the specific impulse along the way: more thrust near departure and arrival, higher Isp in between. For such a variable-Isp rocket the propellant consumed depends only on the integral $J = \int a^2\,dt$ of the squared thrust acceleration. With the power plant sized optimally, the fraction of the initial mass left for payload and structure is

$$\frac{m_{payload}}{m_0} = \left(1 - \sqrt{\frac{J}{2\eta\alpha}}\right)^2 .$$



Optimal variable-I_sp power-limited rendezvous, Earth escape to Mars capture (coplanar circular orbits, planetary spirals excluded); J = ∫a²dt by direct optimisation; payload fraction = (1 − √(J/2ηα))², η = 0.6. Shaded bands are indicative.

FIGURE 42.3 Power-limited Earth–Mars transfers computed for this chapter. (a) Specific power needed for a given trip time when 50 %, 30 % or 10 % of the departing mass arrives as payload and structure. (b) The low-thrust velocity change. Escape and capture spirals at the planets are excluded.

J rises roughly as the inverse cube of trip time. The figure applies this to a one-way trip from Earth escape to Mars capture, optimised numerically for each trip time. A slow spiral needs 5.7 km/s; a 100-day crossing needs 26 km/s and a 40-day crossing 67 km/s. If half the departing mass is to arrive as payload and structure, a 300-day trip needs $\alpha \approx 23$ W/kg and a 190-day trip about 110 W/kg. A 100-day trip needs about 1 kW/kg and a 40-day trip about 17 kW/kg. Solar-electric systems deliver of the order of tens of watts per kilogram at 1 AU; crewed crossings in a month or two would need power systems hundreds of times lighter.

Solar arrays are light and proven, but their output falls with the square of distance from the Sun — at Jupiter to 1/27 of its value near Earth. A reactor gives the same power anywhere, but small reactors manage only a few watts per kilogram and large ones are dominated by their radiators. Solar-electric propulsion suits the inner Solar System; nuclear-electric propulsion the outer planets and heavy cargo.

42.2.2 High-Power Hall Thrusters

Hall thrusters (Chapter 6) used to wear out as ions eroded their ceramic channel walls. **Magnetic shielding**, developed at JPL in the 2010s, shapes the field to keep energetic ions off the walls, extending design lifetimes to tens of thousands of hours. NASA Glenn and Aerojet Rocketdyne (now L3Harris) used it in the 12 kW **Advanced Electric Propulsion System (AEPS)**, which gives about 0.6 N at about 2 600 s. Three AEPS flight thrusters passed acceptance tests in 2025 (Figure 42.4) and were delivered for the **Power and Propulsion Element (PPE)** built for NASA's Gateway station. When NASA paused Gateway in March 2026, it reassigned the PPE as the bus of the nuclear-electric SR-1 Freedom mission described below; AEPS had by then completed environmental qualification, with a life test in preparation. In the laboratory, the three-channel X3 reached 102 kW in 2017 (Table 42.2).

42.2.3 Gridded Ion Engines

Among gridded ion engines, NASA's 6.9 kW **NEXT-C** reaches an Isp of about 4 190 s with 0.24 N of thrust, and a laboratory NEXT engine ran for 51 184 hours between 2005 and 2014. NEXT-C flew on the DART asteroid-deflection mission as a technology demonstration that the mission did not depend on. When it was first fired, in December 2021, it ran for about two hours. But an anomaly in the spacecraft's power electronics led the team to suspend further use (Chapter 26). The faint blue glow of the ion beam is the technology's signature (Figure 42.5).

42.2.4 Magnetoplasmadynamic Thrusters and VASIMR

Above about 100 kW, electromagnetic devices take over. A **magnetoplasmadynamic (MPD)** thruster drives thousands of amperes through a plasma between a central cathode and a surrounding anode, and the magnetic field pushes the plasma out by the Lorentz force $\vec{j} \times \vec{B}$. MPD thrusters handle megawatts in a compact device at 1 500–6 000 s. But their electrodes erode, and they are efficient only at very high power. A small pulsed version was carried by Japan's Space Flyer Unit in 1995.

The **Variable Specific Impulse Magnetoplasma Rocket (VASIMR)** of Ad Astra Rocket Company has no electrodes. Radio waves ionise argon, a second antenna heats the ions at their cyclotron frequency, and a magnetic nozzle turns their gyration into exhaust. The VX-200 prototype

reached 200 kW in 2009 and was measured at 72 % efficiency, about 4 900 s and 5.7 N; in July 2021 the VX-200SS ran for 88 hours at 80 kW. VASIMR has not flown; the fast crewed missions linked to it would need tens of megawatts at specific powers far beyond today's.



FIGURE 42.4 Engineers at NASA Glenn with the third 12 kW AEPS flight thruster during its acceptance-test campaign in 2025. Image: NASA/GRC (NASA/GRC/Jef Janis).

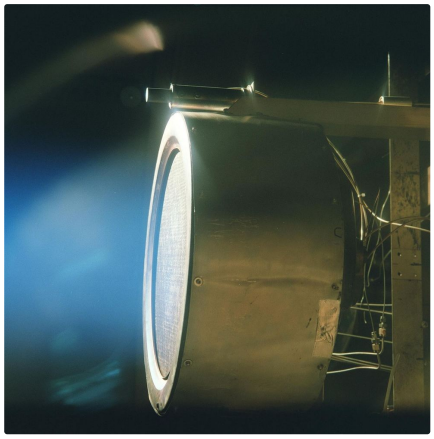


FIGURE 42.5 A prototype of the xenon ion engine flown on Deep Space 1, seen through a port of a JPL vacuum chamber during a long-duration test; the faint blue glow is the ion beam. Image: NASA/JPL (NASA/JPL).

42.2.5 Field-Emission, Electrospray and Air-Breathing Thrusters

At the other extreme, **field-emission electric propulsion (FEEP)** pulls ions out of a liquid metal such as indium with about 10 kV on a sharp emitter, and **electrospray** thrusters do the same with ionic liquids. They give micronewtons to about a millinewton at 2 000–6 000 s or more. Colloid thrusters provided drag-free control for NASA's experiment on ESA's LISA Pathfinder (2015–17). Enpulsion had launched more than 200 indium FEEP systems by 2024.

For very low orbits (Chapter 14), **air-breathing electric propulsion** would collect the thin air as propellant. In 2018 an ESA-led team fired a Sitael thruster on a nitrogen–oxygen mixture representing the atmosphere at about 200 km, the first such test.

TABLE 42.2 Electric thrusters at the frontier of power, efficiency and precision.

Thruster	Type	Power	Isp (s)	Thrust	Status (Oct 2026)
AEPS (NASA, L3Harris)	Magnetically shielded Hall	12 kW	≈ 2 600	≈ 0.6 N	Three flight units delivered 2025
X3 (Univ. of Michigan)	Hall, three channels	102 kW	1 800–2 650	5.4 N	Laboratory record, 2017
NEXT-C (NASA, Aerojet)	Gridded ion	6.9 kW	≈ 4 190	0.24 N	Flown on DART; ≈ 2 h in 2021
VX-200 / VX-200SS (Ad Astra)	Helicon and ion-cyclotron heating	200 kW / 80 kW	≈ 4 900	≈ 5.7 N	Laboratory; 88-hour run, 2021
Applied-field MPD	Electromagnetic	0.1–1 MW	1 500–6 000	1–100 N	Laboratory; small pulsed units flown
Indium FEEP (Enpulsion)	Field emission	tens of W	2 000–6 000	≤ ≈ 1 mN	More than 200 launched

42.3 Nuclear Thermal Propulsion

A **nuclear thermal rocket (NTR)** replaces the combustion chamber with a fission reactor (Figure 42.6). Liquid hydrogen cools the nozzle walls, then flows through hundreds of channels in the hot fuel and leaves at 2 500–2 800 K. The reactor adds energy without adding mass, so the propellant can be the lightest gas of all. Exhaust velocity scales as $\sqrt{T/\mathcal{M}}$, where $\mathcal{M}$ is the exhaust's molar mass; hydrogen at 2 g/mol gives about twice the exhaust velocity of a hydrogen–oxygen engine, whose exhaust is mostly steam. The result is 850–950 s at thrusts of tens to hundreds of kilonewtons.

42.3.1 Reactor Physics in Brief

A uranium-235 nucleus that absorbs a neutron splits, releasing about 200 MeV and two or three neutrons; when on average exactly one of these causes another fission, the reactor is **critical**. Slow ("thermal") neutrons cause fission far more readily than fast ones, so most rocket reactors include a **moderator** — graphite, zirconium hydride or beryllium — that slows them by collisions. **Control drums** around the core, cylinders with a neutron absorber such as boron carbide on one face, rotate to absorb more or fewer neutrons. A **shield** between reactor and tank stops neutrons and gamma rays; elsewhere the spacecraft relies on distance.

The fuel must survive hot hydrogen, which attacks graphite: Rover/NERVA used uranium carbide particles in graphite coated with niobium or zirconium carbide, and today's programmes study carbides and "cermets" of uranium dioxide in tungsten. The 1960s reactors used highly enriched uranium (over 90 % U-235). Current designs use **high-assay low-enriched uranium (HALEU)**, just under 20 %. It is unusable for weapons without further enrichment and far cheaper to secure, at the cost of a larger core.

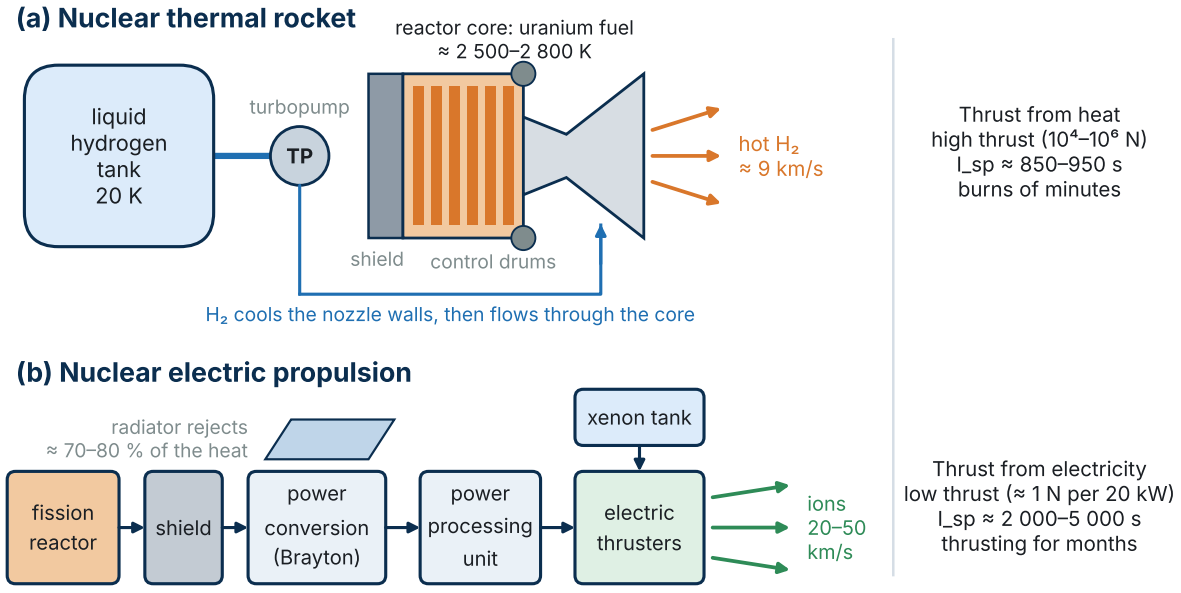


FIGURE 42.6 Principles of nuclear propulsion. (a) In a nuclear thermal rocket, hydrogen cools the nozzle and is then heated in the reactor core. (b) In nuclear electric propulsion, reactor heat is partly converted into electricity for ion or Hall thrusters; radiators reject the rest.

42.3.2 Rover and NERVA

The United States tested about twenty nuclear rocket reactors between 1959 and 1972 in **Project Rover** and the **NERVA** (Nuclear Engine for Rocket Vehicle Application) programme at Jackass Flats, Nevada (Table 42.3). Phoebus-2A ran at 4 082 MW in 1968, still the most powerful nuclear rocket reactor ever operated, and the small Pewee delivered hydrogen at 2 550 K. The XE-Prime engine of 1969, fired downwards in a test stand, was started 28 times and produced about 247 kN. The programme, which had cost about US$ 1.4 billion, was cancelled in January 1973 (Figure 42.7); the Soviet RD-0410 was ground-tested at Semipalatinsk but never flew. Those tests vented exhaust carrying fission products into the open air; a new ground test must capture and filter it, one reason modern programmes are expensive.

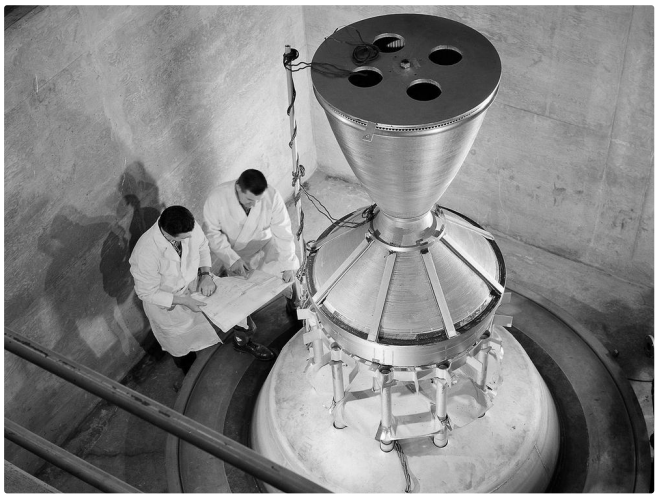


FIGURE 42.7 Technicians at NASA's Lewis (now Glenn) Research Center manufacture a nozzle for the Kiwi B-1-B nuclear rocket engine in a vacuum oven in 1964. Image: NASA/GRC.

TABLE 42.3 Milestones of nuclear thermal propulsion.

Reactor or engine	Year	Reactor power	Milestone
Kiwi-B4E	1964	937 MW	Full-power run and restart
Phoebus-2A	1968	4 082 MW	Most powerful nuclear rocket reactor
Pewee	1968	503 MW	Exit gas 2 550 K, Isp ≈ 850–900 s
NERVA XE-Prime	1969	≈ 1 140 MW	247 kN, 28 starts, flight-like engine
RD-0410 (USSR)	1970s–80s	196 MW	35 kN, 910 s, ground tests only
DRACO (DARPA, NASA)	2023–25	—	HALEU flight demonstration; cancelled 2025

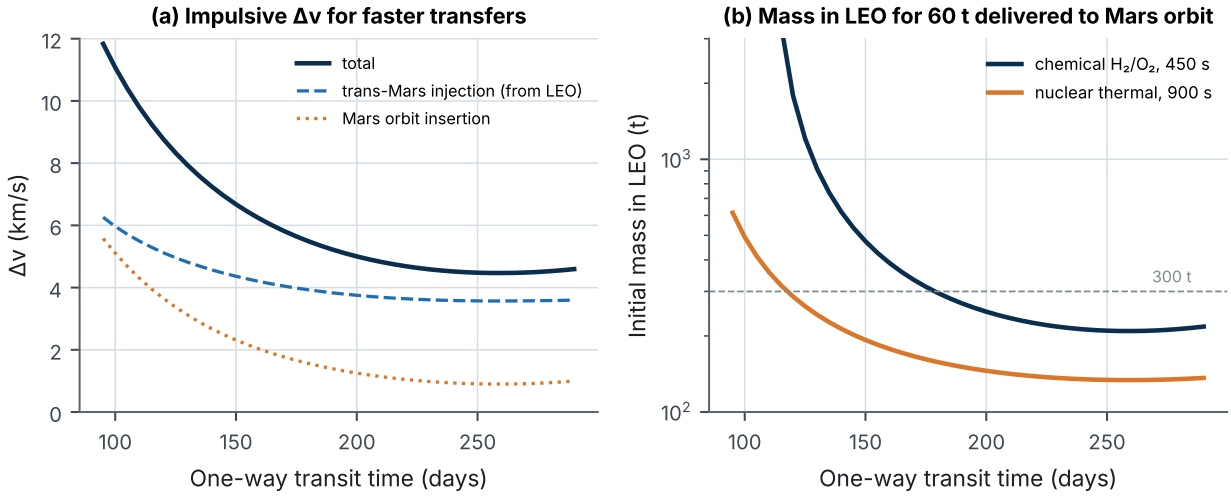
42.3.3 DRACO and After

In 2023 NASA joined DARPA's **DRACO** (Demonstration Rocket for Agile Cislunar Operations), which chose Lockheed Martin to build the spacecraft and BWX Technologies the HALEU reactor, aiming for an in-orbit test as early as 2027. NASA's budget request for 2026 then provided nothing for nuclear thermal or nuclear electric propulsion. In mid-2025 DARPA closed DRACO, citing the experiment's cost, the difficulty of ground testing and the falling cost of conventional launch. Congress nevertheless set aside at least US$ 110 million for nuclear thermal propulsion in NASA's

2026 appropriation, which the administrator told the Senate in April 2026 NASA would spend. Russian and Chinese work, mostly on nuclear-electric systems, is hard to judge from public sources.

42.3.4 Faster to Mars

The prize for crewed missions is time. Figure 42.8 compares single-stage chemical and nuclear thermal vehicles delivering 60 t into a high Mars orbit. Faster transfers need more Δv at both ends: 4.5 km/s for the 259-day minimum-energy transfer, 5.5 km/s for 180 days and 8.8 km/s for 120 days. With 300 t in LEO, the chemical vehicle can make the crossing in about 180 days and the nuclear one in about 120. Shorter transits matter because interplanetary crews receive about 1.8 mSv of cosmic radiation per day (Chapter 30).



Coplanar circular orbits, optimal phasing; 400 km LEO to 250 × 33 793 km Mars orbit. Stage inert mass: chemical 12 % of propellant; nuclear 20 % of propellant + 10 t of engines and shield. No aerocapture; one-way.

FIGURE 42.8 Faster Mars transfers computed for this chapter. (a) Impulsive Δv from a 400 km Earth orbit to a 1-sol (250 × 33 793 km) Mars orbit against transit time. (b) Initial mass in low Earth orbit for chemical and nuclear thermal stages delivering 60 t.

EXAMPLE: SIX MONTHS TO MARS — CHEMICAL OR NUCLEAR?

A transfer vehicle must deliver 60 t (habitat and lander) from a 400 km LEO into a 1-sol Mars orbit in 180 days. The trajectory model gives a trans-Mars injection of 3.93 km/s and an orbit insertion of 1.56 km/s: Δv = 5.49 km/s.

1. **Chemical (Isp 450 s, v_e = 4.41 km/s).** Mass ratio $R = e^{5.49/4.41}$ = 3.47. If tanks, engines and structure weigh 12 % of the propellant, the final mass is $m_f = 60/[1 - 0.12(R - 1)]$ = 85.3 t and the initial mass $m_0 = R\,m_f$ = **296 t**, of which 211 t is propellant.
2. **Nuclear thermal (Isp 900 s, v_e = 8.83 km/s).** $R = e^{5.49/8.83}$ = 1.86. Bulky hydrogen tanks weigh 20 % of the propellant, plus 10 t of engines and shielding: $m_f = (60 + 10)/(1 - 0.20{\times}0.86)$ = 84.6 t and m_0 = **158 t**, with 73 t of propellant.
3. The nuclear vehicle needs about half the mass in LEO. Spending the same 300 t on it instead shortens the trip to about **118 days**, cutting the crew's dose in transit from about 0.32 Sv to 0.21 Sv.
4. The price is hydrogen's low density (71 kg/m³): 73 t fills about 1 000 m³, to be kept at 20 K for months (Chapter 40 discusses boil-off).

42.4 Nuclear Electric Propulsion and Space Reactors

Nuclear electric propulsion (NEP) separates energy from reaction mass completely: reactor heat is partly converted into electricity for ion or Hall thrusters (Figure 42.6 b). Its weak point is waste heat. A closed Brayton converter, in which a helium–xenon gas loop drives a turbine and alternator, is about 20–30 % efficient, so a 1 MWe system at 25 % must reject about 3 MW. A radiator at 500 K with an emissivity of 0.9 sheds only about 3.2 kW per square metre from each face, so it would need some 470 m^2, nearly two tennis courts.

42.4.1 Reactors in Space So Far

The United States has flown one reactor: **SNAP-10A**, launched in April 1965, which produced about 500 W of electricity for 43 days before an electrical failure shut it down. The Soviet Union flew 33: 31 **BES-5 Buk** reactors of about 3 kWe powered radar ocean-reconnaissance satellites (RORSATs) between 1970 and 1988, and two **TOPAZ** thermionic reactors of about 5 kWe flew in 1987 (Table 42.4). RORSATs boosted their reactors into storage orbits about 900 km high after use, but in January 1978 Kosmos 954 failed to and scattered radioactive debris across northern Canada. The accident shaped the United Nations principles on nuclear power sources in space (1992), including the rule that a reactor must not go critical before reaching its operating orbit or interplanetary trajectory.

42.4.2 Kilopower and Fission Surface Power

NASA's modern work began with **Kilopower**, a family of 1–10 kWe concepts with a solid cast uranium core, passive sodium heat pipes and Stirling engines. The **KRUSTY** prototype (Kilopower Reactor Using Stirling Technology) was tested in Nevada from November 2017 to March 2018. The tests ended with a 28-hour full-power run that simulated start-up, failed heat pipes and failed Stirling engines. This was the first test of a new US space-reactor design in decades. In 2022 NASA and the Department of Energy funded three industry designs for a 40 kWe **Fission Surface Power** system for the Moon (Figure 42.9).

Policy then accelerated. A NASA directive in August 2025 called for a lunar reactor of at least 100 kWe ready for launch around 2030. In April 2026 a White House memorandum (NSTM-3) launched a National Initiative for American Space Nuclear Power. It called for reactors in orbit as early as 2028 and on the Moon by 2030. The first lunar unit, **Lunar Reactor-1**, is specified in an August 2026 draft solicitation at no less than 20 kWe — a smaller first step than the 2025 goal. A landing is intended in 2030 to power a lunar base through the 14-day night (Chapter 41).

42.4.3 SR-1 Freedom

The most ambitious near-term project is **Space Reactor-1 (SR-1) Freedom**, announced by NASA Administrator Jared Isaacman in March 2026 as the first spacecraft to use a fission reactor for propulsion beyond Earth orbit. A reactor designed by Idaho National Laboratory, fuelled with HALEU uranium dioxide and cooled by heat pipes, will produce at least 20 kWe through a closed Brayton converter supplied by Sierra Lobo. It sits behind a boron-carbide shield at the end of a boom with radiator fins, and will be switched on only after leaving Earth orbit. It will then drive the xenon Hall thrusters of the repurposed Gateway PPE through an electric-propulsion system of up to 48 kW. The PPE was built by Lanteris Space Systems, part of Intuitive Machines since

January 2026. SR-1 is due to launch in December 2028 and reach Mars about a year later with three Ingenuity-derived helicopters (the SkyFall mission) to scout landing sites. Its reported cost is about US$ 2.1 billion without them. The schedule is tight — the reactor was still being designed in October 2026 — but SR-1 would be the first US reactor launched in more than 60 years. Russia's "Zeus" nuclear-electric tug, once proposed for a first flight around 2030, has shown few public signs of hardware progress since 2023.

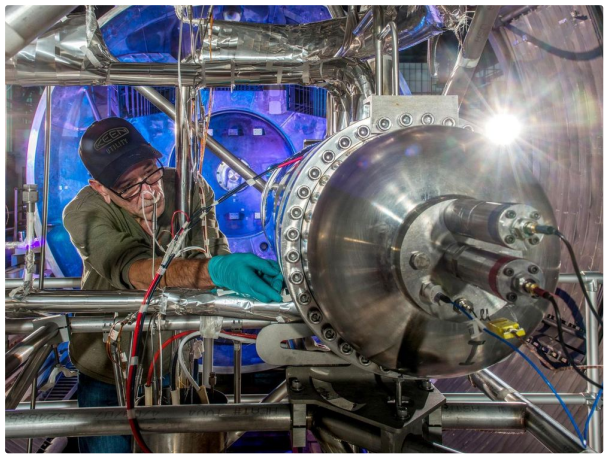


FIGURE 42.9 A 10 kW Stirling power-conversion unit for NASA's Fission Surface Power Technology Demonstration Unit at Glenn Research Center; four such units would turn 180 kW of reactor heat into 40 kW of electricity. Image: NASA/GRC.

TABLE 42.4 Space fission power systems, flown and planned.

System	Country	Years	Electrical power	Conversion	Status
SNAP-10A	USA	1965	≈ 0.5 kW	Thermoelectric	Flown; 43 days
BES-5 Buk	USSR	1970–88	≈ 3 kW	Thermoelectric	31 flown on RORSATs
TOPAZ	USSR	1987	≈ 5 kW	Thermionic	2 flown
KRUSTY (Kilopower)	USA	2017–18	1 kW class	Stirling	Ground test; 28 h at full power
SR-1 Freedom	USA	2028 (planned)	≥ 20 kW	Closed Brayton	In development; drives Hall thrusters
Lunar Reactor-1	USA	2030 (planned)	≥ 20 kW	—	Draft solicitation, August 2026

42.5 Sailing on Light

42.5.1 Radiation Pressure

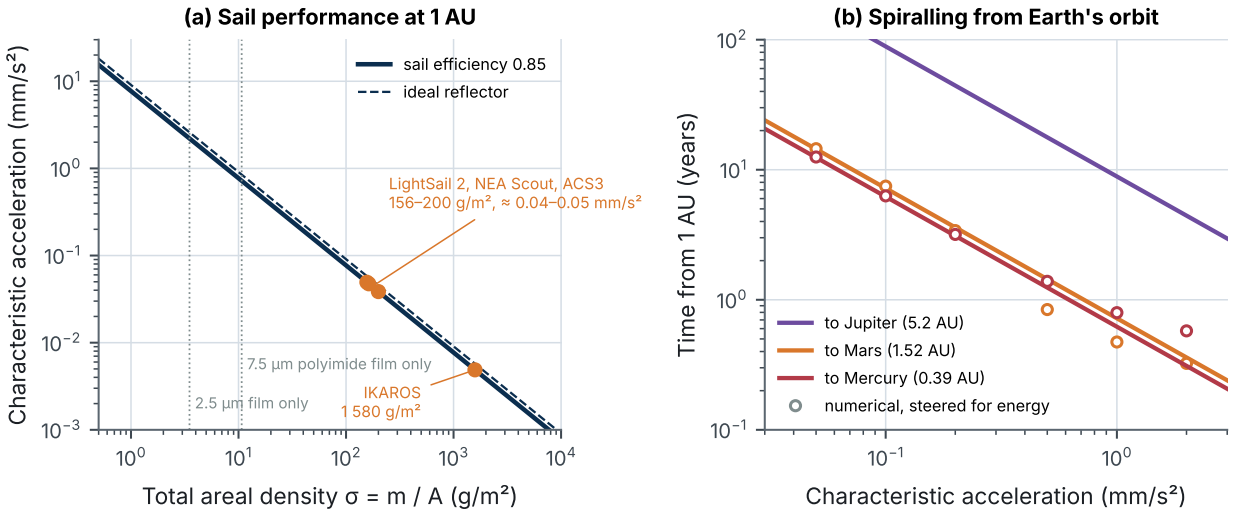
Light carries momentum: a beam of power P pushes on an absorbing surface with a force P/c, and on a perfect mirror facing it with $2P/c$. At 1 AU sunlight delivers 1 361 W/m², a pressure of 4.54 μN/m² on an absorber and 9.08 μN/m² on a perfect mirror. A flat sail of area A, tilted at a **cone angle** α to the sunlight and with an overall efficiency η (about 0.85, allowing for absorption and wrinkles), feels

$$F = \frac{2\eta P_{\odot} A}{c}\cos^2\alpha.$$

Dividing by mass gives the **characteristic acceleration** a_c at normal incidence, which depends only on the **areal density** σ = m/A of the whole spacecraft (Figure 42.10 a, Table 42.5). Sail acceleration and solar gravity both fall with the square of distance, so their ratio — the **lightness number** β — is the same everywhere. Tilting the sail by about 35° gives the largest push along the orbit, raising it in an outward spiral; tilting it the other way spirals inwards. For a slow spiral from radius r_0 to r the time is about

$$T \approx \frac{\sqrt{\mu_\odot / r_0}\;\left|(r/r_0)^{3/2} - 1\right|}{3 \times 0.385\, a_c},$$

where 0.385 is $\cos^2\alpha \sin\alpha$ at the optimum angle (Figure 42.10 b).



a_c = 2ηP/(cσ), P/c = 4.54 µN/m² at 1 AU; σ uses total spacecraft mass. (b) Lines: quasi-circular spiral, cone angle 35.3°, T = √(μ/r₀) |(r/r₀)^1.5 − 1| / (3 × 0.385 a_c). Dots: numerical flat-sail trajectories steered to change orbital energy fastest — time to first reach the distance, without matching the planet's velocity.

FIGURE 42.10 Solar-sail performance computed for this chapter. (a) Characteristic acceleration at 1 AU against the areal density of the whole spacecraft, with the sails flown so far. (b) Time to spiral from Earth's orbit to the distances of Mars, Mercury and Jupiter: lines from the spiral formula, dots from numerical trajectories.

TABLE 42.5 Solar-sail missions. Masses are for the whole spacecraft.

Mission	Operator	Launch	Sail area	Mass	Outcome
IKAROS	JAXA	2010	196 m²	≈ 310 kg	Venus flyby
NanoSail-D2	NASA	2010	10 m²	≈ 4 kg	Deorbit test
LightSail 2	Planetary Soc.	2019	32 m²	≈ 5 kg	Sailed in LEO
NEA Scout	NASA	2022	86 m²	14 kg	Never contacted
ACS3	NASA	2024	80 m²	16 kg	Deployed; boom bent
Solar Cruiser	NASA	—	≈ 1 670 m²	—	Cancelled 2023

42.5.2 Sails Flown So Far

JAXA's **IKAROS**, launched with the Venus orbiter Akatsuki in 2010, was the first interplanetary solar sail. It unfolded a 14 m square of 7.5 µm polyimide film by spinning, carried thin-film solar cells and steered with liquid-crystal panels of switchable reflectivity; JAXA measured a force of 1.12 mN. NASA's **NanoSail-D2** and The Planetary Society's **LightSail 2** followed in Earth orbit; LightSail 2 raised its apogee by about 2 km in its first days of sailing. NASA's 86 m² **NEA Scout**, launched on Artemis I in 2022, was never heard from (Figure 42.11), and the nearly 1 700 m² **Solar Cruiser** was not confirmed for flight and was closed out in 2023. The **Advanced Composite Solar**

Sail System (ACS3), a 16 kg, 12-unit CubeSat, deployed an 80 m² sail on four 7 m carbon-fibre composite booms in August 2024 (Figure 42.12). One boom was later reported bent and the sail was not actively steered, but the deployment proved booms that larger sails will need.



FIGURE 42.11 Engineers at NASA's Marshall Space Flight Center deploy a flight-like engineering unit of the 86 m² NEA Scout solar sail, first unfurled there in August 2016. Image: NASA/MSFC (EMMETT GIVEN/CHARLES BEASON/ JON).



FIGURE 42.12 The ACS3 sail seen by one of the spacecraft's four wide-angle cameras during deployment in orbit on 28 August 2024: reflective quadrants and a composite boom, curved by the lens. Image: NASA (NASA).

42.5.3 Electric and Magnetic Sails

The solar wind, streaming outwards at about 400 km/s, carries far less momentum than sunlight but can be intercepted by structures much lighter than film. Pekka Janhunen's **electric sail** (2004) would spin out thin tethers tens of kilometres long, charged to about 20 kV, whose electric fields repel solar-wind protons over a width far greater than the wires. Because the electron sheaths around the wires widen as the solar wind thins, the thrust falls only as 1/r rather than 1/r². Estonia's ESTCube-1 (2013) and Finland's Aalto-1 (2017) carried tether experiments, but neither managed to unreel its tether. A **magnetic sail** would deflect the solar wind with a large superconducting loop or a plasma-inflated magnetic bubble; none has been tested in flight.

EXAMPLE: SAILING FROM EARTH'S ORBIT TO MARS'S

A sail of 1 600 m² of 2.5 µm aluminised film (about 3.5 g/m², 5.6 kg) is held by 6 kg of composite booms and carries a 40 kg spacecraft: 51.6 kg in all.

1. **Areal density:** σ = 51.6 kg/1 600 m² = 32.2 g/m².
2. **Force at 1 AU:** $F = 2 \times 0.85 \times 4.54\ \mu\text{N/m}^2 \times 1\,600\ \text{m}^2 = \mathbf{12.3\ mN}$, so $a_c = 12.3\ \text{mN}/51.6\ \text{kg} = \mathbf{0.239\ mm/s^2}$ and β = 0.239/5.93 = 0.040.
3. **Δv budget:** at the optimum cone angle the push along the orbit is 0.385 × 0.239 = 0.092 mm/s², or 2.9 km/s per year at 1 AU, falling as the sail moves out.
4. **Spiral time:** with $\sqrt{\mu_\odot / r_0} = 29.8$ km/s and $(1.524)^{3/2} - 1 = 0.881$, $T \approx 29\,800 \times 0.881/(3 \times 0.385 \times 0.000239)\ \text{s} = 9.5 \times 10^7\ \text{s} \approx \mathbf{3.0\ years}$ to Mars's distance. The numerical trajectory gives 3.3 years, against 0.7 years for a chemical Hohmann transfer.
5. By the same formula, a one-year spiral would need $a_c \approx 0.72\ \text{mm/s}^2$, or σ ≈ 11 g/m² for the whole spacecraft — beyond any sail yet built. Sails win where the Δv is enormous, such as stations hovering sunward of L1, where no propellant supply would last.

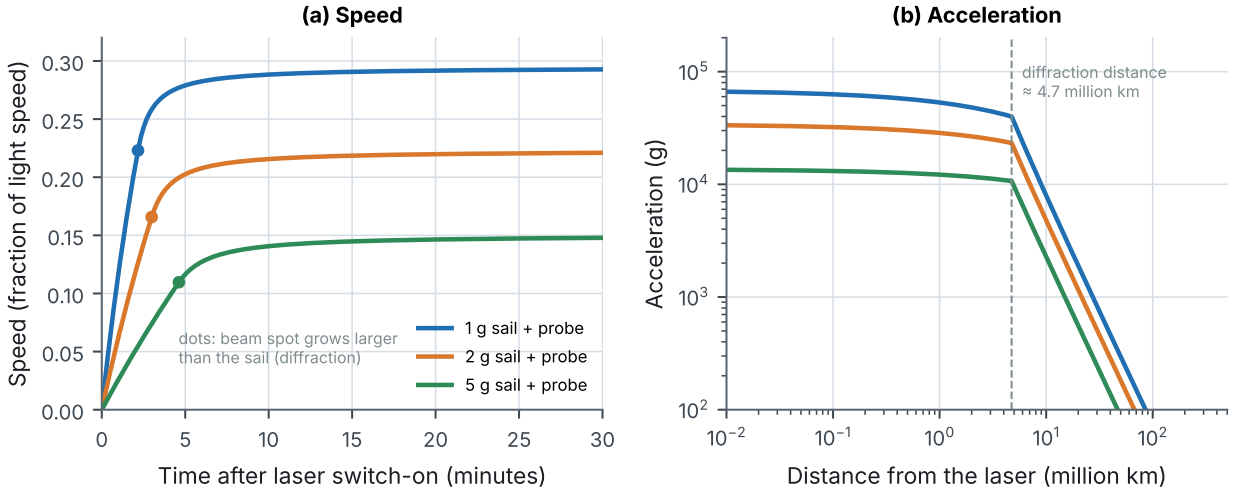
42.6 Beamed Energy and the Interstellar Challenge

The nearest star system, Alpha Centauri, is 4.37 light-years — about 276 000 AU — away. Voyager 1, the fastest object leaving the Solar System at about 17 km/s (3.6 AU per year), would need some 77 000 years to cover that distance. Reaching another star within a lifetime needs a tenth to a fifth of the speed of light.

42.6.1 Breakthrough Starshot

In 2016 the Breakthrough Initiatives announced **Starshot**, with US$ 100 million for research into leaving the engine on the ground. A phased array of about ten million lasers totalling up to 100 GW, spread over about a square kilometre, would focus on a sail a few metres across carrying a gram-scale "StarChip" with cameras, a laser transmitter and a radioisotope power source. Accelerating at about 10 000 g for some minutes, the probe would reach 15–20 % of the speed of light, pass Alpha Centauri about 20 years later and send images home by laser. A cost model by Kevin Parkin put the ground system at about US$ 8 billion, with an optimum 4.1 m sail accelerated for about nine minutes.

The physics is simple and severe (Figure 42.13). A perfectly reflecting sail struck by 100 GW feels $2P/c$ = 667 N; on a 2 g probe that is 34 000 g. But a laser beam spreads by diffraction. An array of diameter D focuses light of wavelength λ into a spot about $2.44\,\lambda L/D$ across at distance L. So a 3 km array keeps a 4.1 m sail fully lit only to about 4.7 million km, some 12 times the distance to the Moon. Beyond that the captured power falls with the square of distance and the relativistic Doppler shift weakens the push, so most of the speed is gained in the first three minutes.



100 GW at 1.064 µm from a 3 km phased array, 4.1 m perfectly reflecting sail; F = (2P/c)(1−β)/(1+β) × captured fraction (1, or $(L_o/L)^2$ beyond the diffraction distance L_o = D d/2.44λ); relativistic motion; light-travel delay neglected.

FIGURE 42.13 Starshot-class laser sails computed for this chapter: 100 GW from a 3 km array at 1.064 µm, a 4.1 m perfectly reflecting sail and total masses of 1, 2 and 5 g. (a) Speed against time. (b) Acceleration against distance from the laser.

The engineering is far harder. It needs a sail under a micrometre thick that reflects almost all of an intense beam and rides it stably, and ten million lasers phased through a turbulent atmosphere. The probe must also survive dust impacts at 60 000 km/s and send data home across 4 light-years. Starshot's first phase judged these problems to have credible conceptual solutions, but the programme was paused in 2024 to seek broader support.

EXAMPLE: THE NUMBERS BEHIND A LASER SAIL

A 2 g probe (1 g sail, 1 g chip) with a 4.1 m perfectly reflecting sail is pushed by 100 GW of 1.064 µm laser light from a 3 km array.

1. **Force and acceleration:** $F = 2P/c$ = 2×10^{11} W/3.00×10^{8} m/s = **667 N**, a pressure of 51 Pa on the 13.2 m² sail; $a = F/m$ = 3.3×10^{5} m/s² ≈ **34 000 g**.
2. **Without diffraction**, reaching 0.2c (6.0×10^{7} m/s) would take v/a = 180 s over $v^2/2a$ = 5.4 million km.
3. **Diffraction distance:** $L_0 = Dd/2.44\lambda$ = $3\,000 \times 4.1/(2.44 \times 1.064 \times 10^{-6})$ m = 4.7×10^{9} m. The simulation in Figure 42.13 reaches 0.17c there after 179 s and 0.22c after 10 minutes, 31 million km out.
4. **Energy:** the laser emits 100 GW × 600 s = 60 TJ; the probe's kinetic energy at 0.2c, $(\gamma - 1)mc^2$, is 3.7 TJ — an efficiency of about 6 %.
5. **Heating:** absorbing just 10^{-5} of the beam, the sail would have to shed 1 MW from 13 m², so it must be an almost perfect dielectric mirror. At 0.2c, Alpha Centauri is about 22 years away.

42.6.2 Fusion and Antimatter

Fusion releases about four times more energy per kilogram than fission, and its charged products could in principle be directed out as exhaust by magnetic fields (Figure 42.14). The British Interplanetary Society's **Project Daedalus** (1973–78) designed an uncrewed two-stage probe of about 54 000 t. Most of this was deuterium–helium-3 pellets, ignited by electron beams 250 times a second. The probe would carry a 450 t payload to Barnard's Star, 5.9 light-years away, at 12 % of the speed of light in about 50 years. Earlier, **Project Orion** (1958–65) studied a ship pushed by small nuclear explosions, until the 1963 Partial Test Ban Treaty and lack of funding ended it.

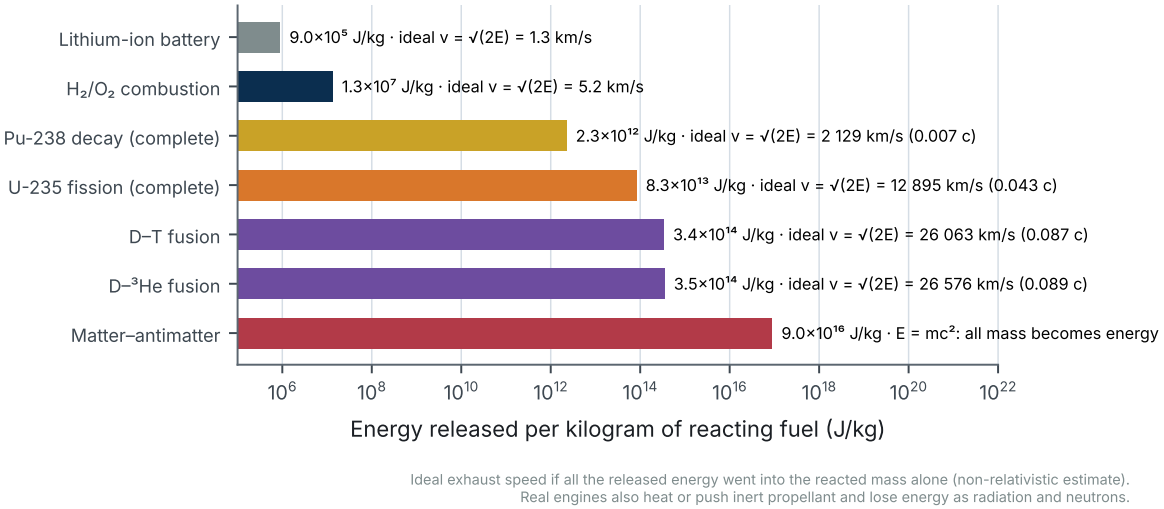


FIGURE 42.14 Energy released per kilogram of reacting fuel and the corresponding ideal exhaust speed, computed from reaction energies. Each step from chemistry to fission, fusion and annihilation multiplies the energy by orders of magnitude.

No fusion device has yet produced more energy than its whole system consumes. The US National Ignition Facility first achieved ignition in December 2022, releasing 3.15 MJ from a capsule heated by 2.05 MJ of laser light, but its lasers drew about 300 MJ from the grid. Compact

propulsion concepts such as the **Direct Fusion Drive**, based on the Princeton field-reversed configuration, are estimated to give 2.5–5 N per megawatt of fusion power at about 10 000 s. They remain small laboratory experiments.

Matter–antimatter annihilation converts all of the mass into energy: 1 g of antimatter meeting 1 g of matter releases 1.8×10^{14} J, as much as a 43-kiloton explosion. But antimatter is made in particle accelerators one particle at a time, at minute efficiency. CERN makes about 10^{12} antiprotons a day, and all it has ever produced amounts to less than 10 nanograms; at that rate a gram would take more than a billion years. Antimatter rockets remain thought experiments.

42.7 Breathing Air and Detonating Fuel

42.7.1 Scramjets and Combined Cycles

A hydrogen–oxygen rocket carries about 6 kg of oxygen for every kilogram of hydrogen. Air-breathing engines avoid that mass and reach fuel-specific impulses of 1 000–4 000 s, but only inside the atmosphere (Chapter 10). Above about Mach 5 the air must stay supersonic through the combustor, the regime of the **scramjet**. NASA's X-43A (Figure 42.15) reached Mach 9.6 for about 10 seconds in November 2004, and the US Air Force's X-51A flew for 210 s at up to Mach 5.1 in 2013. Combined-cycle engines switch between modes. The British SABRE design pre-cooled incoming air from about 1 000 °C to −150 °C in a fraction of a second. This let it run as an air-breathing rocket up to about Mach 5 before switching to pure rocket mode. Its developer, Reaction Engines, tested the pre-cooler at Mach 5 conditions in 2019 but entered administration in October 2024, and its intellectual property was being sold in 2025.

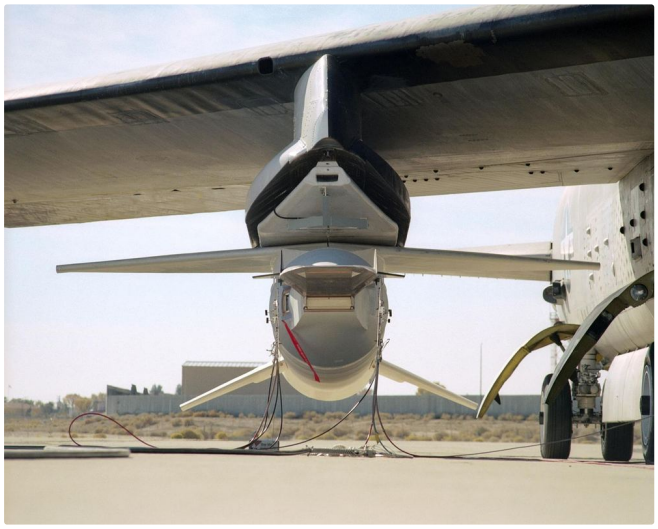


FIGURE 42.15 The X-43A scramjet research vehicle — the small wedge at the nose of its modified Pegasus booster — under the wing of NASA's B-52B on 15 November 2004, the day before its record Mach 9.6 flight. Image: NASA/AFRC (NASA/Tony Landis).

42.7.2 Rotating Detonation Engines

Conventional rocket engines burn propellant at almost constant pressure. In a **rotating detonation engine (RDE)**, propellants injected into an annular channel are consumed by one or more detonation waves — shock waves coupled to combustion. The waves race around the annulus at about 2 km/s, thousands of times a second. Detonation raises the pressure as it burns,

so analyses predict a higher specific impulse for the same propellants, and the thin reaction zone allows a much shorter chamber. In July 2021 Nagoya University and JAXA flew an RDE in space for the first time on the S-520-31 sounding rocket, producing about 500 N for 6 seconds. At NASA's Marshall Space Flight Center a 3D-printed copper-alloy rotating detonation rocket engine (RDRE) produced about 18 kN (4 000 lbf) in 2022. In September 2023 it fired for 251 seconds (Figure 42.16) at more than 25 kN (5 800 lbf). In May 2025 the US company Venus Aerospace reported a short flight of an RDRE-powered test vehicle from Spaceport America. Stabilising the waves and cooling the walls are the main challenges.

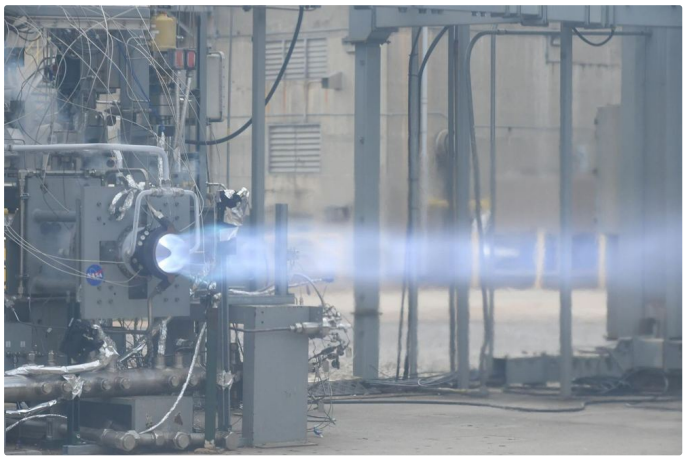


FIGURE 42.16 A 251-second hot fire of a full-scale, 3D-printed rotating detonation rocket engine at NASA's Marshall Space Flight Center in 2023, producing more than 25 kN (5 800 lbf) of thrust. Image: NASA/MSFC.

42.8 Propellantless and Exotic Ideas

42.8.1 Tethers

A long conducting wire moving through Earth's magnetic field acts as a generator. Its **motional EMF** is $\mathcal{E} = vBL$: about 2 kV for a 10 km tether in low orbit, where v = 7.6 km/s and B ≈ 30 µT (Figure 42.17). If the circuit is closed through the ionosphere by plasma contactors, a current flows and the magnetic force $I\vec{L} \times \vec{B}$ opposes the motion. A current of 1 A gives about 0.3 N of drag, enough to deorbit dead satellites without propellant (Chapter 21). Driving the current the other way with solar power turns the tether into a motor that raises the orbit. NASA's Plasma Motor Generator (1993) tested both modes with a 500 m tether. On the TSS-1R mission in February 1996, Space Shuttle Columbia deployed an Italian satellite on a conducting tether to 19.7 km. The system generated about 3 500 V and nearly 0.5 A before arcing burned through the tether and it broke.

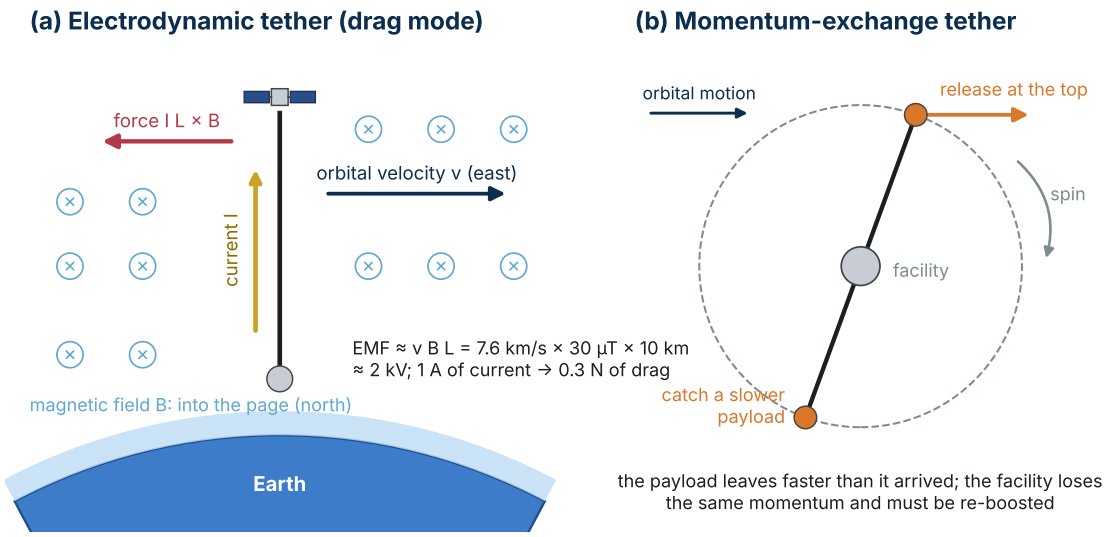


FIGURE 42.17 Tethers. (a) An electrodynamic tether in Earth's magnetic field: the induced current feels a force opposing the motion (or, driven by solar power, along it). (b) A rotating tether catches a slow payload at the bottom of its swing and releases it faster at the top, giving up momentum.

Non-conducting tethers exchange momentum mechanically. In 1993 the SEDS-1 experiment lowered a 26 kg payload on a 20 km tether below a Delta II stage and cut it loose into a re-entry trajectory. In 2007 the student-built YES2 deployed 31.7 km of tether — the longest structure ever deployed in space. The ultimate tether, a **space elevator** from the equator to beyond geostationary orbit, fails on materials. Even with a cable ten times thicker at geostationary altitude than at the ground, it would need a breaking length of about 2 100 km under Earth's gravity. The best commercial fibres offer about 400 km.

Two propellant-free techniques already serve missions: gravity assists (Chapter 24) and atmospheric braking. **Aerocapture** is a single deep pass through a planet's atmosphere inside an aeroshell to brake straight into orbit. It has not yet flown, but could cut the mass of an orbiter for Uranus, Neptune or Titan by a third or more (Chapter 25).

42.8.2 Reactionless Drives and Warp Drives

Every working propulsion system conserves momentum: something — exhaust, photons, the solar wind, a planet's magnetic field — must carry momentum the other way. The **EmDrive**, a closed, tapered microwave cavity proposed around 2001, was claimed to produce thrust without exhaust; a NASA Eagleworks paper in 2016 reported about 1.2 mN per kilowatt. Tests at TU Dresden with new thrust balances and suspensions reproduced the apparent thrust and then made it disappear. It came from thermal expansion of the mountings and from currents in the power cables interacting with Earth's magnetic field. The 2021 results bounded any anomalous force below that of a photon rocket of the same power: 3.3 μN per kilowatt, the push of simply shining the microwaves into space. This bound is hundreds to thousands of times below the earlier claims. "Mach-effect" thrusters tested the same way also failed to show real thrust.

In 1994 Miguel Alcubierre showed that Einstein's equations allow, mathematically, a "warp bubble" that contracts space ahead of a ship and expands it behind, carrying it faster than light without local motion. It needs negative energy density — "exotic matter" unknown in bulk — and the first estimates of the energy required exceeded the mass-energy of the observable universe.

Later designs cut the energy enormously but not to zero, and a bubble could not be steered from inside, because its front edge would lie beyond the reach of any signal from the ship. Warp drives remain an exercise in general relativity, not a propulsion technology.

AI IN ADVANCED PROPULSION

Plasma thrusters are hard to model: electrons in a Hall thruster cross the magnetic field far faster than classical collisions allow, and no first-principles theory yet predicts this "anomalous" transport. Researchers have used Bayesian inference and machine learning on plasma measurements to build data-driven transport models and fast design surrogates. In fusion, reinforcement learning has controlled the shape of a tokamak plasma at EPFL's TCV device (2022). Neural networks also approximate optimal low-thrust control for fast trajectory searches, and Hall-thruster fleets supply telemetry for anomaly detection (Chapters 48, 49 and 52).

KEY TAKEAWAYS

- Mass ratio grows as $e^{\Delta v/v_e}$, and jet power $\tfrac{1}{2}Fv_e$ makes faster exhaust cost power.
- Electric propulsion is scaling up (12 kW AEPS built, 100 kW run), but 100-day Mars trips need about 1 kW/kg.
- Nuclear thermal rockets (850–950 s) roughly halve the LEO mass of a six-month Mars transfer; DRACO was cancelled in 2025.
- Space fission power is returning: SR-1 Freedom (≥ 20 kWe, nuclear-electric) is planned for December 2028, Lunar Reactor-1 for 2030.
- Sails flown so far reach at most about 0.05 mm/s^2; a one-year trip to Mars's distance needs about 11 g/m^2.
- A 100 GW laser could push a gram-scale sail to about 0.2c in minutes, but the engineering is unsolved and Starshot is paused.
- Scramjets, detonation engines and tethers serve niches; reactionless and warp drives fail the tests of physics.

REVIEW QUESTIONS

1. Compare mass ratios for Δv = 10 km/s at v_e = 4.4, 8.8 and 40 km/s. Why not raise v_e without limit?
2. A 50 kW Hall-thruster system works at 60 % efficiency and 2 500 s. What is its thrust, and how long would it take to give a 20 t spacecraft 5 km/s?
3. Define specific power α. Using Figure 42.3, explain why a 40-day crewed trip to Mars is unrealistic with present technology.
4. Why does a nuclear thermal rocket reach twice the Isp of a hydrogen–oxygen engine? What do its moderator, control drums and shield do?
5. Why was DRACO cancelled, and why is ground-testing an NTR harder now than in the 1960s?
6. A 1 MWe NEP system has 25 % efficient converters and radiators at 500 K (emissivity 0.9). Estimate the radiator area.
7. A 4 kg CubeSat carries a 40 m^2 sail. Calculate its characteristic acceleration at 1 AU (η = 0.85) and the time to spiral out to 1.5 AU.
8. Why does a laser sail's push collapse beyond a certain distance, and how does doubling the array diameter change it?
9. How does a rotating detonation engine work? Describe two tests made since 2021.
10. Why do careful experiments rule out the EmDrive? What thrust must any device radiating 1 kW of electromagnetic power produce?